From Homo Economicus to Homo Moralis: A Bewley Theory of the Social Welfare Function*

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We present an aggregation theory that allows the empirical estimation of a Social Welfare Function (SWF) in heterogeneous-agent economies. Individuals are assumed to hold different views on how the government should value others' welfare, giving rise to Individual Welfare Functions (IWFs) shaped by personal life experiences. The SWF is obtained as a politically weighted aggregation of these IWFs. We develop an estimation strategy that identifies both the SWF and IWFs from observed fiscal instruments—capital, consumption, and labor taxes, as well as public debt—extending the inverse-optimal approach to a general equilibrium setting. Applying this framework to France and the United States, we find that France's SWF is more egalitarian, emphasizing low-income welfare, while the US SWF favors low-redistribution. Simulating the US fiscal system under the French SWF reveals that welfare preferences play a central role in shaping fiscal policy and income inequality.

Keywords: Social Welfare Function, Fiscal systems, Heterogeneous agents.

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1 Introduction

Heterogeneous-agent models or more precisely, incomplete-insurance-market models are now widely used to identify new mechanisms and to analyze the effects of policies, such as monetary or fiscal policies. These models allow for investigating intertemporal and distributive trade-offs. A recent literature analyzes optimal policies in these environments to uncover normative implications. As with any model featuring heterogeneous agents, an immediate question arises: Which welfare objective should be adopted? The answer to this question is important both quantitatively and theoretically. Quantitatively, the weights given to different agents will affect the optimal policies, especially if these agents are differently affected by the policy under consideration. Theoretically, the qualitative properties of the allocation, such as the existence of a stationary equilibrium or the use of certain taxes, may also depend on the choice of the welfare objective.

The difficult question of interpersonal utility comparison has a standard solution in the literature: defining an aggregation procedure that combines individual welfare into a social welfare objective, using a Social Welfare Function (henceforth, SWF). Among the class of linear SWFs, the heterogeneous-agent literature often employs the utilitarian SWF for normative analysis following Aiyagari (1995), where equal weights are assigned to all agents. This SWF serves as a useful benchmark, but is only one among many alternatives, each with distinct properties. Some normative analyses use Pareto-Negishi weights, allowing the planner to differentially weight agents, or consider SWFs with inequality aversion to accommodate more general social objectives.

However, the assumption of country-wide social preferences raises challenges, as inequality aversion and the desire for redistribution are heterogeneous among individuals within a given country and may diverge from their strict individual preferences, as evidenced by surveys or experiments. Additionally, some individuals may vote against their direct material interests, guided by their own social preferences, which may thus differ from their strict individual preferences (see the literature review below). As a consequence, a country's social preferences can be viewed as the outcome of a (possibly biased) aggregation of heterogeneous individual social preferences. In this paper, we present a theory of the SWF that explicitly models this aggregation process. The resulting SWF is more general than the utilitarian SWF and sufficiently flexible to be estimated and applied in heterogeneous-agent models.

Our aggregation theory of the SWF rests on four assumptions. First, individuals hold their own view on how the planner should care about the welfare of society's members. This corresponds to so-called ethical preferences (Arrow, 1951; Harsanyi, 1955; Sen, 1977). As Harsanyi (1955) states, ethical preferences reflect "what [an individual] prefers only in those possibly rare moments when he forces a special impartial and impersonal attitude on himself."[1] We represent these ethical preferences using Individual Welfare Functions (IWFs), which are agent-specific and thus heterogeneous. Second, the heterogeneity in IWFs arises from individuals' life experiences. As a result, an agent IWF may evolve as their economic history unfolds. Indeed, following the Bewley tradition, we assume that the relevant life experience shaping IWFs is an agent's economic history. While our framework could accommodate other dimensions of heterogeneity, this representation is sufficiently rich to explore a wide range of political implications. Third, we assume that ethical preferences are represented by a discounted sum of weighted period utilities, where the weights capture how agents perceive the utility of others. This representation is flexible enough to encompass a variety of moral and political concerns—including egalitarian, utilitarian or self-interested motives, or low-redistribution preferences—in the planner's objective.[2] Fourth, the SWF used by the planner to design policies is a (possibly biased) aggregation of the heterogeneous IWFs. We further assume that heterogeneity in agents' ability to influence the planner's decisions – which we term political power – may introduce bias into the aggregation.

We prove that the SWF resulting from this construction is stationary at the steady state, despite the evolution of agents' ethical preferences with their individual histories. For this reason, we refer to this framework as a *Bewley* theory of the SWF. This construction yields a flexible and consistent SWF that can be applied in heterogeneous-agent models. In alternative models featuring ex-ante heterogeneity (e.g., Bassetto, 2014), Pareto-Negishi weights offer a consistent and flexible approach to capturing the relative importance of different groups in designing optimal policies. However, in models where heterogeneity emerges from incomplete insurance markets, agents may transition between rich or poor status depending on the realization of idiosyncratic shocks. Thus, assigning distinct weights to rich and poor agents necessitates weights over idiosyncratic histories—a feature inherent to our construction.

An outcome of our construction is that it generates restrictions enabling the estimation of the SWF from data. The estimation of SWFs relies on revealed social preferences, or, more precisely, the inverse optimal approach (see Heathcote and Tsujiyama, 2021, for a recent example; additional references are provided in the literature review), which we extend to an intertemporal general equilibrium environment. This strategy allows us to link the estimated SWFs to public finance concepts such as Social Marginal Welfare Weights (SMWW) and the Marginal Value of Public Funds (MVPF). Identifying the heterogeneous IWFs underlying the SWFs is more challenging. While experimental or survey data can inform priors, no direct quantitative evidence is currently available. In particular, the high dimensionality of IWFs renders them sensitive to the observer's priors. To address

[1] These ethical preferences have a long tradition. They are the preferences of the "impartial spectator" of Adam Smith (1759) or the preferences under the "veil of ignorance" of Rawls (1971).

[2] We provide in Section 2 the definitions of these concepts in our construction.

this, we develop an estimation strategy that decomposes IWFs into prior-dependent and prior-independent components, each of which can be analyzed separately.

We apply our framework to estimate the SWF for both France and the United States, two countries that exhibit sharp differences in taxation and inequality. First, we demonstrate that a fiscal system comprising a progressive labor tax, a capital tax, a consumption tax, and public debt, when combined with empirically relevant income risk, can closely replicate the observed income and wealth inequality in both countries for the year 2007. We select 2007 to avoid the distorting effects of the global financial crisis and the subsequent COVID-19 shock, which introduced significant transitory changes in fiscal structures. We then consider a heterogeneous-agent model in which agents face income risk and the fiscal system finances a public good. A Ramsey planner sets fiscal policy to maximize an SWF derived from our construction. To solve the intertemporal program in general equilibrium, we employ the truncation method developed by LeGrand and Ragot (2022a, 2023), extending it to accommodate a general disutility of labor that goes beyond the GHH preferences previously analyzed in LeGrand and Ragot (2025). Finally, using the first-order conditions (FOCs) of this Ramsey program, we compute the SWF and the IWFs consistent with the observed allocations and tax systems in France and the US. This approach extends the standard inverse optimal method to an intertemporal general equilibrium framework.

Our estimation of the SWFs produces three main findings. First, the SWFs for France and the US exhibit stark differences. In the US, SWF weights increase with income, assigning the highest weight to high-income agents, followed by middle-class agents and the lowest to low-income agents. This pattern aligns with the decreasing SMWWs documented in the public finance literature. In contrast, the French SWF assigns substantial weight to low-income agents. Notably, the SWF is U-shaped: weights initially decrease with income, with the middle class receiving the lowest weight, before increasing for high-income agents. Thus, the French SWF reflects both egalitarian concerns for the poor and a high valuation of productive individuals. Second, to evaluate the influence of the SWF on inequality, we simulate the optimal US fiscal system under the counterfactual scenario in which the US adopts the French SWF, holding all else constant. We find that the wealth Gini coefficient would decline from 78% to 63% falling below the French level of 68%. This result suggests that social preferences play a crucial role in shaping fiscal systems and, consequently, household (post-tax) inequality.

Third, we decentralize the aggregate SWF into heterogeneous IWFs to evaluate the diversity of social preferences within each country. Following the political economy literature, we use turnout data to estimate political weights in the US and in France. We then identify the set of IWFs consistent with the observed SWF. Our findings indicate that the middle class in the US prefers limited redistribution, whereas in France, it is predominantly egalitarian. Although our framework does not offer a theory explaining why ethical preferences vary across individuals, it provides a quantitative assessment of their role in determining observed inequality.

The paper proceeds as follows. In the next subsection, we relate our construction and methods to the existing literature. We provide the basics of our Bewley construction of SWFs in Section 2 in the context of a simple model. The construction is generalized to an infinite horizon model in Section 3. Section 4 presents the environment in which we will compute the Ramsey program and conduct our estimation of SWF weights. Finally, the quantitative investigation is presented in Section 5, while Section 8 concludes.

Related literature. This paper contributes to the recent literature on optimal policy design in heterogeneous-agent models and on the inverse optimal approach. Early contributions, such as Aiyagari (1995), analyze the general properties of capital tax in these models. Aiyagari and McGrattan (1998) compute the optimal steady-state level of public debt. Dávila et al. (2012) solve for a constrained-efficient allocation and demonstrate that the steady-state capital stock can be too low. Some papers compute the optimal path of relevant instruments using numerical tools (e.g., Conesa et al., 2009; Dyrda and Pedroni, 2022), while others rely on the FOCs of the Ramsey problem to solve for optimal policies (e.g., Bhandari et al., 2021; LeGrand and Ragot, 2022a; Açikgöz et al., 2022). Auclert et al. (2024) use a sequence-space approach to compute long-run fiscal policy in heterogeneous-agent models. They show that a steady-state Ramsey equilibrium may not exist under a utilitarian SWF. These models typically consider the utilitarian SWF as the objective of the planner. However, an important characteristic of Bewley environments is that, even if agents are ex ante identical, their welfare will differ according to the realization of idiosyncratic shocks. Our construction of the IWFs and SWF allows us to rank history-dependent welfare, which cannot be achieved with Pareto-Negishi weights for ex-ante heterogeneous agents. This enables us to derive a sufficiently general formulation that can be estimated.

The inverse optimal approach we use is a common tool in static models of public finance.[3] Considering heterogeneous-agent models, some papers consider dynamic environments introducing Pareto-Negishi weights (as LeGrand and Ragot, 2025, Olivi et al., 2025). Other papers (e.g., LeGrand et al., 2025 and McKay and Wolf, 2025) introduce more general weights, which can be history dependent. Dávila and Schaab (2022) refer to such weights as dynamic-stochastic weights. All these papers implicitly estimate weights to impose optimality at the steady-state, in order to compute the optimal policy following a transitory aggregate shock, starting from a well-defined optimal steady-state. Our construction can be viewed as a micro-foundation for this general form of the SWF,

[3] Heathcote and Tsujiyama (2021) also estimate the SWF in a static environment. Bargain and Keane, 2010; Bourguignon and Amadeo, 2015; Lockwood and Weinzierl, 2016; Hendren, 2020 use an inverse optimal approach in environments without saving and with general tools.

allowing one to explicitly estimate the SWF at the steady state.

2 A Bewley theory of the SWF: Some initial definitions

We present our aggregation theory of the SWF in a simple environment that abstracts from heavy algebra and notation. We first explain how we construct individual welfare functions (IWFs) and then aggregate them to obtain the social welfare function (SWF) (Sections 2.1 and 2.2). We next show how the SWF and the IWFs can be identified from observed allocations (Section 2.3). Finally, we discuss interpretations of the estimates in terms of political and social justice concepts – utilitarian, egalitarian, and low-redistribution principles (Section 2.3).

2.1 The setup

We consider a one-period, one-good economy. The unique good is a final consumption good over which agents have preferences represented by a utility function u, assumed to satisfy standard properties: $u' > 0$, $u'' < 0$, and $u'(0) = \infty$. The economy is populated by two types of agents with total population normalized to one, and each type representing an equal share of 1/2. Types differ only in their endowment. Let $y_1 > y_2$ denote the two endowment levels.

A benevolent planner chooses the best allocation subject to feasibility. Allocations are ranked by a SWF defined below. Feasibility reflects the planner's ability to transfer resources across types at a quadratic redistribution cost scaled by $\kappa > 0$ intended to capture all distortions associated with redistribution, which are micro-founded in the general model below. Implementing consumption c_i for an agent with endowment y_i destroys resources equal to $\frac{\kappa}{2}(c_i - y_i)^2$. Focusing on symmetric allocations in which all agents of type i consume c_i, the resource constraint is:

$$\sum_{i=1}^{2} \left(c_i + \frac{\kappa}{2}(c_i - y_i)^2 \right) \leq \sum_{i=1}^{2} y_i. \tag{1}$$

To ensure interior solutions, we assume redistribution costs are not too high and impose that $\kappa y_i < 1$.

2.2 Individual and social welfare functions

We construct the SWF from individual ethical preferences (IWFs) in three steps – later replicated in the general model (Section 3): (i) each agent's subjective valuation of others' welfare; (ii) the IWFs, representing ethical preferences; (iii) the SWF, representing the planner's preferences.

The Individual Welfare Function (IWF). In this static setting, $u(c_i)$ is agent i's utility from consumption c_i. Beyond individual utility, agent i holds a view about how the planner should count their own welfare and the welfare of others. We model this subjective valuation using loading factors on individual utility. Let $\tilde{\omega}_{ij}$ denote the weight that agent i assigns to agent j; then i's valuation of j's welfare is $\tilde{V}_{ij} = \tilde{\omega}_{ij} u(c_j)$.

Agent i's ethical preferences are constructed by aggregating their valuations of both types. The IWF of agent i is the equally weighted sum of their subjective valuations: $IWF_i := \frac{1}{2}\tilde{V}_{i1} + \frac{1}{2}\tilde{V}_{i2}$, or

$$IWF_i = \frac{1}{2}\tilde{\omega}_{i1} u(c_1) + \frac{1}{2}\tilde{\omega}_{i2} u(c_2). \tag{2}$$

Ethical preferences are heterogeneous because there are two income types; thus, heterogeneity in endowments is accompanied by heterogeneity in IWFs. This aligns with empirical evidence documenting heterogeneity in ethical preferences, partly driven by social position (see Gaertner and Schokkaert, 2012 and Stantcheva, 2021 for recent analyses based on large US surveys).

Two remarks clarify the generality of the weights $(\tilde{\omega}_{ij})_{i,j=1,2}$:

1. *Intensity of preferences.* We impose no normalization on the weights – neither $\sum_j \tilde{\omega}_{ij} = 1$ nor $\tilde{\omega}_{ii} = 1$. Because IWFs will be aggregated, the weights carry both ordinal and cardinal content. Normalization is not innocuous: it would suppress heterogeneity in the *intensity* of ethical preferences (in Arrow's terminology, Arrow, 1951). Consistent with this discussion, we adopt a cardinal representation of IWFs, enabling aggregation via political weights. Our empirical strategy disentangles intensity from political influence.[4]

2. *Possibility of spitefulness or discrimination.* We do not restrict the sign of $(\tilde{\omega}_{ij})_{i,j=1,2}$; weights can be negative. Then, an agent's IWF may decrease with others' utility. Such negative interpersonal concerns have been modeled in asset pricing or macroeconomics to capture "keeping up with the Joneses" (e.g., Abel, 1990; Campbell and Cochrane, 1999) and are supported by experiments (e.g., Fehr et al., 2013), where this trait is termed *spitefulness*.[5] Negative IWF weights can also represent discrimination against certain types (Piacquadio, 2017).

[4] See Section 7 for a longer discussion and our assumption of a linear aggregator.

[5] Fehr and Schmidt (2006): "A spiteful person always values the material payoff of relevant reference agents negatively. Such a person is, therefore, always willing to decrease the material payoff of a reference agent at a personal cost to themself."

The Social Welfare Function (SWF). We aggregate IWFs into the SWF using *political weights*. Following the political-economy literature, agents can differ in their ability to influence the planner – through institutions, lobbying, or voting rules. We capture this heterogeneity by political weights $(\omega_{P,i})_i$, which load IWFs according to type. These weights will be estimated empirically. The SWF is then:

$$SWF = \omega_{P,1} IWF_1 + \omega_{P,2} IWF_2. \tag{3}$$

Substituting (2) yields:

$$SWF = \omega_1 u(c_1) + \omega_2 u(c_2), \tag{4}$$

where the SWF weights are defined (for some $\omega > 0$) by:

$$\omega_1 := \frac{\omega}{2}(\omega_{P,1}\tilde{\omega}_{11} + \omega_{P,2}\tilde{\omega}_{21}), \tag{5}$$

$$\omega_2 := \frac{\omega}{2}(\omega_{P,1}\tilde{\omega}_{12} + \omega_{P,2}\tilde{\omega}_{22}). \tag{6}$$

Unlike IWF weights, SWF weights need not have a cardinal interpretation; we therefore normalize without loss of generality by choosing ω so that $\omega_1 + \omega_2 = 1$.

This construction embeds standard benchmarks. The utilitarian SWF arises with self-interested agents.[6] Such agents value only their own welfare. Their IWF aligns with their utility: $IWF_i = u(c_i)$, and correspond to IWF weights $\tilde{\omega}_{ij} = 1_{i=j}$, where $1_{i=j} = 1$ if $i = j$ and 0 otherwise. Constant political weights, $\omega_{P,i}$, and proper normalization factor ω in (5)–(6) then yields a utilitarian SWF with $\omega_1 = \omega_2 = 1/2$.[7] Moreover, SWF weights need not be positive; negative SWF weights can emerge from the aggregation. In this simple setting, that possibility implies the planner may select allocations that are not Pareto optimal. See also Section 2.3 for further discussion.

2.3 Inverse optimal approach: Identifying the Welfare Functions

The identification of SWF weights. The inverse optimal approach infers social preferences from the observed allocation and available fiscal instruments, assuming the instruments are optimally set by the planner. In our framework, the fiscal system allows the planner to implement any allocation (c_1, c_2) that maximizes the SWF in (4) subject to the resource constraint (1). For given SWF weights (ω_1, ω_2), the optimal allocation satisfies the FOC:

$$\frac{\omega_1 u'(c_1)}{1 + \kappa(c_1 - y_1)} = \frac{\omega_2 u'(c_2)}{1 + \kappa(c_2 - y_2)}, \tag{7}$$

[6] We avoid terms such as "selfish" or "rational" with strong connotations; Sen (1977) uses "self-seeking."

[7] Even with heterogeneous preference intensity of self-interested agents, $\tilde{\omega}_{ij} = \lambda_i 1_{i=j}$ with $\lambda_1 \neq \lambda_2$, we can recover the utilitarian SWF if political weights offset intensity: $\omega_{P,i} = \frac{\lambda_i^{-1}}{\lambda_1^{-1} + \lambda_2^{-1}}$. With $\omega = \lambda_1^{-1} + \lambda_2^{-1}$, we again obtain $\omega_1 = \omega_2 = 1/2$.

together with the constraint (1). The condition $\kappa y_i < 1$ guarantees that the FOC is well-defined.

The inverse optimal approach uses the same FOC but reverses the mapping: it infers the weights from the allocation. For any allocation (c_1, c_2) satisfying the resource constraint (1), the FOC (7) implies:

$$\frac{\omega_1}{\omega_2} = \frac{u'(c_2)}{u'(c_1)} \frac{1 + \kappa(c_1 - y_1)}{1 + \kappa(c_2 - y_2)}, \tag{8}$$

which, together with a normalization, determines the weights (ω_1, ω_2). While one can also compute social *marginal* welfare weights (SMWW) $\omega_i u'(c_i)$, we focus on SWF weights, which more directly characterize the SWF and facilitate political and ethical interpretations – see Section 2.3.

The identification of IWF weights. Our quantitative analysis (Section 5) primarily estimates the SWF. As (8) shows, this step is independent of IWFs. Nonetheless, recovering IWFs consistent with the estimated SWF provides valuable insights into the underlying heterogeneity in social perceptions.

Additional information, such as surveys or experiments, may provide priors on IWFs, denoted $\omega_{i,j}^{prior}$ for $i, j \in \{1, 2\}$. Such evidence is indirect. For example, World Values Survey responses indicate that demand for redistribution varies with income, but it is unclear whether this reflects self-interest or ethical views. To address this ambiguity, we estimate IWFs by decomposing them into two components: one independent of priors and one independent of the SWF.

Specifically, we first estimate political weights $(\omega_{P,i})_i$ for each group using political-economy evidence.[8] For normalization, we set their Euclidean norm to one: $\sqrt{\omega_{P,1}^2 + \omega_{P,2}^2} = 1$. We then define the estimated IWFs as the weights closest (in quadratic norm) to the priors $\omega_{i,j}^{prior}$, subject to consistency with the observed SWF:

$$\begin{aligned}(\tilde{\omega}_{i,j})_{i,j=1,2} =& \operatorname{argmin}_{(\hat{\omega}_{i,j})_{i,j}} \frac{1}{2} \sum_{i,j=1,2} \left(\hat{\omega}_{i,j} - \omega_{i,j}^{prior}\right)^2, \\ &\text{s.t. } \omega_i = \frac{\omega}{2}(\omega_{P,1}\hat{\omega}_{1,i} + \omega_{P,2}\hat{\omega}_{2,i}), \text{ for } i = 1, 2\end{aligned}$$

The solution of this optimization problem is:

$$\tilde{\omega}_{i,j} = F(i,j) + \underbrace{\omega_{P,i}\omega_j}_{\text{CIWF}}, \text{ for } i, j = 1, 2$$

where $F(i,j) := \omega_{i,j}^{prior} - \omega_{P,i}\left(\omega_{P,1}\omega_{1,j}^{prior} + \omega_{P,2}\omega_{2,j}^{prior}\right)$.

The term $F(i,j)$ depends on the priors and political weights but not on the estimated SWF weights (ω_1, ω_2). Consequently, the second term, $\omega_{P,i}\omega_j$, represents the contribution of the estimated SWF to the IWFs, independent of priors. Their interpretation is intuitive:

[8] Empirical measures and robustness checks are in Section 6.

if the aggregate weight on type j is large (high ω_j), then types with greater political weight (high $\omega_{P,i}$) must assign more weight to j to rationalize the aggregate. We will therefore report the term $\omega_{P,i}\omega_j$, which captures the contribution to IWFs (CIWFs) that is independent of priors: $CIWF_{i,j} := \omega_{P,i}\omega_j$. The results are provided in Section 3.

Interpretations of Welfare Functions Given an observed allocation, welfare weights have a natural interpretation in social choice and moral philosophy (Saez and Stantcheva, 2016). We use the following concepts in the quantitative exercise of Section 5:

- *Utilitarian.* Type-i agents assign equal weight to all agents: $\tilde{\omega}_{i1} = \tilde{\omega}_{i2}$. With $\omega_1 = \omega_2$ and $\kappa > 0$, the planner accepts some inequality ($c_1 > c_2$) due to redistribution costs.

- *Egalitarian.* Type-i agents oppose inequality, placing greater weight on poorer agents: $\tilde{\omega}_{i1} < \tilde{\omega}_{i2}$. The planner reduces c_1 and raises c_2, lowering inequality at the cost of total consumption.

- *Low-redistribution.* Type-i agents favor limited redistribution: $\tilde{\omega}_{i1} > \tilde{\omega}_{i2}$. Motivations include maximizing output (minimizing distortions), merit-based views of endowments, or a preference for lower distribution or a smaller state.

In a two-type economy, these three categories partition the space of welfare weights, imply different redistribution levels, and align with distinct moral norms – an advantage of this representation. Each agent type, and the planner, belongs to exactly one category. If agent types fall in different categories, the planner's classification depends on the interaction of political weights and preference intensity. This partition does not extend to economies with more than two types; we return to this in Section 3.

What about Pareto deviations? Our SWF construction is not restricted to Pareto-optimal SWFs. In the simple setup, non-Pareto optimal allocations can arise with negative SWF weights. In richer settings, such allocations may occur even with positive SWF weights. For example, in the presence of individual risk, a low-redistribution planner may be reluctant to implement insurance even if it is ex ante individually optimal for all agents. As shown by Sen (1970), the SWF need not satisfy the Pareto principle if the planner considers factors beyond individual utility, such as ethical, moral, or political concerns. Kaplow and Shavell (2001) further show that Pareto can fail when departing from welfarism, i.e., when the SWF goes beyond agents' intertemporal welfare. Since our goal is to rationalize observed fiscal systems and explore their political implications, we relax welfarism rather than restrict the generality of the SWF.

3 A Bewley theory of the SWF: The intertemporal case

We extend the previous construction of the SWF to a general intertemporal framework, which is a main contribution of this paper. Our approach relies on the sequential representation of the heterogeneous-agent model, a formulation particularly well suited to normative analysis.[9] Before constructing the SWF in Section 3.2, we present the setup, which coincides with the quantitative model used in the subsequent sections.

3.1 The setup

We consider an infinite-horizon economy with incomplete financial markets. Time is discrete, indexed by $t \geq 0$. The population consists of a continuum of ex-ante identical agents of measure one.

Risk structure. Idiosyncratic risk is modeled as an uninsurable individual productivity shock y_t that can take Y distinct values in the finite set $\mathcal{Y}$. The productivity risk follows a first-order Markov chain with transition matrix $(\Pi_{yy'})_{y,y' \in \mathcal{Y}}$. This matrix is assumed to be irreducible and aperiodic, ensuring a unique stationary distribution $(\pi_y)_{y \in \mathcal{Y}}$, satisfying $\sum_{y \in \mathcal{Y}} \pi_y = 1$.

Let $y^t = \{\ldots, y^t_{t-1}, y^t_t\}$ denote a one-sided infinite sequence of productivity levels representing an agent's history up to date t. The set of all such histories is $\mathcal{Y}^\infty$. Since we track the evolution of histories over time, we retain time subscripts for clarity. For a given history $y^t \in \mathcal{Y}^\infty$, we use: (i) $y^t_\tau \in \mathcal{Y}$ for the productivity level at date $\tau \leq t$; (ii) $y^{s,t}$ for the truncation of y^t at date $s \leq t$, such that $y^{s,t}$ and y^t coincide for all $\tau \leq s$, i.e., $y^{s,t}_\tau = y^t_\tau$. Distinct histories are indicated with decorators, e.g., $\tilde{y}^t$ and y^t.

Initial distribution. To simplify notation, we impose two assumptions on the initial distribution: (i) all agents start the economy with an initial infinite history in $\mathcal{Y}^\infty$; (ii) initial wealth depends only on an agent's initial history. This assumption encompasses, as special cases, economies in which all agents share identical wealth or start from the steady-state wealth distribution. Because our analysis focuses on stationary allocations and distributions, this restriction entails no loss of generality.

[9] While some studies (e.g., Chang et al., 2018) have considered social weights depending on endogenous variables, such as consumption or wealth, this approach can introduce inconsistencies and multiple equilibria. When social weights vary with the allocation, the planner must account for the feedback between the allocation and the weights, which complicates the optimization problem and may undermine the stability of the solution. To address these issues, we instead allow social weights to depend on idiosyncratic histories rather than on wealth or other endogenous variables. This choice ensures analytical tractability and preserves a clear separation between individual choices and social valuation.

The sequential representation. A subtlety of infinite-horizon models is that the set of histories has the cardinality of the continuum (neither finite nor countable). Hence, the probability space over histories must be defined with a general measure. We construct the probability space $(\mathcal{Y}^\infty, \mathcal{F}, \mu)$, where $\mathcal{F}$ is a suitable σ-algebra and μ a measure (see Appendix A.1). For any measurable set $B \in \mathcal{F}$, $\mu(B) \geq 0$ is the measure of agents currently in histories $y^t \in B$.[10] Normalizing population size to one implies $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$.[11]

Transitions across histories are defined as follows. Consider $y^{t+1}, \tilde{y}^t \in \mathcal{Y}^\infty$. The probability of moving from $\tilde{y}^t$ in the current period to y^{t+1} in the next period equals the probability of switching from $\tilde{y}^t_t$ to y^{t+1}_{t+1} if $y^{t,t+1}$ and $\tilde{y}^t$ coincide, and zero otherwise. Denote this conditional probability by $\mu_1(\cdot \mid \tilde{y}^t)$, formally: $\mu_1(dy^{t+1} \mid \tilde{y}^t) = \Pi_{\tilde{y}^t_t y^{t+1}_{t+1}} \delta_{\tilde{y}^t}(dy^{t,t+1})$, where $\delta_{\tilde{y}^t}$ is the Dirac delta at $\tilde{y}^t$.[12]

By induction, the probability of moving from $\tilde{y}^t$ to a future history y^{t+s} in $s \geq 1$ periods is $\mu_s(dy^{t+s} \mid \tilde{y}^t) = \prod_{k=0}^{s-1} \Pi_{y^{t+s}_{t+k} y^{t+s}_{t+k+1}} \times \delta_{\tilde{y}^t}(dy^{t,t+s})$. In words, transitioning from $\tilde{y}^t$ to y^{t+s} requires that the histories coincide up to period t, followed by the cumulative probability of the sequence $y^{t+s}_{t+1}, \ldots, y^{t+s}_{t+s}$.

Individual intertemporal welfare. For a given allocation, let $U(y^t)$ denote the period utility of an agent with history y^t. To lighten notation, we suppress explicit dependence on the allocation. In the benchmark case with private consumption and labor supply (as in our quantitative application), $U(y^t) := u(c(y^t)) - v(l(y^t))$, where $c : \mathcal{Y}^\infty \to \mathbb{R}^+$ and $l : \mathcal{Y}^\infty \to \mathbb{R}^+$ are the policy functions for consumption and labor.

Intertemporal welfare is time separable and of the expected-utility form. It is thus defined as the discounted sum of expected period utilities over all future dates:

$$V(y^t) = \sum_{s=0}^{\infty} \beta^s \int_{\tilde{y}^{t+s} \in \mathcal{Y}^\infty} U(\tilde{y}^{t+s}) \mu_s(d\tilde{y}^{t+s} | y^t). \tag{9}$$

As we consider a steady-state allocation, the intertemporal welfare can be written recursively as:

$$V(y^t) = U(y^t) + \beta \mathbb{E}_{\tilde{y}^{t+1}} \left[V(\tilde{y}^{t+1}) | y^t \right], \tag{10}$$

where $\mathbb{E}_{\tilde{y}^{t+1}} [V(\tilde{y}^{t+1}) | y^t] = \int_{\tilde{y}^{t+1} \in \mathcal{Y}^\infty} V(\tilde{y}^{t+1}) \mu_1(d\tilde{y}^{t+1} | y^t)$ is a conditional expectation. See Appendix A.3.

[10] A single history is an event of measure zero in $\mathcal{F}$. Therefore, statements that hold for all histories y^t should be understood as holding almost surely.

[11] In finite time, we would write $\sum_{y^t \in \mathcal{Y}^t} \mu_t(y^t) = 1$ instead of $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$. The finite-time intuition carries over directly to the infinite-horizon framework.

[12] The measure μ_1 satisfies the standard properties of a conditional probability: $\mu_1 \geq 0$, $\int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_1(dy^{t+1} \mid \tilde{y}^t) = 1$, and $\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(dy^{t+1} \mid \tilde{y}^t) \mu(d\tilde{y}^t) = \mu(dy^{t+1})$. See Appendix A.2.

3.2 Constructing the SWF

We now extend the SWF construction of Section 2 to the intertemporal framework. The construction proceeds in three steps. The first step differs slightly from the static case. In the simple framework, agents of type i held subjective views on how the utility of agents of type j should be valued by the planner. Here, because histories are the sole source of heterogeneity, perceptions of others' situations are perceptions of other histories. Thus, an agent with history y^t forms a subjective valuation of the utility of agents with history $\tilde{y}^t$.

The second and third steps parallel the static case. In the second step, we aggregate perceptions over the entire distribution μ of histories to obtain the Individual Welfare Function (IWF) for agents with history y^t. In the third and final step, we aggregate IWFs across histories to obtain the Social Welfare Function (SWF). We now formalize these steps.

For clarity, we first state the assumption underlying the aggregation procedure. Issues regarding cardinal comparability of utilities and the role of linear aggregation are discussed in Section 7.

Step 1: Constructing the subjective valuation of the utility of another agent. Consider two agents characterized by their histories $y^t \in \mathcal{Y}^\infty$ and $\tilde{y}^t \in \mathcal{Y}^\infty$ at date t; the allocation is taken as given. Under our preference specification, the $\tilde{y}^t$-agent values the allocation associated with any future history $\tilde{y}^{t+s}$ by its period utility $U(\tilde{y}^{t+s})$. The y^t-agent may perceive the same future allocation differently. Specifically, the y^t-agent assigns to the utility $U(\tilde{y}^{t+s})$ of the $\tilde{y}^t$-agents a corrective factor $\hat{\omega}(y^t, \tilde{y}^{t+s})$, such that $\hat{\omega}(y^t, \tilde{y}^{t+s})U(\tilde{y}^{t+s})$ represents the valuation of the history $\tilde{y}^{t+s}$ by the y^t-agent.[13]

Summing these weighted valuations $\hat{\omega}(y^t, \tilde{y}^{t+s})U(\tilde{y}^{t+s})$ over all future periods and histories – using the appropriate conditional probabilities – yields the total valuation that the y^t-agent assigns to the $\tilde{y}^t$-agent's lifetime utility. Denoting this by $\hat{V}(y^t, \tilde{y}^t)$, we obtain:

$$\hat{V}(y^t, \tilde{y}^t) = \sum_{s=0}^{\infty} \int_{\tilde{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \tilde{y}^{t+s}) U(\tilde{y}^{t+s}) \mu_s(d\tilde{y}^{t+s}|\tilde{y}^t). \tag{11}$$

This expression mirrors the definition of the intertemporal utility $V(\tilde{y}^t)$ but includes the weighting terms $\hat{\omega}(y^t, \tilde{y}^{t+s})$. In words, $\hat{V}(y^t, \tilde{y}^t)$ represents the expected intertemporal welfare of an agent currently in history $\tilde{y}^t$, as perceived by another agent whose current history is y^t. Consequently, two agents with different histories today, such as y^t and $y^{*,t}$, can disagree on the welfare of agents in history $\tilde{y}^t$. For instance, unemployed and employed individuals may hold different views about the welfare of a firm manager.

This construction embeds an assumption of stable moral values. Although the weights

[13] This corrective term parsimoniously captures habit formation or preference heterogeneity: agents with distinct histories may value the same consumption bundle or labor differently.

that an agent in history y^t assigns to future history $\tilde{y}^{t+s}$ evolve as their own history changes, they evaluate the future using the values attached to their current history y^t. In other words, they do not internalize that their subjective valuations may change over time.[14]

Step 2: Constructing the Individual Welfare Function (IWF). The IWF for agents with history y^t is obtained by aggregating their subjective valuations $\hat{V}(y^t, \tilde{y}^t)$ over possible histories $\tilde{y}^t$:

$$IWF(y^t) = \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \hat{V}(y^t, \tilde{y}^t)\mu(d\tilde{y}^t). \tag{12}$$

The IWF represents the ethical preferences of agents with history y^t – that is, how these agents think the welfare of all others should be accounted for by the planner. As in the simple model, no weight normalization is imposed at this stage.

Step 3: Aggregating IWFs to obtain the Social Welfare Function. We assume that the planner observes IWFs in the population and aggregates them according to weights assigned to each agent. Agents need not have the same importance for the planner; they differ in their *political weights*, as in the simple model. Formally, the IWF of agents with history y^t receives planner weight $\omega_P(y^t)$, which captures their influence in the political process and hence their ability to have their IWF accounted for by the planner:

$$SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) IWF(y^t)\mu(dy^t). \tag{13}$$

Special cases. We illustrate the construction with two special cases.

First, suppose agents identically value all other histories. Formally, for agent y^t, $\hat{\omega}(y^t) := \hat{\omega}(y^t, \tilde{y}^{t+s})$ for all $\tilde{y}^{t+s}$. Then the ethical preferences are represented by an IWF proportional to the utilitarian SWF: $IWF(y^t) = \hat{\omega}(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} V(\tilde{y}^t)\mu(d\tilde{y}^t)$. All agents

[14] An alternative specification would allow agents to anticipate potential changes in ethical values and to evaluate future welfare using those future ethical perspectives. In that case, the intertemporal valuation would be:

$$\begin{aligned}\hat{V}(y^t, \tilde{y}^t) &= \sum_{s=0}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \int_{\tilde{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \hat{\omega}(y^{t+s}, \tilde{y}^{t+s}) U(\tilde{y}^{t+s})\mu_s(d\tilde{y}^{t+s}|\tilde{y}^t)\mu_s(dy^{t+s}|y^t), \\ &= \sum_{s=0}^{\infty} \int_{\tilde{y}^{t+s} \in \mathcal{Y}^\infty} \beta^s \underbrace{\left(\int_{y^{t+s} \in \mathcal{Y}^\infty} \hat{\omega}(y^{t+s}, \tilde{y}^{t+s})\mu_s(dy^{t+s}|y^t)\right)}_{=\tilde{\hat{\omega}}(y^t, \tilde{y}^{t+s})} U(\tilde{y}^{t+s})\mu_s(d\tilde{y}^{t+s}|\tilde{y}^t).\end{aligned}$$

This is equivalent to our specification with weights $\tilde{\hat{\omega}}(y^t, \tilde{y}^{t+s})$ expressed as the expectation of future weights. As a consequence, only well-defined average weights matter. This property is related to the additive expected utility specification. Indeed, it is akin to the property of ambiguity neutrality in the additive expected utility framework: when there is ambiguity about future probabilities, only the average probability—not the full distribution—affects utility.

place the same weights in their ethical preferences; hence there is no disagreement about the ordering of allocations. The SWF reflects this absence of disagreement and is also proportional to the utilitarian SWF: $SWF = \left(\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \omega_P(\tilde{y}^t)\hat{\omega}(\tilde{y}^t)\mu(d\tilde{y}^t)\right) \int_{y^t \in \mathcal{Y}^\infty} V(y^t)\mu(dy^t)$.[15]

Second, consider self-interested agents who only care about histories they can possibly experience. Formally, $\hat{\omega}(y^t, \tilde{y}^{t+s}) := \delta_{y^t}(\tilde{y}^{t,t+s})\hat{\omega}(y^t)$. Then ethical preferences coincide with individual preferences and the IWF is proportional to intertemporal utility: $IWF(y^t) = \hat{\omega}(y^t)V(y^t)$. In this case the SWF is expressed as a weighted sum of individual intertemporal utilities: $SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t)\hat{\omega}(y^t)V(y^t)\mu(dy^t)$, corresponding to a weighted additive SWF consistent with the welfarist approach. It reduces to the utilitarian SWF when the product $\omega_P(\cdot)\hat{\omega}(\cdot)$ is constant.

3.3 Properties of the SWF

An explicit expression of the SWF. We state the following proposition.

Proposition 1 *The SWF (13) admits the following expression:*

$$SWF = \sum_{t=0}^{\infty} \int_{y^t \in \mathcal{Y}^\infty} \beta^t \omega(y^t) U(y^t) \mu(dy^t), \tag{14}$$

where the weights ω are given by:

$$\omega(y^{t+s}) = \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \omega_P(\tilde{y}^t)\hat{\omega}(\tilde{y}^t, y^{t+s})\mu(d\tilde{y}^t). \tag{15}$$

Proposition 1 shows that there exist period weights ω depending on the current history such that the SWF equals the discounted sum, across dates and histories, of period utilities weighted by ω. In other words, the utilitarian SWF is "twisted" by history-dependent weights – the utilitarian case corresponds to constant and identical ω). The weight $\omega(y^{t+s})$ in (15) can be interpreted as the *average* weight given to history y^{t+s} by all agents, where agents are themselves weighted by their political leverage ω_P. See Appendix A.4 for the details of the proof.

The sequential representation (14) also admits a recursive form:

$$SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U(y^t) \mu(dy^t) + \beta \cdot SWF,$$

which extends the recursive representation of the utilitarian SWF to history-dependent weights. Under aggregate dynamics, period utility U_t is time dependent (through the allocation). The SWF then satisfies $SWF_t = \int_{y^t \in \mathcal{Y}^\infty} \omega(y^t) U_t(y^t) \mu(dy^t) + \beta \cdot SWF_{t+1}$. We use this form when solving the Ramsey problem.

[15] This property is known at least since Aiyagari and McGrattan (1998) as a justification for using the utilitarian SWF under the veil of ignorance.

Weight restriction. As in the simple framework, we do not require the SWF to satisfy the Pareto principle. We do, however, impose a weaker restriction. To state it, we make the dependence on the allocation explicit. Let $U : \mathcal{Y}^\infty \times \mathcal{A} \to \mathbb{R}$ denote period utility and $SWF : \mathcal{A} \to \mathbb{R}$ the social welfare function, where $\mathcal{A}$ is the set of allocations. Period utility under history y^t and allocation A is $U(y^t, A)$. In our quantitative application, $U(y^t, A) := u(c(y^t)) - v(l(y^t))$, where A collects the policy functions (c, l). We can now define the property.

Definition 1 *A SWF $SWF : \mathcal{A} \to \mathbb{R}$, associated with a period utility $U : \mathcal{Y}^\infty \times \mathcal{A} \to \mathbb{R}$, is said to be element-wise monotone if for any two allocations A and A' such that $U(y^t, A) \geq U(y^t, A')$ for all y^t, we have $SWF(A) \geq SWF(A')$.*

This definition states that if one allocation yields weakly higher period utility in every history and date, then it is weakly preferred by the SWF. The property parallels element-wise monotonicity for utility functions. It is weaker than the Pareto principle, which would require A to be preferred to A' in terms of intertemporal (not merely period) utility.[16]

Proposition 2 *A SWF fulfills element-wise monotonicity iff the weights ω defined in (15) are non-negative.*

In our quantitative work we may impose $\omega \geq 0$, corresponding to an element-wise monotone SWF: the SWF cannot increase when the welfare of any agent is reduced; everybody (positively) counts. The proof is direct and it can be found in Appendix A.5.

3.4 Identification of the weights

The SWF expression in Proposition 1 is general and not immediately suited for estimation. We therefore introduce a tractability assumption that allows us to compute the SWF weights and the IWFs. We assume that agents with the same current productivity level value future histories identically, and that this valuation depends only on the current productivity level of the future history under consideration. We impose a parallel restriction on political weights, which also depend only on current productivity. In words, current productivity is a sufficient statistic for both IWFs and political power, and agents with the same productivity place the same welfare weight. The primary motivation is empirical: this restriction renders the SWF identifiable in the data. More general specifications deliver similar results (see Section 7). Formally:

[16] This helps explain why the planner may choose not to implement an insurance mechanism even when it is individually desirable. Such a mechanism typically raises utility in some states (the "bad" ones) and lowers it in others (the "good" ones), which is not covered by element-wise monotonicity of Definition 1.

Assumption A *There exist two functions, denoted (with slight abuse) $\tilde{\omega} : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ and $\omega_P : \mathcal{Y} \to \mathbb{R}$ such that for all histories $y^t, \tilde{y}^{t+s} \in \mathcal{Y}^\infty$, we have:*

$$\begin{cases} \tilde{\omega}(y^t, \tilde{y}^{t+s}) := \hat{\omega}_{y_t^t, \tilde{y}_{t+s}^{t+s}}, \\ \omega_P(y^t) := \omega_{P, y_t^t}, \end{cases}$$

where we recall that y_t^t is the current productivity level in history y^t.

Assumption A reduces dimensionality by defining weights on finite sets. We interpret these as loading factors on productivity levels. To avoid heavy notation, we maintain the same symbols but use subscripts for productivity dependence. The weights ω in (15) then satisfy for all $y \in \mathcal{Y}$, $\omega_y = \sum_{\tilde{y} \in \mathcal{Y}} \pi_{\tilde{y}} \, \omega_{P,\tilde{y}} \, \tilde{\omega}_{\tilde{y},y}$, where $\pi_{\tilde{y}}$ is the share of agents with productivity $\tilde{y}$. Simple relationships between the SWF, political weights, and IWFs follow.

Proposition 3 (IWFs identification) *1. If all groups are self-interested ($\tilde{\omega}_{\tilde{y}y} = 1_{y=\tilde{y}}$), then the SWF is pinned down solely by political weights: $\omega_y = \pi_y \omega_{P,y}$ for all y.*

2. If all agents agree on the same SWF, $\tilde{\omega}_{\tilde{y}y} = \omega_y$ for all $y, \tilde{y}$, then the SWF is consistent with constant political weights: $\omega_{P,y} = 1$ for all y.

Proposition 3 follows directly from the aggregation identity $\omega_y = \sum_{\tilde{y} \in \mathcal{Y}} \pi_{\tilde{y}} \omega_{P,\tilde{y}} \tilde{\omega}_{\tilde{y},y}$. It highlights two extreme identification strategies. First, imposing self-interest implies that political weights fully determine the SWF (Item 1), so the SWF is informative only about political power. Second, imposing away political heterogeneity by assuming common IWFs (Item 2) implies constant political weights, so the SWF fully summarizes agents' welfare functions. Neither extreme is consistent with evidence on altruism or spitefulness in experiments (see Fehr and Schmidt, 2006; Andreoni et al., 2007) or with heterogeneity in welfare and policy views documented in experiments (Almås et al., 2020) and in surveys (Stantcheva, 2021). Our identification therefore proceeds in three steps: (i) estimate the SWF weights $(\omega_y)_{y \in \mathcal{Y}}$ from the model; (ii) estimate political weights $(\omega_{P,y})_{y \in \mathcal{Y}}$ from the data; (iii) infer the IWF weights $(\tilde{\omega}_{\tilde{y}y})_{y,\tilde{y} \in \mathcal{Y}}$ from the SWF weights and data as the smallest deviation from observer's priors about IWFs. The main focus of the paper is on the first step and the second and third steps are relegated to Section 6.

Without loss of generality, we impose the normalization $\sum_y \pi_y \omega_y = 1$.

Identification of SWF weights. We identify the SWF weights by finding $(\omega_y)_{y \in \mathcal{Y}}$ such that a given fiscal system (typically the observed one) can be rationalized as the solution to a Ramsey problem. Formally, the FOCs of the Ramsey planner imply some linear constraints for the SWF weights $(\omega_y)_{y \in \mathcal{Y}}$. The number of constraints, n, depends on the number of the planner's instruments. In general, the system (the n linear constraints plus the normalization) is weakly underdetermined because $|\mathcal{Y}| \geq n+1$ (for any set X, $|X|$ denotes cardinality).[17] Hence there are $p = |\mathcal{Y}| - n - 1$ degrees of freedom.

Following Heathcote and Tsujiyama (2021), we resolve underdetermination by assuming a parametric specification for ω_y with exactly $n+1$ parameters. Under mild conditions on the functional form, the weights are then identified uniquely as the solution to a nonlinear system. We formalize this as follows.

Definition 2 *Given $1 \leq n \leq |\mathcal{Y}| - 1$ linear constraints represented by the matrix $(L_{k,y})_{k=1,\ldots,n;\, y\in\mathcal{Y}}$ and a set of parametric functions $f_y : \mathbb{R}^{n+1} \to \mathbb{R}$ ($y \in \mathcal{Y}$) for the weights, the estimated SWF weights are $(\omega_y)_{y\in\mathcal{Y}} = (f_y(\theta))_y$, where $\theta \in \mathbb{R}^p$ solves*

$$0 = \sum_{y\in\mathcal{Y}} L_{k,y} f_y(\theta) \text{ for all } k = 1, \ldots, n, \tag{16}$$

$$1 = \sum_{y\in\mathcal{Y}} \pi_y f_y(\theta). \tag{17}$$

As a robustness check, we also consider a nonparametric estimation. In that case, the system (16)–(17) is underdetermined, and we select the solution with the lowest variance across productivity levels. Both approaches deliver quantitatively similar weights, which supports our identification strategy. See Appendix F.2 for definitions and results in the nonparametric case.

4 The general model and the Ramsey program

We now develop the macroeconomic model that underpins the estimation of the SWF and the IWFs. The economy features a unit mass of ex-ante identical agents facing idiosyncratic productivity risk y – the risk structure is the same as in Section 3. There are two goods: a final consumption good c and labor l. The remaining elements of the environment are as follows: the planner's fiscal instruments and the production side in Section 4.1; households' behavior and the competitive equilibrium in Section 4.2; the Ramsey program and its FOCs in Sections 4.3 and 4.4; the SWF expression in Section 4.5; and the solution method for the Ramsey program in Section 4.6.

4.1 Production and government

Production. In each period t, a production technology with constant returns to scale transforms capital K_{t-1} and labor L_t into $F(K_{t-1}, L_t)$ units of output. The production function is smooth in K and L and satisfies the standard Inada conditions. Depreciation is subsumed into F. Labor L_t is measured in efficiency units. Output is produced by a

[17] In our quantitative exercise, there are typically 10 idiosyncratic states and 4 planner instruments.

representative profit-maximizing firm. Let $\tilde{w}_t$ denote the real before-tax wage and $\tilde{r}_t$ the real before-tax rental rate of capital. Profit maximization implies, for $t \geq 1$,

$$\tilde{r}_t = F_K(K_{t-1}, L_t) \quad \text{and} \quad \tilde{w}_t = F_L(K_{t-1}, L_t). \tag{18}$$

Government. A benevolent government finances a path of public spending (G_t) with several instruments. First, it issues default-free one-period public debt B_t. With no aggregate risk, public debt and capital are perfect substitutes and earn the same pre-tax return $\tilde{r}_t$. Second, it levies distortionary taxes on consumption, labor income, and capital income. Consumption and capital taxes are linear and denoted by τ_t^c and τ_t^K. For labor income, if an agent with productivity y supplies labor l, pre-tax labor income is $\tilde{w}_t yl$. The labor income tax, $\mathcal{T}_t$, is non-linear and potentially time varying, as in Heathcote et al. (2017) (HSV, henceforth):

$$\mathcal{T}_t(\tilde{w}yl) := \tilde{w}yl - \kappa_t(\tilde{w}yl)^{1-\tau_t}, \tag{19}$$

where κ_t controls the level of taxation and τ_t its progressivity – both are planner instruments. When $\tau_t = 0$, labor taxes are linear with rate $1 - \kappa_t$; when $\tau_t = 1$, all agents earn the same post-tax income κ_t. This functional form, together with a linear capital tax, reproduces salient features of the US tax system and its progressivity.[18]

These three taxes imply a total governmental revenue equal to $\tau_t^c C_t + \int_i \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t})\ell(di) + \tau_t^K \tilde{r}_t(K_{t-1} + B_{t-1})$, where C_t is aggregate consumption and $A_{t-1} := K_{t-1} + B_{t-1}$ denotes aggregate savings at $t-1$ – with capital income $\tilde{r}_t A_{t-1}$ at t. With these elements, the governmental budget constraint can be written as follows:

$$G + (1 + \tilde{r}_t)B_{t-1} = \tau_t^c C_t + \int_i \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t})\ell(di) + \tau_t^K \tilde{r}_t A_{t-1} + B_t. \tag{20}$$

We define the post-tax rates r_t and w_t as follows:

$$r_t := (1 - \tau_t^K)\tilde{r}_t, \quad w_t := \kappa_t(\tilde{w}_t)^{1-\tau_t}. \tag{21}$$

Using constant returns-to-scale and (21), we can rewrite the government budget constraint (20) as:

$$G + (1 + r_t)B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t}\ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t) + B_t. \tag{22}$$

[18] The literature uses either the combination of a linear tax and a lump-sum transfer (e.g., Dyrda and Pedroni, 2022; Açikgöz et al., 2022) or the HSV structure (see Ferriere and Navarro, 2023). Heathcote and Tsujiyama (2021) show that the HSV structure is quantitatively more relevant.

4.2 Households' program

Period utility. Agents have separable period utility over private consumption and labor:

$$U(c,l) := u(c) - v(l). \tag{23}$$

The function $u : \mathbb{R}_+ \to \mathbb{R}$ is twice continuously differentiable, strictly increasing, and strictly concave, with $u'(0) = \infty$, while $v : \mathbb{R}_+ \to \mathbb{R}$ is twice continuously differentiable, strictly increasing, and strictly convex, with $v'(0) = 0$.

Agents' program. Agents' resources consist of labor income and savings payoffs. For agent i with productivity $y_{i,t}$ and labor $l_{i,t}$, post-tax labor income is $\tilde{w}_t y_{i,t} l_{i,t} - \mathcal{T}_t(\tilde{w}_t y_{i,t} l_{i,t}) = w_t (y_{i,t} l_{i,t})^{1-\tau_t}$. Since public debt and capital are perfect substitutes, savings pay $(1+r_t)a_{i,t-1}$. The agent chooses consumption, labor, and next-period assets to solve:

$$\max_{\{c_{i,t}, l_{i,t}, a_{i,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(u(c_{i,t}) - v(l_{i,t})\right), \tag{24}$$

$$(1+\tau_t^c) c_{i,t} + a_{i,t} \leq w_t (y_{i,t} l_{i,t})^{1-\tau_t} + (1+r_t) a_{i,t-1}, \tag{25}$$

$$a_{i,t} \geq -\bar{a}, c_{i,t} > 0, l_{i,t} > 0, \tag{26}$$

where $\mathbb{E}_0$ is an expectation operator over idiosyncratic risk, and where the initial state $(y_{i,0}, a_{i,-1})$ is given.

At date 0, agents decide their consumption $(c_{i,t})_{t\geq 0}$, their labor supply $(l_{i,t})_{t\geq 0}$, and their saving plans $(a_{i,t})_{t\geq 0}$ that maximize their intertemporal utility of equation (24), subject to a budget constraint (25) and a previous borrowing limit (26), taking prices as given. Decisions depend on the initial asset $a_{i,-1}$ and the idiosyncratic history y_i^t. To simplify notation, we denote optimal objects with subscripts (i,t) rather than writing explicit functional dependence on histories as in Section 3. Aggregation over agents replaces aggregation over histories: $\int_i X_{i,t} \ell(di)$ on the population interval J with measure ℓ.

The FOCs associated with the agents' program (24)–(26) are:

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[(1+r_{t+1}) \frac{1+\tau_t^c}{1+\tau_{t+1}^c} u'(c_{i,t+1}) \right] + \nu_{i,t}, \tag{27}$$

$$v'(l_{i,t}) = \frac{1-\tau_t}{1+\tau_t^c} w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \tag{28}$$

where $\beta^t \nu_{i,t}$ is the Lagrange multiplier on agent i's credit constraint at t.

Market clearing and resource constraints. The clearing conditions for capital and labor markets can be written as follows:

$$\int_i a_{i,t}\ell(di) = A_t = B_t + K_t, \quad \int_i y_{i,t}l_{i,t}\ell\,(di) = L_t. \tag{29}$$

Equilibrium definition. A formal definition is in Appendix B. Intuitively, given a fiscal policy $(\tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t)_t$, a competitive equilibrium is a collection of individual decisions $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t,i}$, of aggregate quantities $(K_t, L_t, Y_t)_t$, and of prices $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_t$ consistent with: (i) agents' optimality (24)–(26); (ii) market clearing (29); and (iii) factor pricing (18) and (21). A steady-state competitive equilibrium has time-invariant quantities.

4.3 The Ramsey problem and the identification of weights

We use Proposition 1 to construct the SWF and impose Assumption A for identification. With separable period utility, the date-0 SWF is:

$$SWF_0 := \mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t\int_i \omega_{y_{i,t}}\left(u(c_{i,t}) - v(l_{i,t})\right)\ell(di)\right], \tag{30}$$

where the weights $(\omega_y)_{y\in\mathcal{Y}}$ depend solely on current productivity due to Assumption A.

The Ramsey planner chooses a fiscal policy that corresponds to a competitive equilibrium maximizing aggregate SWF (30) subject to the government budget constraint. A Ramsey equilibrium consists of a policy, prices, individual allocations, and aggregates solving the program; a Ramsey steady state is a time-invariant Ramsey equilibrium. Formally,

$$\max_{\left(w_t,r_t,\tilde{w}_t,\tilde{r}_t,\tau_t^c,\tau_t^K,\tau_t,\kappa_t,B_t,G_t,K_t,L_t,(c_{i,t},l_{i,t},a_{i,t},\nu_{i,t})_i\right)_{t\geq 0}} W_0, \tag{31}$$

$$G_t + (1+r_t)B_{t-1} + w_t\int_i (y_{i,t}l_{i,t})^{1-\tau_t}\ell(di) + r_tK_{t-1} = \tau_t^cC_t + F(K_{t-1}, L_t, s_t) + B_t, \tag{32}$$

$$\text{for all } i\in\mathcal{I}\text{: } (1+\tau_t^c)c_{i,t} + a_{i,t} = (1+r_t)a_{i,t-1} + w_t(y_{i,t}l_{i,t})^{1-\tau_t}, \tag{33}$$

$$a_{i,t} \geq -\bar{a},\ \nu_{i,t}(a_{i,t}+\bar{a}) = 0,\ \nu_{i,t}\geq 0, \tag{34}$$

$$u'(c_{i,t}) = \beta\mathbb{E}_t\left[(1+r_{t+1})\frac{1+\tau_t^c}{1+\tau_{t+1}^c}u'(c_{i,t+1})\right] + \nu_{i,t}, \tag{35}$$

$$v'(l_{i,t}) = \frac{(1-\tau_t)}{1+\tau_t^c}w_ty_{i,t}(y_{i,t}l_{i,t})^{-\tau_t}u'(c_{i,t}), \tag{36}$$

$$K_t + B_t = \int_i a_{i,t}\ell(di),\ L_t = \int_i y_{i,t}l_{i,t}\ell(di), \tag{37}$$

$$\tilde{r}_t = F_K(K_{t-1}, L_t) \quad\text{and}\quad \tilde{w}_t = F_L(K_{t-1}, L_t), \tag{38}$$

$$r_t = (1-\tau_t^K)\tilde{r}_t, \quad w_t = \kappa_t(\tilde{w}_t)^{1-\tau_t}. \tag{39}$$

plus positivity of c and l and given initial conditions.

Because the Ramsey planner selects a competitive equilibrium, the constraints include individual budget constraints (33), borrowing constraints (34), Euler equations (35), labor FOCs (36), and market clearing (37). Moreover, the fiscal policy must also fulfill the governmental budget constraint (32).

The Ramsey program can be simplified. First, with a linear capital tax, a progressive labor tax, and one-period public debt, consumption taxes are redundant. Second, following Chamley (1986), we can work with post-tax prices only. Under these steps, the planner's instruments are post-tax wage and interest (W_t, R_t), labor tax progressivity, and public debt – chosen to satisfy (22). The other choice variables are individual and aggregate allocations satisfying competitive-equilibrium conditions: (25), (26), (27)–(28), and (29). Proposition 4 in Appendix C.1 states the reformulated program.

A *Ramsey allocation* solves (31)–(39). A *Ramsey steady state* is a Ramsey allocation with constant prices and aggregates, which can be understood as the long-run Ramsey allocation.

4.4 Interpreting the Ramsey FOCs

The planner's trade-offs can be identified in the Ramsey FOCs that can be found in Appendix C.2.[19] We summarize the economic content using convenient statistics.

Fix a fiscal instrument $(I_t)_t$ that directly impacts solely the current-period budget constraint. In our context, this can be capital taxation (or post-tax R_t), the level of labor taxation (or post-tax W_t), or labor tax progressivity. Public debt, which affects current and next-period budget constraints, is handled separately below. Consider raising resources via I_t, which reduces consumption for all agents. The Lagrangian is

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big(\mathcal{W}_t + \mu_t \mathcal{B}_t \Big).$$

Here $\mathcal{W}_t$ is the date-t augmented welfare, comprising the pure welfare term $\omega_{y_{i,t}}(u(c_{i,t}) - v(l_{i,t}))$ and as well as the general-equilibrium effects implied by individual decisions about savings (i.e., Euler equation) and labor supply (i.e., FOC labor supply).[20] It depends on the fiscal instrument I_t, and on all consumption levels $(c_{i,t})_i$ of date t. The term $\mathcal{B}_t$ is the government budget constraint and $\beta^t \mu_t$ its multiplier. As shown in Appendix C.2, $\mathcal{B}_t$ depends on I_t but not on $(c_{i,t})_i$. The FOC with respect to I_t is:[21]

$$\mu_t = \int_i \underbrace{\left.\frac{\partial \mathcal{W}_t}{\partial c_{i,t}}\right|_{I_t}}_{=SVL_{i,t}} \underbrace{\frac{-\frac{\partial c_{i,t}}{\partial I_t}}{\frac{\partial \mathcal{B}_t}{\partial I_t} + \frac{1}{\mu_t} \left.\frac{\partial \mathcal{W}_t}{\partial I_t}\right|_{c_{i,t}}}}_{=MVPF^I_{i,t}} \ell(di), \tag{40}$$

[19] Technical issues in Lagrangian formulations are discussed in LeGrand and Ragot (2025).
[20] The consumption tax is redundant and as such does not imply any specific FOC.
[21] $\left.\frac{\partial \mathcal{W}_t}{\partial c_{i,t}}\right|_{I_t}$ is the partial derivative of $\mathcal{W}_t$ with respect to $c_{i,t}$ holding I_t fixed; $\left.\frac{\partial \mathcal{W}_t}{\partial I_t}\right|_{c_{i,t}}$ holds all $c_{i,t}$ fixed.

or

$$\mu_t = \int_i SVL_{i,t} \times MVPF^I_{i,t}\, \ell(di). \tag{41}$$

Thus, the instrument is set where the shadow value of revenue (μ_t) equals the total marginal cost across agents. For agent i, the marginal cost measures the welfare cost of instrument I and equals the product of two terms: the welfare impact of a marginal consumption variation ($SVL_{i,t}$) and the consumption cut for agent i implied by instrument I. Following Finkelstein and Hendren (2020) and Hendren and Sprung-Keyser (2020), we interpret this marginal cost as the (negative) *bang for the buck* of one unit of resources raised by the planner from agents i.

The *social value of liquidity*, $SVL_{i,t}$ (denoted $\psi_{i,t}$ in Appendix C) measures the total welfare loss from a one-unit reduction in agent i's consumption. This notion is independent of the instrument generating the consumption cut. It includes three components:

$$\begin{aligned} SVL_{i,t} = \underbrace{\omega_{i,t} u'(c_{i,t})}_{\text{direct effect}} - \underbrace{(\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u''(c_{i,t})}_{\text{externality on savings incentives}} \\ + \underbrace{\lambda_{l,i,t}(1-\tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} u''(c_{i,t})}_{\text{externality on labor supply incentives}}, \end{aligned} \tag{42}$$

where $\beta^t \lambda_{c,i,t}$ and $\beta^t \lambda_{l,i,t}$ are multipliers on the Euler equation and labor FOC. The first term, $\omega_{i,t} u'(c_{i,t})$, reflects the direct welfare effect of a consumption variation. This is related to the concept of *social marginal welfare weight* (SMWW) used in the public finance literature (e.g., Ferey et al., 2024, among many others). The second and third terms capture induced changes in savings and labor. The welfare impact due to savings channels through the Euler equation, while the one due to labor supply channels through the FOC on labor supply. This makes these effects proportional to the Lagrange multipliers.[22]

The term $MVPF^I_{i,t}$ measures the agent i's consumption cut caused by a one-unit revenue raised through I_t, net of fiscal externalities. We call it the *marginal value of public fund*, as it extends the same static public finance concept, used in Finkelstein and Hendren (2020) and Hendren and Sprung-Keyser (2020). In equation (40), $\frac{\partial \mathcal{B}_t}{\partial I_t}$ is the direct revenue effect and equals the tax base of I: asset payoffs for capital taxes, labor supply for labor taxes. The term $\frac{1}{\mu_t} \frac{\partial \mathcal{W}_t}{\partial I_t}\Big|_{c_{i,t}}$ is the fiscal externality of I_t that channels through the distortion of savings incentives. This fiscal externality reflects to which extent the instrument I_t is distortionary.

With an available aggregate lump-sum transfer T, the MVPF for T would simply be 1. There is no externality, $\frac{\partial \mathcal{W}_t}{\partial T_t}\Big|_{c_{i,t}} = 0$, and the flow of resources from the transfer to agents involves $-\frac{\partial c_{i,t}}{\partial T_t} = \frac{\partial \mathcal{B}_t}{\partial T_t}$. The planner would set T_t such that $\mu_t = \int_i SVL_{i,t} \ell(di)$. With

[22] This SVL is similar to the quantity $\hat{g}$ defined in Ferey et al. (2024) that they describe as "the social marginal welfare weights augmented with the fiscal impact of income effects" and which represents "the full social value of marginally increasing the disposable income of [an] individual".

individual lump-sum transfers T^i, each would satisfy $\mu_t = SVL_{i,t}$. The gap $SVL_{i,t} - \mu_t$ can be viewed as capturing the cost of relying on distortionary rather than lump-sum instruments for agent i.[23]

Public debt is the sole instrument that does not admit a FOC of the form (41). Public debt issued at t relaxes the budget constraint $\mathcal{B}_t$ contemporaneously and tightens $\mathcal{B}_{t+1}$ via repayment. Public debt also has no direct utility effect: $\frac{\partial \mathcal{W}_t}{\partial B_s} = 0$. The FOC is $\mu_t = \beta\mu_{t+1}(1 + r_{t+1})$. This Euler-like condition states that debt smooths the marginal cost of public funds over time.

4.5 Expression of the SWF

Fiscal policy is composed of five instruments $(\tau^K, \tau^c, B, \kappa, \tau)$, but these five instruments actually impose only two constraints on social weights. Indeed, consumption taxes τ^c are redundant, as explained above (see Appendix C.1 for the details). Second, the public debt FOC, provided in equation (78) of Appendix C.2, imposes a steady-state value on the before-tax real interest rate $1 + \tilde{r} = 1/\beta$, but does not restrict the social weights. Therefore, this means that the three remaining instruments (τ^K, κ, τ) yield three FOCs. One of them determines the Lagrange multiplier of the governmental budget constraint μ_t, while the two remaining ones provide two linear constraints on the SWF weights.

We apply then the identification strategy in Definition 2 with three constraints: two linear constraints from Ramsey FOCs plus the normalization. A parametric specification with three free parameters delivers exact identification. We adopt the following functional form, which naturally extends the one in Heathcote and Tsujiyama (2021):

$$\forall y \in \mathcal{Y},\ \log \omega_y := \bar{\omega}_0 + \bar{\omega}_1 \log(y) + \bar{\omega}_2 (\log(y))^2, \tag{43}$$

where $(\bar{\omega}_i)_{i=0,1,2}$ are free parameters.

4.6 Solution method and algorithm

The Ramsey problem discussed in Section 4.3 involves a joint distribution across wealth and Lagrange multipliers. This high-dimensional object raises a number of difficulties for the resolution of the Ramsey program. We rely on the truncation method, which has already been used in recent papers (LeGrand and Ragot, 2022b, 2023, 2025). We here improve on previous work to allow for the estimation of the SWF with a utility function separable into consumption and labor, instead of the GHH case considered in previous papers.

[23] LeGrand and Ragot (2025) introduce the notion of SVL. In this section, we generalize their analysis by introducing the concepts of SMWW and MVPF, thereby connecting our results to the public finance literature.

The idea is to construct a consistent finite state-space approximation, compute the Ramsey FOCs in that environment, and then apply the inverse optimal approach to estimate the SWF. More precisely, the solution method is based on the following steps.

1. Simulate the heterogeneous-agent model at the steady state using realistic fiscal instruments and income-wealth inequality. Standard techniques deliver the steady-state wealth distribution $\Lambda(a, y)$ for each idiosyncratic state $y \in \mathcal{Y}$ and asset holding a, and policy rules for assets, consumption, and labor, $g_a(a, y)$, $g_c(a, y)$, and $g_l(a, y)$.

2. Fix a finite set $\mathcal{H}$ of histories with Markovian transitions. An intuitive choice is the set of all histories of length N, yielding Y^N truncated histories when there are Y idiosyncratic states.

3. For a truncated history $y^N := \{y_1, \ldots, y_N\} \in \mathcal{H}$, representing agents who experienced y^N over the last N periods, use $\Lambda(\cdot\,, y_1)$ and the policy rules to compute the wealth distribution $\tilde{\Lambda}(\cdot, y^N)$ for the truncated histories.

4. Aggregate over $\tilde{\Lambda}$ to construct truncated-history equations. The size of y^N is $S_{y^N} = \int_0^\infty \tilde{\Lambda}(da, y^N)$; per-capita consumption is $c_{y^N} = \int_0^\infty g_c(a, y^N)\tilde{\Lambda}(da, y^N)/S_{y^N}$; average marginal utility satisfies $\int_0^\infty u'(g_c(a, y^N))\tilde{\Lambda}(da, y^N)/S_{y^N} := \xi^u_{y^N} u'(c_{y^N})$, where $\xi^u_{y^N}$ captures nonlinearity and within-y^N wealth heterogeneity. This yields Y^N budget constraints, Euler equations, and labor choices (see equations (92)–(94) in Appendix D.1). This defines the *truncated model*.

5. Derive the FOCs of the planner in the truncated model (see Appendix D.2).

6. Extract from those FOCs two linear constraints on the SWF weights and feed them into (16) within Definition 2.

7. Use the parametric form $\omega_y = e^{\bar{\omega}_0 + \bar{\omega}_1 \log(y) + \bar{\omega}_2 (\log(y))^2}$ with parameters $(\bar{\omega}_i)_{i=0,1,2}$ and apply the identification strategy in Definition 2. This is a direct generalization of Heathcote and Tsujiyama (2021)'s parametric form, expanding from two to three degrees of freedom.

The detailed derivations of these steps are performed in Appendix D. We consider 10 idiosyncratic states and $N = 5$ as a benchmark, and thus 10^5 possible histories. The estimation process takes less than 3 minutes on a standard laptop, and we have checked that the results are robust to an increase in N. We now turn to the quantitative exercise and further discuss the choice of political weights in Section 2.3.

5 Estimating SWFs

We begin by calibrating the model to reproduce the tax systems and the wealth distributions of the United States and France over 1995–2007.

5.1 The French and the US fiscal systems

We document the tax systems of the US and France, focusing on average values from 1995 to 2007, before the Great Recession and the COVID-19 period, which introduced (so far) transitory changes.[24] We construct country-level averages using Trabandt and Uhlig (2011), reported in Table 1, which also includes inequality measures.

	Total taxes (% GDP)	$\tau^K(\%)$	$\tau^L(\%)$	$\tau^c(\%)$	B (% GDP)	G (% GDP)	Gini before redist.	Gini after redist.	Gini wealth
France	40	35	46	18	60	24	0.48	0.28	0.68
United States	26	36	28	5	63	15	0.48	0.40	0.77

Table 1: Summary of fiscal systems and inequalities in the US and in France.
Total taxes, public debt B and public spending G in percentage of GDP; tax rates τ^K, τ^L and τ^c in percent; Gini indices unitless.

The first column reports the total mandatory levies as a share of GDP for the two countries. Following the literature, total levies are divided into three linear components: capital tax (τ^K), labor tax (τ^L), and consumption tax (τ^c). Since Mendoza et al. (1994), this decomposition is widely used to compare the tax structure across countries (OECD, Eurostat). Columns 2–4 report these implicit tax rates: capital tax receipts over the capital stock, labor tax receipts over aggregate labor income, and consumption tax receipts over consumption.

To compare redistribution, Table 1 reports the average Gini index between 1995 and 2007, for income before and after taxes. The data come from the OECD Income Inequality Database.[25] Before-tax income Ginis are similar across countries. For France, the higher pre-tax Gini reflects the treatment of public pensions as transfers rather than income, which mechanically raises pre-tax inequality. After-tax income Gini indices diverge: transfers in France reduce the Gini by about 20 points, compared with less than 10 points in the United States. France's fiscal system is thus more redistributive than the US one.

The last column reports wealth Gini indices. For France, we use the 2010 Household Finance and Consumption Survey (HFCS), which is the closest wave to our benchmark window. The Gini index is stable across waves. For the US, we use the PSID in 2006.[26] Wealth inequality is substantially higher than post-tax income inequality in both countries (by more than 35 points), and remains about 10 points higher in the US than in France.

Although Table 1 lists a *linear* labor tax, our model uses a *progressive* labor income tax consistent with both countries' systems. We estimate the HSV functional form for the

[24] Extending the sample to 1995–2021 does not alter the main tax results. For consistency with the steady-state analysis, we use a pre-shock window.

[25] See `https://stats.oecd.org/index.aspx?queryid=66670`.

[26] The 2007 SCF yields a wealth Gini of 0.78, close to the PSID value.

labor tax:

$$\text{Tax:} \qquad T(I_c) = I_c - \kappa I_c^{1-\tau}, \tag{44}$$

$$\text{Disposable income:} \qquad D(I_c) = \kappa I_c^{1-\tau}, \tag{45}$$

where I_c denotes labor income in country c, τ is progressivity, and κ governs the level of taxation. This formulation enables cross-country comparisons despite a complex combination of heterogeneous schedules and multiple deductions in each country.

We estimate labor tax progressivity using the Luxembourg Income Study (LIS) database for France and the US in 2005.[27] Regressing the log of disposable income on the log of labor income (i.e., the log of (45)) yields the estimates in Table 2, where $\hat{\tau}$ is the estimated labor tax progressivity and SE is the associated standard error. France exhibits substantially greater progressivity. Our US estimate $\hat{\tau} = 0.16$ is close to values in the literature. It is slightly below Heathcote et al. (2017)'s value of 0.181, since we focus on labor income only instead of labor and capital income.

	$\hat{\tau}$	SE	Obs.	R^2
France	0.23	0.0056	5289	0.855
United States	0.16	0.0019	38111	0.942

Table 2: Estimates $\hat{\tau}$ of the progressivity of the labor income tax in the US and in France for 2005.
We regress the log of equation (45) using the LIS database: SE is the standard error of $\hat{\tau}$, Obs. is the number of observations, R^2 is the R^2 of this regression.

5.2 Calibration of the model

We now calibrate the model to reproduce the inequality and fiscal aggregates described above. The model period is a quarter across all calibrations.

The US calibration

The estimation parameters are gathered in Table 3, and we detail below our calibration strategy.

Preference parameters. We set $\beta = 0.992$ to match a realistic capital-output ratio. Period utility is $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ and $v(l) = \frac{1}{\chi}\frac{l^{1/\varphi+1}}{1/\varphi+1}$. We set $\gamma = 1.8$ to match wealth

[27] We restrict our attention to the heads of households and their spouses aged between 25 and 60 who were employed. We define *labor income* as the sum of wage income, self-employment income, and private transfers. Using Trabandt and Uhlig (2011) to infer capital tax, we compute the amount of capital income tax. We then subtract from total LIS income taxes the capital tax amount, to estimate labor income tax. *Disposable income* is then *labor income* minus the *labor income tax*.

inequality at the targeted capital-output ratio, and $\varphi = 0.5$ per Chetty et al. (2011). The scaling parameter $\chi = 0.0477$ normalizes aggregate hours to 0.3.

Technology. The production function, with depreciation embedded, is $F(K, L, s) = sK^{\alpha}L^{1-\alpha} - \delta K$. We set $\alpha = 0.36$ and quarterly depreciation rate $\delta = 2.5\%$ (10% annualized). These values are standard.

Idiosyncratic labor risk. Productivity follows $\log y_t = \rho_y \log y_{t-1} + \varepsilon_t^y$ with $\varepsilon_t^y \sim \mathcal{N}(0, \sigma_y^2)$. We set $\rho_y = 0.99$ and $\sigma_y = 0.0995$, close to Krueger et al. (2018), and discretize via Tauchen (1986) using 10 states. The calibration reproduces a post-tax income Gini index of 0.40, as in Table 1.

Taxes and government budget constraint. We adopt Trabandt and Uhlig (2011)'s values for fiscal parameters $\tau^K = 36\%$ and $\tau^c = 5\%$. For the progressivity of the labor tax, we use our estimate $\tau = 0.16$ from Table 2. We choose κ to match a public-spending-to-GDP ratio of 15%, obtaining $\kappa = 0.85$, close to Ferriere and Navarro (2023). The model implies a public-debt-to-GDP ratio of 63%, as in Table 1.

The model replicates key macro ratios. Consumption over GDP is 58% in the model, while it is 60% in the data for 1995-2007. Investment-to-GDP ratio generated by the model amounts to 27%, close to the empirical value of 25%. Inequality targets are also close to empirical data. The post-tax income Gini index is 0.40 in the model and in the data, while the wealth Gini index is 0.78 in the model and 0.77 in the data. Table 4 collects model-data comparisons. These implications show that our tax system provides a good approximation of the income and wealth distribution in the US, and hence of the redistributive effects of the US tax system. This confirms the results of Heathcote et al. (2017) and Dyrda and Pedroni (2022).

French calibration

The French calibration mirrors the US one: same period and same functional forms. Parameters appear in Table 3.

Preference parameters. We set $\beta = 0.996$ and $\varphi = 0.5$, and choose $\chi = 0.0228$ to normalize aggregate hours to 0.3. As in the US, $\gamma = 1.8$ matches inequality targets.

Technology and TFP shock. We keep the same production function: $F(K, L, s) = sK^{\alpha}L^{1-\alpha} - \delta K$ with $\alpha = 0.36$ and $\delta = 2.5\%$, which fits well French aggregates.

		US		France	
Parameter	Description	Value	Target or ref.	Value	Target or ref.
Preference parameters					
β	discount factor	0.992	$K/Y = 2.7$	0.996	$K/Y = 3.1$
u	utility function	$\cdot$	$\gamma = 1.8$	$\cdot$	$\gamma = 1.8$
φ	Frisch elasticity	0.5	Chetty et al. (2011)	0.5	Chetty et al. (2011)
χ	hours worked	0.33	Penn World Table	0.29	Penn World Table
α	capital share	36%	Profit Share, NIPA	36%	Profit Share, INSEE
δ	depreciation rate	2.5%	Chetty et al. (2011)	2.5%	Own calc., INSEE
Productivity parameters					
σ^y	std. err. prod.	0.10	Gini for income	0.06	Fonseca et al. (2023)
ρ^y	autocorr. prod.	0.99	Gini for income	0.99	Fonseca et al. (2023)

Table 3: Parameter values.

Idiosyncratic risk. The AR(1) productivity process uses $\rho_y = 0.99$ and $\sigma_y = 0.0646$, which are consistent with Fonseca et al. (2023). As for the US, we discretize with 10 states.

Taxes and government budget constraint. We follow Table 1. We consider a capital tax of $\tau^K = 35\%$, a progressivity parameter of $\tau = 0.23$, and a consumption tax of $\tau^c = 18\%$. This tax system has realistic implications for the model, as reported in Table 4. Choosing $\kappa = 0.728$ matches the empirical public-spending-to-GDP of 24% and implies a public-debt-to-GDP ratio of 60%, as in Table 1. The model generates an aggregate private consumption equal to 44% of GDP, which is close to the value of 45% estimated by Trabandt and Uhlig (2011) for the period 1995-2007, while investment amounts to 31% of GDP, equal to its empirical counterpart. The Gini index is 0.28 for post-tax income and 0.68 for wealth, matching the data of Table 1. Again, this confirms that the tax system is empirically relevant.

5.3 Estimation of the SWFs

We estimate the SWF weights using the algorithm in Section 4.3 and the truncation algebra in Appendix D. We set the truncation length to $N = 5$ but results are robust to larger N. With 10 idiosyncratic productivity levels, the number of truncated histories is $10^5 = 100000$.

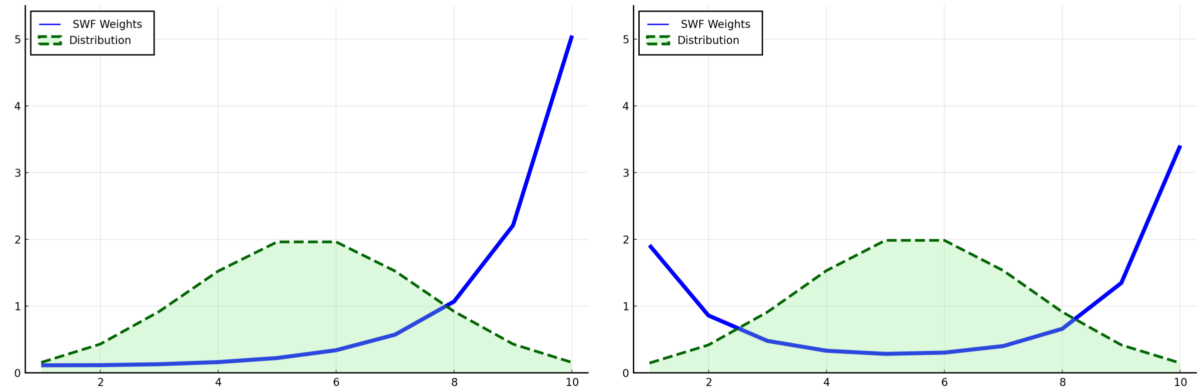
As in Section 4.5, we recover weights such that the Ramsey FOCs are exactly satisfied. Applying the algorithm of Section 4.6, we obtain the following parametric forms:

$$\begin{aligned}\log \omega(y)^{us} &= -0.25 + 1.06 \log(y) + 0.22(\log(y))^2, \\ \log \omega(y)^{fr} &= -0.51 + 0.62 \log(y) + 1.44(\log(y))^2.\end{aligned}$$

Parameter	Description	US		France	
		Model	Data	Model	Data
Public finance aspects					
B/Y	Public debt (%GDP)	63%	63%	60%	60%
G/Y	Public spending (%GDP)	15%	15%	25%	24%
Aggregate quantities					
C/Y	Aggregate consumption (%GDP)	58%	60%	44%	45%
I/Y	Aggregate investment (%GDP)	27%	25%	31%	31%
Inequality measures					
	Gini for post-tax income	40%	40%	28%	28%
	Gini for wealth	78%	77%	68%	68%

Table 4: Model implications for key variables.
Empirical values are discussed in Section 5.1 and summarized in Table 1.

Figure 1 plots the resulting period SWF weights by productivity index. The US weights rise with productivity, and agents with the highest weight in the population are those with the highest productivity. In France, weights are U-shaped, assigning relatively high weight to the lowest and highest productivity groups and lower weight in the middle. The high productivity agents have the highest weights, but the low-productivity agents come next.

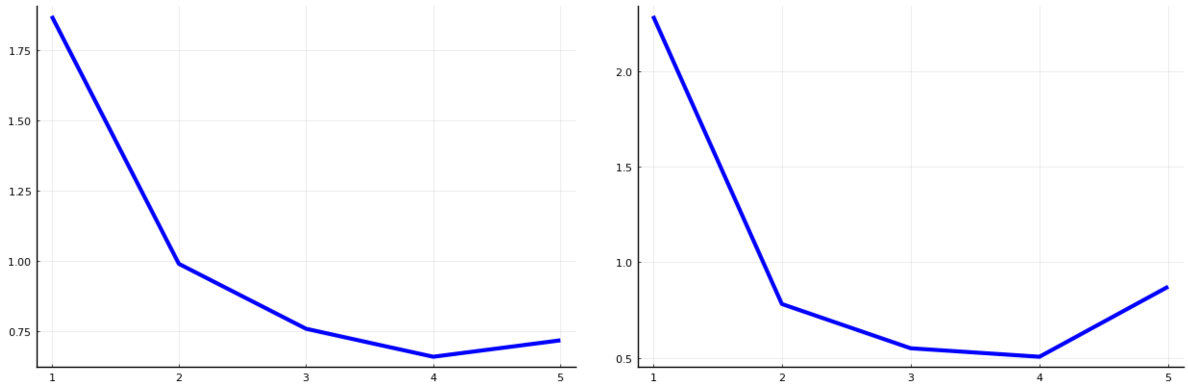


(a) United States (b) France

Figure 1: Parametric period weights as a function of productivity for the US and France.

Implied marginal weights

As discussed in Section 4.4, public finance analyses often consider *social marginal welfare weights* (SMWWs), $\omega_i \bar{u'}_i$, i.e., SWF weights multiplied by average marginal utility for group i. Although the relevant object in our environment is the SVL (equation (42)), SMWWs provide a useful benchmark (see Hendren, 2020 for US evidence). Figure 2 reports mean SMWWs by income quintile, normalized to average one.



(a) United States (b) France

Figure 2: Mean SMWWs per quintile as a function of the quintile of labor income in the United States and in France.

First, despite differences in SWF weights (Figure 1), the SMWWs display broadly decreasing patterns in both countries because average marginal utilities $\bar{u'}_i$ decline sharply with income. Hence, similar SMWW shapes can be consistent with very different SWF profiles.

Second, the shape of US SMWWs is close to the one estimated in Hendren (2020): decreasing in income with a slight uptick at the top. Our first-quintile value (about 1.8) exceeds Hendren's (1.2), reflecting that our fiscal system, although relevant for macroeconomics, is very simple compared to the actual transfer scheme in the US. It does not include some transfers that reduce marginal utility among the poorest in administrative data. Even so, the overall similarity supports the consistency of heterogeneous-agent models with public finance estimates.

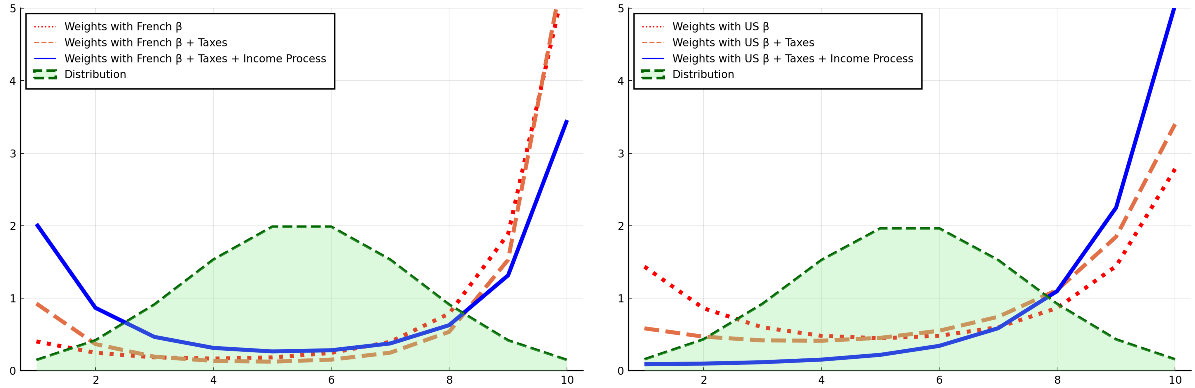
5.4 Investigating the drivers of the weight differences between the US and France

We next explore why estimated weights differ across countries by isolating three dimensions in which calibrations differ: (i) the discount factor β; (ii) the tax system; and (iii) the productivity process.

Discount factor. In panel (a) of Figure 3, the red dotted line shows US weights when replacing the US discount factor by France's. We observe slightly higher weights for low-productivity agents, with the overall increasing shape preserved. In panel (b) of Figure 3, adopting the US discount factors in France reduces weights at the bottom and raises them at the top. Thus, greater patience, for both agents and the planner, tends to tilt the SWF toward the bottom of the distribution and to decrease weights at the top.

Tax system. In panel (a) of Figure 3, the orange dashed line shows US weights under the French tax system and French discount factor β. SWF weights shift toward the bottom at the expense of the top. Conversely, in panel (b) of Figure 3, applying the US tax system (and the US discount factor) to France shifts top weight upward, but bottom weight downward. Hence, tax progressivity and the level of redistribution play a central role in shaping the SWF. Higher progressivity, and a greater inclination to reduce inequality, contributes to increasing the weights at the bottom at the expense of the top.

Productivity process. Finally, we replace the US income process with France's, together with French discount factor and taxes. The resulting SWF weights exactly reproduce the French weights (blue line in panel (a) of Figure 3). The same holds for France (panel (b)). This mechanical check verifies that the three ingredients together account for the cross-country differences.



(a) United States (b) France

Figure 3: Impact of different preference parameters, tax systems, and income processes on the US and French period weights.

We further analyze replacing the US tax system with the French one (and conversely). Figure 7 (Appendix E) shows that adopting the French tax system in the US reduces labor income for high-productivity agents, because it reduces labor supply incentives. Thus, it also reduces the utility of these high-productivity agents. Conversely, the progressive tax system boosts the consumption and the utility of low-productivity agents, and this is the result of their larger SWF weights. This experiment demonstrates that changes in the tax system can explain changes in weights. Adopting a more progressive labor tax would shift US weights toward the bottom at the expense of the top.

5.5 A world where the US has the French SWF

We now compute the US fiscal system that makes its SWF weights as close as possible to France's. The exercise isolates the role of social preferences in shaping policy, holding technology and preferences fixed.

We proceed as follows. Consider a "core" US economy with US preferences, technology, and productivity process. We then jointly choose the capital tax and the labor-tax progressivity to minimize the Euclidean distance between the resulting SWF weights and the French SWF weights. Because SWF weights are defined on each country's productivity grid, we interpolate both sets of weights onto a common grid before computing the distance. Finally, we adjust the parameter κ, driving the labor tax level, to keep the spending-to-output ratio equal to the US benchmark. Hence, at the solution, the main macro ratios (capital-to-output, investment-to-output, aggregate consumption to output, and public spending to output) match US values by construction.

The outputs of this estimation are a fiscal system and SWF weights. Resulting SWF weights of the "US with French SWF" closely match the French profile, as can be seen in Figure 4.

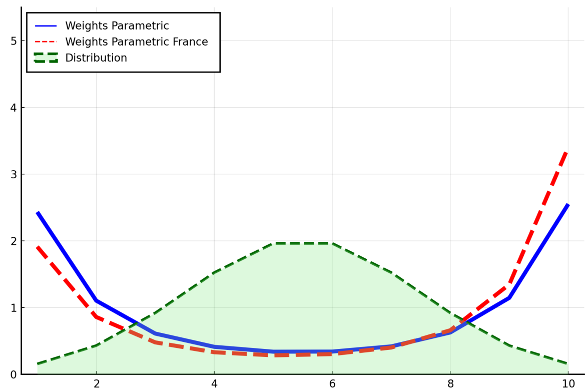


Figure 4: SWF weights for France (red dashed line) and for the US with the tax system that minimizes the distance to the French weights (blue line). The x-axis corresponds to the 10 productivity levels in France.

Table 5 reports the implied fiscal system for the "US with French SWF" alongside the baseline US and French systems. Inequality statistics in the "US with French SWF" are now close to France's and are substantially lower than in the US benchmark.

	Public debt (%GDP)	τ_k (%)	τ (%)	κ (%)	Gini post-tax income	Gini wealth
US	63	36	16	85	40	78
France	60	35	23	73	28	68
US with French SWF	299	9	57	71	27	63

Table 5: Comparison between the benchmark economies and the US economy with the French SWF.

Mechanisms and macro implications. This reduction in inequality is primarily driven by the higher weight placed on low-productivity (low-income) agents in the French

SWF profile. The optimization selects a much higher labor-tax progressivity (from 16% to 57%), which mechanically redistributes toward the bottom. Crucially, the other "core" parameters and the main macro ratios (C/Y, G/Y, I/Y, and—by construction—K/Y) are held at their US benchmark values by appropriate adjustment of κ, and the pre-tax interest rate remains pinned down by $1/\beta$.

Higher progressivity reduces labor supply at the top (via a lower net-of-tax wage), compressing earnings dispersion. Given the maintained US productivity process, the reduction in effective hours for high-productivity agents lowers the efficient labor input. With K/L anchored with the pre-tax interest rate, the implied capital stock adjusts to preserve the steady-state factor price conditions. At the same time, higher social weights at the bottom increase the planner's willingness to trade-off efficiency for redistribution, raising disposable income for low-productivity agents and supporting their consumption. Because the model keeps the aggregate ratios and productivity at US levels, the saving demand is higher than in France. This requires an expansion in public debt—from 63% to 299% of GDP—so that the resource constraint and the targeted G/Y continue to hold.

The higher progressivity is detrimental to high-productivity agents. However, they have a quite a large weight in the French SWF. To partly offset the large progressivity increase for those agents, the capital tax is lowered. This lower capital tax also tends to boost aggregate savings, and hence also contributes to increasing the public debt. Ultimately, the middle class suffers from the higher progressivity and does not benefit much from the lower capital tax, which is why they end up with the lowest weights in the population.

6 Estimation of Individual Welfare Functions

This section develops an identification strategy for Individual Welfare Functions (IWFs) using the estimated SWF. Our quantitative analysis (Section 5) primarily estimates the SWF, which is independent of IWFs. Nonetheless, recovering IWFs consistent with the estimated SWF provides valuable insights into the underlying heterogeneity in social perceptions.

Identifying IWF weights is a high-dimensional problem: we must recover the $|\mathcal{Y}| \times |\mathcal{Y}|$ parameters $\hat{\omega}_{\tilde{y}y}$ from the $|\mathcal{Y}|$ SWF weights. Additional information, such as surveys or experiments, may provide priors on IWFs, denoted $\omega^{prior}_{\tilde{y},y}$ for $(\tilde{y}, y) \in \mathcal{Y}^2$. However, such evidence is indirect. For example, World Values Survey responses indicate that demand for redistribution varies with income, but it is unclear whether this reflects self-interest or ethical views. To address this ambiguity, we propose an estimation strategy that allows us to decompose IWFs into two parts: one independent of SWF, and another independent of priors.

We begin with empirical estimates of political weights $\omega_{P,y}$. Without loss of generality, we normalize using the Euclidean norm: $\sqrt{\sum_{y\in\mathcal{Y}}\pi_y(\omega_{P,y})^2} = 1$. Our estimation strategy consist in choosing IWF weights $\tilde{\omega}$ that are closest to given priors subject to consistency with the estimated SWF. This procedure offers an intuitive decomposition of the IWF into two components, including one independent of the prior, generalizing the results in Section 2.3.

We first formalize the estimation strategy, then present political-weight estimates, and finally report quantitative results.

Definition 3 *Given: SWF weights $(\omega_y)_{y\in\mathcal{Y}}$; political weights $(\omega_{P,y})_{y\in\mathcal{Y}}$ normalized such that $\sum_{y\in\mathcal{Y}}\pi_y(\omega_{P,y})^2 = 1$; and priors on IWFs $\omega^{prior}_{\tilde{y},y}$ for $(\tilde{y},y)\in\mathcal{Y}^2$, the estimated IWF weights are the matrix $(\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y}$ solving*

$$(\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y} \; = \arg\min_{(\hat{\omega}_{\tilde{y}y})_{\tilde{y},y}} \sum_{(\tilde{y},y)\in\mathcal{Y}^2} \pi_{\tilde{y}}\Big(\hat{\omega}_{\tilde{y}y} - \omega^{prior}_{\tilde{y},y}\Big)^2,$$

$$s.t.\ \omega_y = \sum_{\tilde{y}\in\mathcal{Y}} \pi_{\tilde{y}}\omega_{P,\tilde{y}}\hat{\omega}_{\tilde{y}y} \quad \text{for all } y\in\mathcal{Y}.$$

The solution satisfies, for all $(\tilde{y},y)\in\mathcal{Y}^2$,

$$\tilde{\omega}_{\tilde{y}y} = F(\tilde{y},y) + \omega_{P,\tilde{y}}\omega_y, \tag{46}$$

where $F(\tilde{y},y)$ depends on the prior $\omega^{prior}_{\tilde{y},y}$ but not on ω_y.

Equation (46) has a transparent interpretation (see Appendix A.6 for the details of the algebra). The first term captures the contribution of the prior and is independent of the estimated SWF weights. The second term, $\omega_{P,\tilde{y}}\omega_y$, is the contribution to IWFs implied by the observed SWF and is prior-free. Intuitively, if the aggregate social weight on type y is high, then agents with greater political influence (high $\omega_{P,\tilde{y}}$) must assign more weight to type y to rationalize the aggregate; thus, the estimated valuation increases with both ω_y and $\omega_{P,\tilde{y}}$.

To report results that are independent of priors, we focus on this adjustment term,

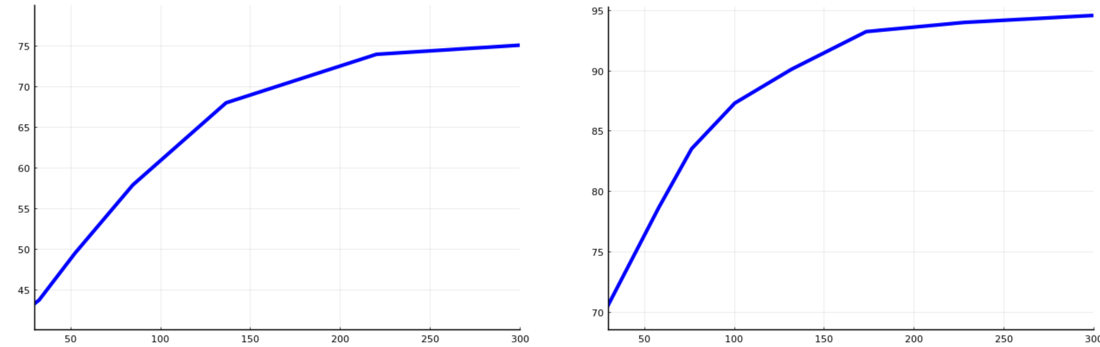
$$CIWF_{\tilde{y},y} := \omega_{P,\tilde{y}}\omega_y,$$

which we label the *contribution in IWFs* (CIWF).

For completeness, Appendix H reports estimated IWFs under a standard *self-interested* prior, $\omega^{prior}_{\tilde{y},y} = 1_{y=\tilde{y}}$, where agents only value their own productivity type. This allows us to illustrate how priors affect the prior-specific term $F(\tilde{y},y)$.

6.1 Political weights

The political-economy literature offers several proxies for political power across income groups: turnout, donations, and representation biases (see Cagé, 2024 for a survey). All suggest that political weights increase with income. Because donation regimes differ substantially between France and the United States, we use turnout-based measures and discuss robustness in Section 7.[28]



a) United States b) France

Figure 5: Participation rates (in percent) as a function of annual individual income (expressed as a percentage of average income). US data refer to the 2008 presidential election. Data for France are for the 2007 presidential election. The 100 on the x-axis is the average income in each country.

Figure 5 plots participation rates by relative income for the US (panel a) and France (panel b). Relative income is computed as a percentage of the ratio of the individual annual income and the average annual individual income in the country.[29] Average individual income is \$51726 in the US and 31093 € in France (both 2007). We observe that participation rises with income in both countries. Our identification strategy is to infer relative political weights from relative participation:

$$\frac{\omega_{P,y}}{\omega_{P,\tilde{y}}} = \frac{Part_y}{Part_{\tilde{y}}},$$

where $Part_y$ is the participation rate of income group y, interpolated from Figure 5. Thus, political weights inherit the shape of participation profiles. Cross-country differences in average participation levels do not directly map into political weights under this normalization.

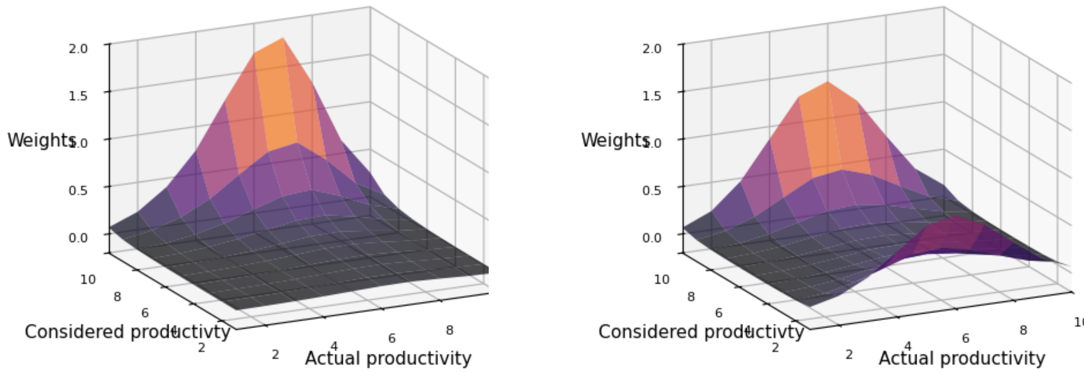
6.2 Results

We report the CIWF, $CIWF_{y,\tilde{y}} = \omega_{P,y}\, \omega_{\tilde{y}}$, for the United States and France. Figure 6 displays, for each country, the politically weighted factors from *Actual productivity* y (valuer) to *Considered productivity* $\tilde{y}$ (recipient): panel (a) shows the US and panel (b)

[28]In France, individual donations are capped at a relatively low level (7500€), complicating cross-country comparisons. Moreover, evaluating the causal effectiveness of donations on policy is difficult. Our identification strategy is robust to alternative measures that generate increasing political weights.

[29]US: Current Population Survey, November 2008, Table 8 (US Census Bureau). France: IPSOS participation by occupation combined with DADS 2007 to map occupations to annual income.

France. The z-axis reports $\omega_{P,y}\,\omega_{\tilde{y}}$ for all $y, \tilde{y} = 1, \ldots, 10$. As discussed in Section 6, these factors indicate how political influence on the valuer side loads onto social valuation on the recipient side.



(a) United States

(b) France

Figure 6: CIWF weights for the US and France

Note: The CIWF weights are measured as $(\omega_{P,y}\omega_{\tilde{y}})_{y,\tilde{y}}$. *Actual productivity* is the productivity y of the agents, while *Considered productivity* is the productivity $\tilde{y}$ of the agents under consideration.

Three patterns emerge. First, in both countries, CIWF increases with the recipient's productivity $\tilde{y}$, reflecting larger SWF weights at higher productivity; the gradient is steeper in the US, where the most productive types receive the largest valuation. Second, along the valuer dimension y, the US exhibits a stronger gradient: groups with greater political weight load more onto all recipients, generating a concentrated peak around intermediate y and mid-high $\tilde{y}$. Third, combining these dimensions, CIWF is largest where politically influential valuers (y) align with socially emphasized recipients ($\tilde{y}$). Political power translates therefore directly into higher social valuation of productive types. Hence, shifts in political influence among high-y groups disproportionately affect the social valuation of high-$\tilde{y}$ recipients, with clear distributional implications. This dynamic is particularly pronounced in the US. In France, weights are more evenly distributed across y, indicating a more balanced mapping from political influence to social valuation.

7 Discussion of identification and cardinality assumptions

Linear aggregation assumption. We consider the linear aggregation procedure at the individual level (concerning IWFs), which enables a transparent and tractable aggregation from IWFs to SWFs. To our knowledge, this approach was first proposed in Aiyagari and McGrattan (1998) for incomplete-insurance models and has since become standard because it is consistent with the expected-utility framework.[30]

Departing from this linear aggregation would introduce additional concavity that would be difficult to identify and estimate empirically. As emphasized by Saez and Stantcheva (2016), the linear SWF can be understood as a first-order Taylor expansion of a more general SWF. This is also the approach followed in Olivi et al. (2025). SWF weights can thus be interpreted as marginal valuations in the neighborhood of the observed allocation, of a more general SWF. Hence, our estimated SWF weights can be understood as *marginal social weights*.

The cardinality of the utility function and estimated weights. In the static setting, utility functions are ordinal concepts defined up to a monotonic transformation. Indeed, u and $\phi \circ u$ represent the same preferences whenever ϕ is strictly increasing. However, in the presence of risk and in the context of expected utility, the function u is unique up to a positive affine transformation only – and not to a general increasing transformation. This is also the case in a multi-period setting. In particular, changing the period utility function u to $\phi \circ u$ with a general ϕ changes the agents' preferences: it alters the elasticity of substitution and the risk aversion.

However, our SWF weights are invariant to affine transformations. Therefore, the cardinality assumption we are making is actually subsumed in the expected utility framework and does not affect our SWF weights.

Identification of the SWF. In our baseline specification, we adopt the functional form (43) for SWF weights. The estimation strategy of Definition 2 introduces a parametric solution to address the underdetermination of weights, which is required since the number of unknowns exceeds the number of Ramsey constraints. We verify that results are robust to relaxing this parametric form. In particular, we estimate nonparametric weights by selecting the variance-minimizing weights that satisfy the Ramsey constraints. These weights can also be interpreted as the closest to the utilitarian benchmark. The resulting nonparametric estimates, reported in Appendix F.1, are very similar to those in the main text: SWF weights increase with productivity in the US and display a U-shape in France.

Weights depending on histories. The identification of SWF weights relies on the assumption that weights depend only on current productivity (Assumption A). As discussed in Section 3, weights cannot depend on endogenous variables such as wealth or consumption, since the planner would then need to account for the effect of allocations on its own weights, potentially generating inconsistencies and multiple equilibria. However, in a more general setting, weights could depend on the full idiosyncratic history. We

[30] More precisely, Aiyagari and McGrattan (1998) show that the utilitarian SWF is consistent with expected individual welfare when state variables are drawn from the equilibrium distribution.

interpret the restriction of weights dependence to current productivity as a practical identification assumption with negligible quantitative cost.

Indeed, relaxing this assumption and assigning separate weights to each truncated history implies Y^N weights instead of Y (where Y is the number of productivity levels). We estimate these as the variance-minimizing weights satisfying the Ramsey constraints. The resulting estimates are nearly identical to the benchmark ones. In particular, the average of history-dependent weights by initial productivity is very close to the nonparametric weights discussed above. Further details are provided in Appendix F.2.

Political weights. Political weights pose a different normalization issue, as there is no clear convention in the literature. Following Section 6, we rely on voter participation rates in the US and France, since other measures (e.g., lobbying resources or donations) are not comparable across countries or only loosely related to our concept of political influence. Because we use ratios of participation rates, our strategy is invariant to any linear transformation of political weights. While nonlinear transformations could in principle affect the planner's problem, our robustness checks suggest that such effects are quantitatively small.

Specifically, we explore ad hoc variations in the profiles of political weights – flat, linearly increasing, or convex (instead of concave) – using the participation data plotted in Figure 5. None of these alternatives substantially affects the estimated IWFs. The results under these alternative profiles are presented in Appendix G. Hence, although the exact measurement of political weights remains an open issue, our identification strategy appears robust.

8 Conclusion

We propose a methodology to identify the Social Welfare Function (SWF) and the Individual Welfare Functions (IWFs) from a country's empirical wealth and income distributions and its observed tax system. We apply this approach to France and the United States. Using four fiscal instruments—consumption, capital, and progressive labor taxes, together with public debt—we estimate the SWF in each country and show that their structures differ markedly. The French SWF assigns greater weight to low-productivity agents and exhibits less heterogeneity, whereas the US SWF increases with productivity, placing larger weights on higher-productivity agents. The United States thus appears more inclined toward *low-redistribution*, while France is more *egalitarian*, particularly at lower income levels.

These results open new avenues for research, especially regarding the stability of social preferences over time. A key next step is to examine the role of the SWF in shaping fiscal responses to economic shocks, particularly in business cycle stabilization.

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Online Appendix

A Proofs related to the SWF construction

A.1 Construction of the measure on the set of idiosyncratic histories $\mathcal{Y}^\infty$

We can construct a probability space related to the set of infinite idiosyncratic histories, $\mathcal{Y}^\infty$. We summarize here the construction, and further details can be found in LeGrand and Ragot (2022a, Appendix B.3). Consistently with the main text, we will typically denote by y^t an element of $\mathcal{Y}^\infty$. Such an element $\tilde{y}^t \in \mathcal{Y}^\infty$ can be described as a left-infinite sequence:

$$\tilde{y}^t = (\ldots, y_{-k}(\tilde{y}^t), \ldots, y_{-1}(\tilde{y}^t), y_0(\tilde{y}^t)),$$

where each $y_{-k} : \mathcal{Y}^\infty \to \mathcal{Y}$ is a coordinate function returning the idiosyncratic state k periods ago. For the sake of simplicity, and as in the main text, we will denote by $y_s^t := y_{-(t-s)}(y^t)$ for any $s \leq t$ the state at date s, which is $t-s$ periods ahead of date t.

We define $L(y^t)$ as the past of history y^t – which discards the current state y_t^t:

$$L(y^t) = (\ldots, y_{t-k}^t, \ldots, y_{t-1}^t).$$

Consistently with the main text, we also denote by $y^{t-1,t}$ the past history of y^t: $y^{t-1,t} = L(y^t)$.

We can then define the cylinder sets $C_k(A)$ for any $k \geq 1$ and any $A \subset \mathcal{Y}^k$ as follows:

$$C_k(A) = \{y^t \in \mathcal{Y}^\infty : (y_{t-k+1}^t, \ldots, y_t^t) \in A\}.$$

The cylinder set $C_k(A)$ is the subset of $\mathcal{Y}^\infty$ containing all idiosyncratic histories whose truncation of length k belongs to A. We then define $\mathcal{C}_0$ as the set of all cylinder sets, which can be shown to be a field (Billingsley, 2012, Section 2). We denote by $\mathcal{F} := \sigma(\mathcal{C}_0)$ the cylindrical σ-algebra generated by $\mathcal{C}_0$, and we define the set function $\mu : \mathcal{C}_0 \to \mathbb{R}$ from the transition matrix Π and the stationary vector π, such that for any $k \geq 2$ and any $A \subset \mathcal{Y}^k$:

$$\begin{cases} \mu(C_k(A)) = \sum_{(y_{-k+1},\ldots,y_0)\in A} \pi_{y_{-k+1}} \Pi_{y_{-k+1}y_{-k+2}} \ldots \Pi_{y_{-1}y_0} & \text{for any } k \geq 2 \text{ and } A \subset \mathcal{Y}^k, \\ \mu(C_1(A)) = \sum_{y_0 \in A} \pi_{y_0} & \text{for any } A \subset \mathcal{Y}. \end{cases} \tag{47}$$

Finally, we can state the following lemma.

Lemma 1 *The triplet $(\mathcal{Y}^\infty, \mathcal{F}, \mu)$ is a probability space.*

Proof. A proof can be found in Billingsley (2012, Section 2). The key part of the proof is to extend the measure μ defined on $\mathcal{C}_0$ to a measure defined on $\sigma(\mathcal{C}_0) = \mathcal{F}$. $\blacksquare$ A consequence of Lemma 1 is that $\mu(\mathcal{Y}^\infty) = 1$, or $\int_{y^t \in \mathcal{Y}^\infty} \mu(dy^t) = 1$.

A.2 The conditional measure

To lighten formulas, for any $(y_{-k+1}, \dots y_0) \in \mathcal{Y}^k$, we define $y_{-k+1:0} := (y_{-k+1}, \dots, y_0)$ as the vector of length k, containing elements whose indices range from $-k+1$ to 0. Similarly, for any $\tilde{y}^t \in \mathcal{Y}^\infty$, $\tilde{y}^t_{(t-k+1:t)} = (\tilde{y}^t_{t-k+1}, \dots, \tilde{y}^t_t)$ is the vector of the last k realization of $\tilde{y}^t$. We define μ_1 by:

$$\mu_1(C_1(Y_1)|\tilde{y}^t) = \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}^t_t y_{t+1}}, \text{ for any } Y_1 \subset \mathcal{Y}, \tag{48}$$

$$\forall k \geq 2,\ \mu_1(C_k(Y_k)|\tilde{y}^t) = \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{t-k+2:t}}, \text{ for any } Y_k \subset \mathcal{Y}^k, \tag{49}$$

where for any elements $x, \tilde{x}$ of the same set, $1_{x=\tilde{x}} = 1$ if $x = \tilde{x}$ and $1_{x=\tilde{x}} = 0$ otherwise. Intuitively, the expression in (49) sums, over all possible vectors $y_{(t-k+2):t+1}$ of length k, the probability to switch from $\tilde{y}^t$ to a history ending up in $y_{(t-k+2):t+1}$. The latter probability is equal to the probability to switch from $\tilde{y}^t_t$ to y_{t+1}, provided that $\tilde{y}^t$ and $y_{(t-k+2):t}$ are compatible (i.e., the $k-1$ last realization of $\tilde{y}^t$ equals $y_{(t-k+2):t}$). Note that we could also write: $1_{y_{(t-k+2:t)} = \tilde{y}^t_{(t-k+2:t)}} = \prod_{j=0}^{k-2} 1_{y_{t-j} = \tilde{y}^t_{t-j}}$.

Lemma 2 *For any $\tilde{y}^t \in \mathcal{Y}^\infty$, the set function $C \in \mathcal{C}_0 \mapsto \mu_1(C|\tilde{y}^t)$ is a pre-measure.*

Proof.

In the remainder of the proof, we set $\tilde{y}^t \in \mathcal{Y}^\infty$. As a preliminary, we state two properties that we will use extensively below:

- For all $k' \geq k \geq 1$, for all $Y_k \subset \mathcal{Y}^k$, for all $Y'_{k'-k} \subset \mathcal{Y}^{k'-k}$:

$$\begin{aligned}
&\sum_{y_{(t-k'+2):t+1} \in Y'_{k'-k} \times Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}^t_{(t-k'+2):t}} = \\
&\sum_{y_{(t-k+2):t+1} \in Y_k} \Bigg(\Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{(t-k+2):t}} \times \\
&\sum_{y_{(t-k'+2):(t-k+1)} \in Y'_{k'-k}} 1_{y_{(t-k'+2):(t-k+1)} = \tilde{y}^t_{(t-k'+2):(t-k+1)}} \Bigg)
\end{aligned} \tag{50}$$

- For all $k' \geq k \geq 0$:

$$\sum_{y_{t-k':t-k} \in \mathcal{Y}^{k'-k+1}} 1_{y_{t-k':t-k} = \tilde{y}^t_{t-k':t-k}} = 1. \tag{51}$$

In the remainder these will be referred to by their equation numbering. The proof of (51) is straightforward and comes from the fact that $(\tilde{y}^t_{t-k'}, \dots, \tilde{y}^t_{t-k})$ is a unique element of the set $\mathcal{Y}^{k'-k+1}$. For the proof of (50), we denote by $S_{k,k'}$ the left-hand side. We have:

$$S_{k,k'} = \sum_{y_{(t-k+2):t+1}\in Y_k} \sum_{y_{(t-k'+2):(t-k+1)}\in Y'_{k'-k}} \Bigg(\Pi_{\tilde{y}_t^t y_{t+1}} \times \\ 1_{y_{(t-k+2):t}=\tilde{y}^t_{(t-k+2):t}} 1_{y_{(t-k'+2):(t-k+1)}=\tilde{y}^t_{(t-k'+2):(t-k+1)}} \Bigg), \tag{52}$$

where we have used the properties of a sum on a product space and the fact that $(x, y) = (x', y')$ iff $x = x'$ and $y = y'$ (where (x, x') and (y, y') are two pairs of vectors of the same length). Then we can factorize $\Pi_{\tilde{y}_t^t y_{t+1}}$ and $1_{y_{(t-k+2):t}=\tilde{y}^t_{(t-k+2):t}}$ in (52), as they are independent from the sum over $y_{(t-k'+2):(t-k+1)} \in Y'_{k'-k}$. We then readily obtain (50).

We now go back to the proof of Lemma 2. Three properties need to hold: (i) well-defined, (ii) (countably) additive, (iii) $\mu_1(\mathcal{Y}^\infty|\tilde{y}^t) = 1$ for all $\tilde{y}^t \in \mathcal{Y}^\infty$.

For Point (i), we need to check that $\mu_1(C_k(Y_k)|\tilde{y}^t) = \mu_1(C_{k'}(\mathcal{Y}^{k'-k} \times Y_k)|\tilde{y}^t)$ for all $k' \geq k$. We have for $k \geq 2$:

$$\begin{aligned}\mu_1(C_{k'}(\mathcal{Y}^{k'-k} \times Y_k)|\tilde{y}^t) &= \sum_{y_{(t-k'+2):t+1}\in\mathcal{Y}^{k'-k}\times Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k'+2):t}=\tilde{y}^t_{(t-k'+2):t}}, \\ &= \sum_{y_{(t-k+2):t+1}\in Y_k} \Pi_{\tilde{y}_t^t y_{t+1}} 1_{y_{(t-k+2):t}=\tilde{y}^t_{(t-k+2):t}} = \mu_1(C_k(Y_k)|\tilde{y}^t), \end{aligned} \tag{53}$$

where the first and last equalities use the definition of μ_1, and the second the combination of properties (50) and (51).

We need to prove the result for $k = 1$. We have for $k' \geq 1$:

$$\begin{aligned}\mu_1(C_{k'}(\mathcal{Y}^{k'-1} \times Y_1)|\tilde{y}^t) &= \sum_{y_{t+1}\in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} \Bigg(\sum_{y_{(t-k'+2):t}\in\mathcal{Y}^{k'-1}} 1_{y_{(t-k'+2):t}=\tilde{y}^t_{(t-k'+2):t}} \Bigg), \\ &= \sum_{y_{t+1}\in Y_1} \Pi_{\tilde{y}_t^t y_{t+1}} = \mu_1(C_1(Y_1)|\tilde{y}^t),\end{aligned}$$

where the first equality combines the definition of μ_1 and (50), the second uses the property (51), and the last the definition of μ_1.

For point (ii), we consider two disjoint cylinders, $C_k(Y_k)$ and $C_{k'}(Y'_{k'})$ ($k' \geq k$, $Y_k \subset \mathcal{Y}^k$ and $Y'_{k'} \subset \mathcal{Y}^{k'}$). Since both cylinders are disjoint, then $\mathcal{Y}^{k'-k} \times Y_k$ and $Y'_{k'}$ are disjoint too.

We deduce that:

$$\begin{aligned}
\mu_1(C_k(Y_k) \cup C_{k'}(Y'_{k'})|\tilde{y}^t) &= \mu_1(C_{k'}((\mathcal{Y}^{k'-k} \times Y_k) \cup Y'_{k'})|\tilde{y}^t), \\
&= \sum_{y_{(t-k'+2):t+1} \in (\mathcal{Y}^{k'-k} \times Y_k) \cup Y'_{k'}} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}^t_{(t-k'+2):t}}, \\
&= \sum_{y_{(t-k'+2):t+1} \in (\mathcal{Y}^{k'-k} \times Y_k)} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}^t_{(t-k'+2):t}} \\
&+ \sum_{y_{(t-k'+2):t+1} \in Y'_{k'}} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}^t_{(t-k'+2):t}}, \\
&= \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{(t-k+2):t}} \\
&+ \sum_{y_{(t-k'+2):t+1} \in Y'_{k'}} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k'+2):t} = \tilde{y}^t_{(t-k'+2):t}}, \\
&= \mu_1(C_k(Y_k)|\tilde{y}^t) + \mu_1(C_{k'}(Y'_{k'})|\tilde{y}^t),
\end{aligned}$$

where the first equality uses the algebra property of cylinder sets, the second the definition of μ_1, the third the property that $\mathcal{Y}^{k'-k} \times Y_k$ and $Y'_{k'}$ are disjoint, the fourth the combination of properties (50) and (51), and the last the definition of μ_1 twice. We have thus proved that $\mu_1(\cdot|\tilde{y}^t)$ is finitely additive.

For point (iii), let $k \geq 2$. We have:

$$\begin{aligned}
\mu_1(C_k(\mathcal{Y}^k)|\tilde{y}^t) &= \sum_{y_{t+1} \in \mathcal{Y}} \Pi_{\tilde{y}^t_t y_{t+1}} \sum_{y_{(t-k+2):t} \in \mathcal{Y}^{k-1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{(t-k+2):t}}, \\
&= \sum_{y_{t+1} \in \mathcal{Y}} \Pi_{\tilde{y}^t_t y_{t+1}} = 1,
\end{aligned}$$

where the first equality uses the definition of μ_1 and (50), the second property (51), and the third the property of the transition matrix Π. We thus deduce that $\mu_1(\mathcal{Y}^\infty|\tilde{y}^t) = 1$.

We have proven that $\mu_1(\cdot|\tilde{y}^t)$ is a finitely additive probability measure on the algebra $\mathcal{C}_0$. Billingsley (2012, Theorem 2.3) states that any finitely additive probability measure on the cylinder algebra is countably additive. We thus conclude that μ_1 is a countably additive probability measure on $\mathcal{C}_0$ and is therefore a pre-measure on $\mathcal{C}_0$. ■

We then prove the following lemma. We recall that $\mathcal{F}$ is the cylindrical σ-algebra, $\sigma(\mathcal{C}_0)$, generated by $\mathcal{C}_0$.

Lemma 3 *For all $\tilde{y}^t \in \mathcal{Y}^\infty$, the function $\mu_1(\cdot|\tilde{y}^t)$ uniquely extends to a measure on $\mathcal{F}$.*

The proof is similar to the one showing the extension of μ as a pre-measure on $\mathcal{C}_0$ to a measure on $\mathcal{F}$. It relies on the Hahn-Kolmogorov theorem (Billingsley 2012, Theorem 3.1). See LeGrand and Ragot (2022a, Lemma 3 in Section B.3).

Finally we can state the following lemma, showing that μ_1 is a conditional measure.

Lemma 4 *For all $\tilde{y}^t \in \mathcal{Y}^\infty$ and for all $F \in \mathcal{F}$:*

$$\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(F|\tilde{y}^t)\mu(d\tilde{y}^t) = \mu(F). \tag{54}$$

Proof. We first prove (54) for F being a cylinder set.

First, let $Y_1 \subset \mathcal{Y}$ and consider $C_1(Y_1)$. We have:

$$\begin{aligned}
\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(C_1(Y_1)|\tilde{y}^t)\mu(d\tilde{y}^t) &= \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}^t_t y_{t+1}} \mu(d\tilde{y}^t), \\
&= \sum_{\tilde{y}^t_t \in \mathcal{Y}} \pi_{\tilde{y}^t_t} \sum_{y_{t+1} \in Y_1} \Pi_{\tilde{y}^t_t y_{t+1}}, \\
&= \sum_{y_{t+1} \in Y_1} \sum_{\tilde{y}^t_t \in \mathcal{Y}} \pi_{\tilde{y}^t_t} \Pi_{\tilde{y}^t_t y_{t+1}}, \\
&= \sum_{y_{t+1} \in Y_1} \pi_{y_0} = \mu(C_1(Y_1)),
\end{aligned}$$

where the first equality comes from the definition (48) of μ_1, the second from the fact that the integral is actually carried over a cylinder set of the form $C_1(\mathcal{Y})$ and from the definition (47) of μ, the third from the permutation of finite sums, the fourth from the fact that π is stationary ($\sum_{y \in \mathcal{Y}} \pi_y \Pi_{yy'} = \pi_{y'}$), and the last (as in the line four) from the definition (47) of μ.

Second, let $Y_k \subset \mathcal{Y}^k$ and consider $C_k(Y_k)$. We have:

$$\begin{aligned}
&\int_{\tilde{y}^t \in \mathcal{Y}^\infty} \mu_1(C_k(Y_k)|\tilde{y}^t)\mu(d\tilde{y}^t) \\
&= \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{(t-k+2):t}} \mu(d\tilde{y}^t), \\
&= \sum_{\tilde{y}^t_{(t-k+2):t} \in \mathcal{Y}^{k-1}} \pi_{\tilde{y}^t_{t-k+2}} \Pi_{\tilde{y}^t_{t-k+2} \tilde{y}^t_{t-k+3}} \dots \Pi_{\tilde{y}^t_{t-1} \tilde{y}^t_t} \sum_{y_{(t-k+2):t+1} \in Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{(t-k+2):t}}, \\
&= \sum_{y_{(t-k+2):t+1} \in Y_k} \sum_{\tilde{y}^t_{(t-k+2):t} \in \mathcal{Y}^{k-1}} \pi_{\tilde{y}^t_{t-k+2}} \Pi_{\tilde{y}^t_{t-k+2} \tilde{y}^t_{t-k+3}} \dots \Pi_{\tilde{y}^t_{t-1} \tilde{y}^t_t} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t} = \tilde{y}^t_{(t-k+2):t}}, \\
&= \sum_{y_{(t-k+2):t+1} \in Y_k} \pi_{y_{t-k+2}} \Pi_{y_{t-k+2} y_{t-k+3}} \dots \Pi_{y_{t-1} y_t} \Pi_{y_t y_{t+1}} = \mu(C_k(Y_k)),
\end{aligned}$$

where the first equality comes from the definition (49) of μ_1, the second from the fact that the integral is actually carried over a cylinder set of the form $C_{k-1}(\mathcal{Y}^{k-1})$ and from the definition (47) of μ, the third from the permutation of finite sums, the fourth from the fact that all terms in the sum over $\tilde{y}^t_{(t-k+2):t} \in \mathcal{Y}^{k-1}$ are zero except the one for $\tilde{y}^t_{(t-k+2):t} = y_{(t-k+2):t}$, and the last from the definition (47) of μ. This proves (54) on $\mathcal{C}_0$. Since cylinder sets are a generating family of $\mathcal{F}$ and a π-system, the equality (54) also holds on $\sigma(\mathcal{C}_0) = \mathcal{F}$. ■

We finally prove the following lemma. It justifies that in the main text we define μ_1 as $\mu_1(dy^{t+1}|\tilde{y}^t) = \Pi_{\tilde{y}^t_t y^{t+1}_{t+1}} \delta_{\tilde{y}^t}(dy^t)$.

Lemma 5 *For all $\tilde{y}^t \in \mathcal{Y}^\infty$ and for all $C \in \mathcal{C}_0$, we have:*

$$\mu_1(C|\tilde{y}^t) = \int_{y^{t+1}\in C} \Pi_{\tilde{y}^t_t y^{t+1}_{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})),$$

where $\delta_{\tilde{y}^t}$ is the Dirac mass in $\tilde{y}^t$.

Proof. Let $Y_k \subset \mathcal{Y}^k$ for some $k \geq 1$.

For $k = 1$, we have:

$$\begin{aligned}\int_{y^{t+1}\in C_1(Y_1)} \Pi_{\tilde{y}^t_t y^{t+1}_{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})) &= \int_{(y^t,y_{t+1})\in\mathcal{Y}^\infty\times Y_1} \Pi_{\tilde{y}^t_t y_{t+1}} \delta_{\tilde{y}^t}(dy^t),\\ &= \sum_{y_{t+1}\in Y_1} \Pi_{\tilde{y}^t_t y_{t+1}} \int_{y^t\in\mathcal{Y}^\infty} \delta_{\tilde{y}^t}(dy^t),\\ &= \sum_{y_{t+1}\in Y_1} \Pi_{\tilde{y}^t_t y_{t+1}} = \mu_1(C_1(Y_1)|\tilde{y}^t),\end{aligned}$$

where the first equality comes from using $C_1(Y_1) = \mathcal{Y}^\infty \times Y_1$, the second from using the Fubini theorem (we consider σ-finite measure spaces and $(\tilde{y}^t, y_{t+1}) \in \mathcal{Y}^\infty \times Y_1 \mapsto \Pi_{\tilde{y}^t_t y_{t+1}} \delta_{\tilde{y}^t}(dy^t)$ is integrable), the third the property of Dirac mass, and the fourth the definition of μ_1.

For $k \geq 2$, we have:

$$\int_{y^{t+1}\in C_k(Y_k)} \Pi_{\tilde{y}^t_t y^{t+1}_{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})) \tag{55}$$

$$= \int_{(y^{t-k+1},y_{(t-k+2):t+1})\in\mathcal{Y}^\infty\times Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t}=\tilde{y}^t_{(t-k+2):t}} \delta_{L^{(k-1)}(\tilde{y}^t)}(dy^{t-k+1}), \tag{56}$$

where $L^{(k-1)}(\cdot)$ is k-iterate of the shift operator L: $L^{(k-1)}(\tilde{y}^t) = (\ldots, \tilde{y}^t_{t-k}, \tilde{y}^t_{t-k+1})$. To write equality (55), we used that $C_k(Y_k) = \mathcal{Y}^\infty \times Y_k$ and the fact that $\delta_{\tilde{y}^t}(Y_{k-1} \times C) = \sum_{y_{(t-k+2):t}\in Y_{k-1}} 1_{y_{(t-k+2):t}=\tilde{y}^t_{(t-k+2):t}} \delta_{L^{(k-1)}(\tilde{y}^t)}(C)$ for all $C \in \mathcal{C}_0$ and $Y_{k-1} \subset \mathcal{Y}^{k-1}$. Hence the two measures coincide on $\mathcal{C}_0$ and are also σ- finite. Using Fubini and the property of a Dirac mass, we obtain from (55):

$$\begin{aligned}\int_{y^{t+1}\in C_k(Y_k)} \Pi_{\tilde{y}^t_t y^{t+1}_{t+1}} \delta_{\tilde{y}^t}(L(dy^{t+1})) &= \sum_{y_{(t-k+2):t+1}\in Y_k} \Pi_{\tilde{y}^t_t y_{t+1}} 1_{y_{(t-k+2):t}=\tilde{y}^t_{(t-k+2):t}},\\ &= \mu_1(C_k(Y_k)|\tilde{y}^t),\end{aligned}$$

where the last equality comes from the definition of μ_1. This concludes the proof. ■

A.3 Proof of Equality 10

We consider an agent whose individual preferences are represented by a utility function denoted by $V : \mathcal{Y}^\infty \to \mathbb{R}$. The allocation is subsumed. Let $y^t \in \mathcal{Y}^\infty$. The representation result of equation (9) becomes, after splitting the sum for $s = 0$ and $s \geq 1$:

$$V(y^t) = U(y^t) + \beta \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \mu_s(dy^{t+s}|y^t). \tag{57}$$

Using the definition of μ_s given in Section 3.1 and Bayes' rule, we have, for all $s \geq 1$:

$$\mu_s(dy^{t+s}|y^t) = \int_{y^{t+1} \in \mathcal{Y}^\infty} \mu_{s-1}(dy^{t+s}|y^{t+1}) \mu_1(dy^{t+1}|y^t). \tag{58}$$

In words, it means that the probability of transitioning from y^t to y^{t+s} in s periods is equal to the product of the probabilities of transitioning from y^t to y^{t+1} (in 1 period) and from y^{t+1} to y^{t+s} (in $s - 1$ periods), summed over all possible histories y^{t+1}. Using (58), the expression of V in (57) becomes:

$$\begin{aligned} V(y^t) = U(y^t) + \beta \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} & \beta^{s-1} U(y^{t+s}) \\ \times \int_{y^{t+1} \in \mathcal{Y}^\infty} & \mu_{s-1}(dy^{t+s}|y^{t+1}) \mu_1(dy^{t+1}|y^t). \end{aligned}$$

Since all functions are integrable, we can use the Fubini theorem twice, to swap the order of integrals and then of the sum and of the integral on $y^{t+1} \in \mathcal{Y}^\infty$. We obtain:

$$\begin{aligned} V(y^t) &= U(y^t) + \\ &+ \beta \int_{y^{t+1} \in \mathcal{Y}^\infty} \left\{ \sum_{s=1}^{\infty} \int_{y^{t+s} \in \mathcal{Y}^\infty} \beta^{s-1} U(y^{t+s}) \mu_{s-1}(dy^{t+s}|y^{t+1}) \right\} \mu_1(dy^{t+1}|y^t). \end{aligned} \tag{59}$$

The term between curly braces is $V(y^{t+1})$ using (9) applied to $t+1$. The previous equality becomes:

$$V(y^t) = U(y^t) + \beta \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1}|y^t).$$

which gives the representation (10) using the notation $\mathbb{E}_{y^{t+1}} \left[V(y^{t+1}) | y^t \right] = \int_{y^{t+1} \in \mathcal{Y}^\infty} V(y^{t+1}) \mu_1(dy^{t+1}|y^t)$ for the conditional expectation.

A.4 Proof of Proposition 1

Using the definition (12), the expression (13) of the SWF becomes:

$$SWF = \int_{y^t \in \mathcal{Y}^\infty} \omega_P(y^t) \int_{\tilde{y}^t \in \mathcal{Y}^\infty} \hat{V}(y^t, \tilde{y}^t) \mu(d\tilde{y}^t) \mu(dy^t),$$

which can be further simplified using the expression (11) of $\hat{V}$:

$$
\begin{aligned}
SWF &= \int_{y^t\in\mathcal{Y}^\infty} \omega_P(y^t) \int_{\tilde{y}^t\in\mathcal{Y}^\infty} \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s}\in\mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s}|\tilde{y}^t)\mu(d\tilde{y}^t)\mu(dy^t),\\
&= \int_{y^t\in\mathcal{Y}^\infty} \omega_P(y^t) \sum_{s=0}^{\infty} \int_{\tilde{y}^t\in\mathcal{Y}^\infty} \int_{\hat{y}^{t+s}\in\mathcal{Y}^\infty} \beta^s \hat{\omega}(y^t, \hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu_s(d\hat{y}^{t+s}|\tilde{y}^t)\mu(d\tilde{y}^t)\mu(dy^t).
\end{aligned}
$$

Since all functions under consideration are positive and measurable and since it is assumed that $SWF < \infty$, we can use the Fubini theorem to permute the order of integrals and obtain:

$$
\begin{aligned}
SWF = \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s}\in\mathcal{Y}^\infty} \beta^s & \left[\int_{y^t\in\mathcal{Y}^\infty} \omega_P(y^t)\hat{\omega}(y^t, \hat{y}^{t+s})\mu(dy^t)\right] U(\hat{y}^{t+s}) \\
& \times \left[\int_{\tilde{y}^t\in\mathcal{Y}^\infty} \mu_s(d\hat{y}^{t+s}|\tilde{y}^t)\mu(d\tilde{y}^t)\right].
\end{aligned}
$$

A straightforward extension of Lemma 4 for μ_s (for any $s \geq 1$) yields: $\int_{\tilde{y}^t\in\mathcal{Y}^\infty} \mu_s(d\hat{y}^{t+s}|\tilde{y}^t)\mu(d\tilde{y}^t) = \mu(d\hat{y}^{t+s})$. Using the definition (15) of the weights ω, we deduce:

$$
SWF = \sum_{s=0}^{\infty} \int_{\hat{y}^{t+s}\in\mathcal{Y}^\infty} \beta^s \omega(\hat{y}^{t+s}) U(\hat{y}^{t+s}) \mu(d\hat{y}^{t+s}),
$$

which proves Proposition 1.

As a final remark, observe that by splitting the sum over s into $s = 0$ and a sum for $s \geq 1$, we also obtain the following expression for SWF: $SWF = \int_{y^t\in\mathcal{Y}^\infty} \omega(y^t)U(y^t, A)\mu(dy^t) + \beta\, SWF$.

A.5 Proof of Proposition 2

We recall the expression of the SWF when the allocation is explicit:

$$
SWF(A) = \sum_{t=0}^{\infty} \beta^t \int_{y^t\in\mathcal{Y}^\infty} \omega(y^t)U(y^t, A)\mu(dy^t).
$$

Assume that the weights are non-negative and consider two allocations A and A' such that A element-wise dominates A'. We thus have $U(y^t, A) \geq U(y^t, A')$ for all y^t. The non-negativity of weights implies that: $\beta^t\omega(y^t)U(y^t, A) \geq \beta^t\omega(y^t)U(y^t, A')$ for all y^t, which after integration and sum yields $SWF(A) \geq SWF(A')$.

Let us assume that $SWF(A) \geq SWF(A')$ for any pair of allocations A and A' such that A element-wise dominates A'. Let us assume that the weights are strictly negative on a subset $\mathcal{X} \subset \mathcal{Y}^\infty$ of positive measure. We consider an allocation A'. We construct the allocation A such that A and A' coincide on $\mathcal{Y}^\infty \setminus \mathcal{X}$ and A strictly dominates A' on $\mathcal{X}$. We thus have $U(y^t, A) = U(y^t, A')$ for all $y^t \in \mathcal{Y}^\infty \setminus \mathcal{X}$ and $U(y^t, A) > U(y^t, A')$ for all $y^t \in \mathcal{X}$: A element-wise dominates A'. We thus deduce that:

$$\begin{aligned}\int_{y^t\in\mathcal{Y}^\infty}\omega(y^t)(U(y^t,A)-U(y^t,A'))\mu(dy^t) &= \int_{y^t\in\mathcal{Y}^\infty\setminus\mathcal{X}}\omega(y^t)(U(y^t,A)-U(y^t,A'))\mu(y^t)\\ &+\int_{y^t\in\mathcal{X}}\omega(y^t)(U(y^t,A)-U(y^t,A'))\mu(y^t),\\ &=\int_{y^t\in\mathcal{X}}\omega(y^t)(U(y^t,A)-U(y^t,A'))\mu(y^t),\\ &<0,\end{aligned}$$

where the first equality is a split of the integral over two disjoint sets, the second comes from $U(y^t,A) = U(y^t,A')$ on $\mathcal{Y}^\infty \setminus \mathcal{X}$, and the third from $U(y^t,A) > U(y^t,A')$ and $\omega(y^t) < 0$ on $\mathcal{X}$.

Summing the previous discounted inequality implies $SWF(A) < SWF(A')$, which is a contradiction. We must thus have positive weights. This concludes the proof.

A.6 Proof of equation (46) in Definition 3

The program in Definition 3 is:

$$\begin{aligned}(\tilde{\omega}_{\tilde{y}y})_{\tilde{y},y} =&\text{argmin}_{(\hat{\omega}_{\tilde{y}y})_{\tilde{y},y}}\sum_{(y,\tilde{y})\in\mathcal{Y}^{\infty 2}}\pi_{\tilde{y}}\left(\hat{\omega}_{\tilde{y}y}-\omega^{prior}_{\tilde{y},y}\right)^2,\\ &\text{s.t. } \omega_y = \sum_{\tilde{y}\in\mathcal{Y}^\infty}\pi_{\tilde{y}}\omega_{P,\tilde{y}}\hat{\omega}_{\tilde{y}y}\ (y\in\mathcal{Y}).\end{aligned}\tag{60}$$

We denote by $2\lambda_y$ the Lagrange multiplier to the constraint of equation (60) for $y \in \mathcal{Y}$. We obtain the following Lagrangian:

$$\mathcal{L} = \frac{1}{2}\sum_{(y,\tilde{y})\in\mathcal{Y}^{\infty 2}}\pi_{\tilde{y}}\left(\hat{\omega}_{\tilde{y}y}-\omega^{prior}_{\tilde{y},y}\right)^2 - \sum_{y\in\mathcal{Y}}\lambda_y\left(\sum_{\tilde{y}\in\mathcal{Y}^\infty}\pi_{\tilde{y}}\omega_{P,\tilde{y}}\hat{\omega}_{\tilde{y}y}-\omega_y\right).$$

Computing the derivative with respect to $\hat{\omega}_{\tilde{y}y}$ yields the following FOC:

$$\hat{\omega}_{\tilde{y}y} = \omega^{prior}_{\tilde{y},y} + \lambda_y\omega_{P,\tilde{y}}.$$

Using the constraint of equation (60) and $\sqrt{\sum_{\tilde{y}\in\mathcal{Y}^\infty}\pi_{\tilde{y}}(\omega_{P,\tilde{y}})^2} = 1$ we deduce:

$$\lambda_y = \omega_y - \sum_{\tilde{y}\in\mathcal{Y}^\infty}\pi_{\tilde{y}}\omega_{P,\tilde{y}}\omega^{prior}_{\tilde{y}y},$$

This finally implies equation (46), with

$$F(y,\tilde{y}) := \omega^{prior}_{\tilde{y},y} - \omega_{P,\tilde{y}}\sum_{\hat{y}\in\mathcal{Y}^\infty}\pi_{\hat{y}}\omega_{P,\hat{y}}\omega^{prior}_{\hat{y}y}$$

B Competitive equilibrium

We provide a formal definition of a competitive equilibrium.

Definition 4 (Competitive equilibrium) *A sequential competitive equilibrium is a collection of individual allocations* $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t\geq 0, i\in\mathcal{I}}$*, of aggregate quantities* $(K_t, L_t, Y_t)_{t\geq 0}$*, of price processes* $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_{t\geq 0}$*, and of fiscal policies* $(\tau_t^c, \tau_t^K, \tau_t, \kappa_t, B_t)_{t\geq 0}$*, such that, for initial conditions and initial values of capital stock and public debt satisfying* $K_{-1} + B_{-1} = \int_i a_{i,-1}\ell(di)$*, we have:*

1. *given prices, the functions* $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t\geq 0, i\in\mathcal{I}}$ *solve the agents' optimization program in equations (24)–(26);*

2. *financial, labor, and goods markets clear at all dates: for any* $t \geq 0$*, equation (29) holds;*

3. *the government's budget is balanced at all dates: equation (22) holds for all* $t \geq 0$*;*

4. *factor prices* $(w_t, r_t, \tilde{w}_t, \tilde{r}_t)_{t\geq 0}$ *are consistent with the condition (18) and post-tax definitions (21).*

A steady-state competitive equilibrium is a competitive equilibrium in which the joint distribution of agents' decisions (c, l, a, ν), aggregate quantities K, L, Y, prices $w, r, \tilde{w}, \tilde{r}$, and fiscal policy $(\tau^c, \tau^K, \tau, \kappa, B)$ are time-invariant.

C The Ramsey program 4

C.1 Reformulating the Ramsey program

We now reformulate the Ramsey problem. We define the following variables:

$$\tilde{a}_{i,t} := \frac{a_{i,t}}{1+\tau_t^c}, \tag{61}$$

$$W_t := \frac{w_t}{1+\tau_t^c}, \tag{62}$$

$$R_t := \frac{(1+r_t)(1+\tau_{t-1}^c)}{1+\tau_t^c}, \tag{63}$$

which represent the asset choices in (61), the wage rate in (62), and the interest rate in (63). With this notation, the agent's budget and credit constraints become:

$$c_{i,t} + \tilde{a}_{i,t} = W_t (y_{i,t} l_{i,t})^{1-\tau_t} + R_t \tilde{a}_{i,t-1}, \tag{64}$$

$$\tilde{a}_{i,t} \geq -\frac{\overline{a}}{1+\tau_t^c} := -\tilde{\overline{a}}. \tag{65}$$

Since taxes and prices are considered as given by agents, we can equivalently state their optimization program using the notation (61)–(63) and the constraints (64) and (65), rather than the original notation and the constraints (25) and (26). This modifies the Euler equations (27)–(28) as follows:

$$u'(c_{i,t}) = \beta \mathbb{E}_t \Big[R_{t+1} u'(c_{i,t+1}) \Big] + \nu_{i,t},$$
$$v'(l_{i,t}) = (1-\tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}).$$

We now turn to the government budget constraint. We further define:

$$\tilde{B}_t := \frac{B_t}{(1+\tau_t^c)}, \tag{66}$$
$$\tilde{A}_t := \frac{A_t}{1+\tau_t^c}, \tag{67}$$

and

$$\hat{B}_t := (1+\tau_t^c)\tilde{B}_t - \tau_t^c \tilde{A}_t. \tag{68}$$

With these new definitions, the financial market equilibrium given by (37) holds, since we have $\tilde{A}_t = \int_i \tilde{a}(i) l(di)$.

Using the government budget constraint defined in (22), we have:

$$G_t + (1+r_t)B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} = \tau_t^c C_t + F(K_{t-1}, L_t, s_t) + B_t.$$

Using the resource constraint $C_t + G_t + K_t = F(K_{t-1}, L_t, s_t) + K_{t-1}$, we obtain:

$$G_t + (1+r_t)B_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + r_t K_{t-1} =$$
$$\tau_t^c (F(K_{t-1}, L_t, s_t) - G_t - (K_t - K_{t-1})) + F(K_{t-1}, L_t, s_t) + B_t.$$

Divide both sides of the equation above by $(1+\tau_t^c)$ and obtain:

$$G_t + \frac{1+r_t}{1+\tau_t^c} B_{t-1} + \frac{w_t}{1+\tau_t^c} \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t}{1+\tau_t^c} K_{t-1} =$$
$$- \frac{\tau_t^c}{1+\tau_t^c}(K_t - K_{t-1}) + F(K_{t-1}, L_t, s_t) + \frac{B_t}{1+\tau_t^c}.$$

Using the definitions (62), (63), and (66):

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t}{1+\tau_t^c} K_{t-1} = -\frac{\tau_t^c}{1+\tau_t^c}(K_t - K_{t-1}) + F(K_{t-1}, L_t, s_t) + \tilde{B}_t.$$

We now substitute the expressions of K_t and K_{t-1}. From (66) and (67), we have $K_{t-1} =$

$A_{t-1} - B_{t-1} = (1 + \tau^c_{t-1})(\tilde{A}_{t-1} - \tilde{B}_{t-1})$ and:

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \frac{r_t(1+\tau^c_{t-1})}{1+\tau^c_t}(\tilde{A}_{t-1} - \tilde{B}_{t-1}) =$$
$$\frac{\tau^c_t(1+\tau^c_{t-1})}{1+\tau^c_t}(\tilde{A}_{t-1} - \tilde{B}_{t-1}) + F(K_{t-1}, L_t, s_t) - \tau^c_t(\tilde{A}_t - \tilde{B}_t) + \tilde{B}_t.$$

Observe from (63) and (68) that $\frac{r_t(1+\tau^c_{t-1})}{1+\tau^c_t} = R_t - \frac{1+\tau^c_{t-1}}{1+\tau^c_t}$ and $-\tau^c_t(\tilde{A}_t - \tilde{B}_t) + \tilde{B}_t = \hat{B}_t$. This yields:

$$G_t + R_t \tilde{B}_{t-1} + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - (1 + \tau^c_{t-1}))(\tilde{A}_{t-1} - \tilde{B}_{t-1}) =$$
$$F(K_{t-1}, L_t, s_t) + \hat{B}_t.$$

Finally, using (68) in period $t-1$ (i.e., $\hat{B}_{t-1} = (1+\tau^c_{t-1})\tilde{B}_{t-1} - \tau^c_{t-1}\tilde{A}_{t-1}$) we get:

$$G_t + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - 1)\tilde{A}_{t-1} + \hat{B}_{t-1} =$$
$$F(K_{t-1}, L_t, s_t) + \hat{B}_t.$$

Since the public debt can be freely chosen by the planner, it is equivalent for the planner to choose $\hat{B}_t$ rather than B_t.

The reformulated Ramsey program. We reformulate the Ramsey program (69)–(75) using the variables $\tilde{a}_{i,t}$, W_t, R_t, $\tilde{A}_t$, $\hat{B}_t$ introduced in (61)–(63) and (67)–(68). The program can be expressed in post-tax prices R_t and W_t – taxes and pre-tax factor prices can be deduced from the allocation and the post-tax price definitions. The following proposition summarizes the reformulation of the Ramsey program.

Proposition 4 *The Ramsey program (69)–(75) can be rewritten as:*

$$\max_{\left(W_t, R_t, \tau_t, \hat{B}_t, \tilde{A}_t, K_t, L_t, (c_{i,t}, l_{i,t}, \tilde{a}_{i,t}, \nu_{i,t})_i\right)_{t\geq 0}} SWF_0, \tag{69}$$

$$G + W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (R_t - 1)\tilde{A}_{t-1} + \hat{B}_{t-1} = F(K_{t-1}, L_t, s_t) + \hat{B}_t, \tag{70}$$

for all $i \in \mathcal{I}$:
$$c_{i,t} + \tilde{a}_{i,t} = W_t (y_{i,t} l_{i,t})^{1-\tau_t} + R_t \tilde{a}_{i,t-1}, \tag{71}$$

$$\tilde{a}_{i,t} \geq -\bar{\tilde{a}}, \ \nu_{i,t}(\tilde{a}_{i,t} + \bar{\tilde{a}}) = 0, \ \nu_{i,t} \geq 0, \tag{72}$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[R_{t+1} u'(c_{i,t+1}) \right] + \nu_{i,t}, \tag{73}$$

$$v'(l_{i,t}) = (1 - \tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \tag{74}$$

$$K_t + \hat{B}_t = \tilde{A}_t = \int_i \tilde{a}_{i,t} \ell(di), \ L_t = \int_i y_{i,t} l_{i,t} \ell(di). \tag{75}$$

C.2 The Lagrangian, the FOCs of the Ramsey program and equilibrium definitions

The Lagrangian associated with the Ramsey program (69)–(75) can be written as:

$$\begin{aligned}
\mathcal{L} = &\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t}(u(c_{i,t}) - v(l_{i,t}))\ell(di) \\
&- \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1})\, u'(c_{i,t})\ell(di) \\
&- \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \lambda_{l,i,t} \left(v'(l_{i,t}) - (1-\tau_t)W_t y_{i,t}(y_{i,t}l_{i,t})^{-\tau_t} u'(c_{i,t})\right) \ell(di) \\
&- \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \left(G_t + (1-\delta)\hat{B}_{t-1} + (R_t - 1 + \delta) \int_i \tilde{a}_{i,t-1}\ell(di) + W_t \int_i (y_{i,t}l_{i,t})^{1-\tau_t}\ell(di) - Y_t - \hat{B}_t \right),
\end{aligned}$$

where the value of ν_{it} is given by the complementary slackness conditions (72) and (73), and where we have:

$$c_{i,t} = -\tilde{a}_{i,t} + R_t \tilde{a}_{i,t-1} + W_t (y_{i,t}l_{i,t})^{1-\tau_t}, \tag{76}$$

$$Y_t = \Big(\int_i \tilde{a}_{i,t-1}\ell(di) - \hat{B}_{t-1}\Big)^{\alpha} \Big(\int_i y_{i,t}l_{i,t}\ell\,(di)\, \Big)^{1-\alpha}. \tag{77}$$

Consequently, the instruments are: $\tilde{a}_{i,t}$, $l_{i,t}$, W_t, R_t, τ_t, and $\hat{B}_t$. Using the two previous equations to substitute $c_{i,t}$ and Y_t, the planner's program is:

$$\max_{\left(W_t, R_t, \tau_t, \hat{B}_t, (l_{i,t}, \tilde{a}_{i,t})_i\right)_{t\geq 0}} \mathcal{L}.$$

We now provide the first-order conditions of the planner, and we present an alternative interpretation of the Lagrangian in the next section.

FOC with respect to public debt $\hat{B}_t$.

$$\mu_t = \beta \mathbb{E}_t \left[(1 + \tilde{r}_{t+1})\mu_{t+1} \right]. \tag{78}$$

FOC with respect to saving choices $\tilde{a}_{i,t}$. We define the marginal social value of liquidity for agent i at date t as:

$$\psi_{i,t} := \omega_{i,t}u'(c_{i,t}) - \left(\lambda_{c,i,t} - R_t \lambda_{c,i,t-1} - \lambda_{l,i,t}(1-\tau_t)W_t (y_{i,t})^{1-\tau_t}(l_{i,t})^{-\tau_t} \right) u''(c_{i,t}), \tag{79}$$

and $\hat{\psi}_{i,t} := \psi_{i,t} - \mu_t$ as the marginal social value of liquidity net of the planner's resource cost. Using (78), we obtain:

$$\hat{\psi}_{i,t} = \beta \mathbb{E}_t \left[R_{t+1} \hat{\psi}_{i,t+1} \right]. \tag{80}$$

FOC with respect to labor supply $l_{i,t}$. We define:

$$\psi_{l,i,t} := \omega_{i,t} v'(l_{i,t}) + \lambda_{l,i,t} v''(l_{i,t}).$$

The FOC with respect to labor supply $l_{i,t}$ is:

$$\psi_{l,i,t} = (1-\tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \hat{\psi}_{i,t} + \mu_t F_{L,t} y_{i,t} - (1-\tau_t) W_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \lambda_{l,i,t} \tau_t \frac{u'(c_{i,t})}{l_{i,t}}.$$

FOC with respect to the wage rate W_t.

$$0 = \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{l,j,t} (1-\tau_t) u'(c_{j,t})/l_{j,t} \right) \ell(dj).$$

FOC with respect to the interest rate R_t.

$$0 = \int_j (\hat{\psi}_{j,t} \tilde{a}^j_{t-1} + \lambda_{c,j,t-1} u'(c_{j,t})) \ell(dj). \tag{81}$$

FOC with respect to progressivity τ_t.

$$\begin{aligned} 0 = & \int_j (y_{j,t} l_{j,t})^{1-\tau_t} (\hat{\psi}_{j,t} + \lambda_{l,j,t} (1-\tau_t) (u'(c_{j,t})/l_{j,t})) \ln(y_{j,t} l_{j,t}) \ell(dj) \\ & + \int_j \lambda_{l,j,t} (y_{j,t} l_{j,t})^{1-\tau_t} (u'(c_{j,t})/l_{j,t}) \ell(dj). \end{aligned} \tag{82}$$

A *Ramsey allocation* is a competitive equilibrium in which the fiscal policy and allocations satisfy the first-order conditions of the planner (78)-(82).

A *Ramsey steady state* is a Ramsey allocation in which fiscal policy $(\tau^c, \tau^K, \tau, \kappa, B)$, aggregate variables (K, A, B) and prices r, w are constant.

C.3 Expression of the Lagrangian using a public finance representation

The Lagrangian can be written as:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\mathcal{W}_t + \mu_t \mathcal{B}_t \right),$$

with:

$$\begin{aligned} \mathcal{W}_t := & \int_i (\omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) - (\lambda_{c,i,t} - R_t \lambda_{c,i,t-1}) u'(c_{i,t}) \\ & - \lambda_{l,i,t} \left(v'(l_{i,t}) - (1-\tau_t) W_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}) \right) \Big) \ell(di), \\ \mathcal{B}_t := & Y_t - \hat{B}_t - G_t - (1-\delta) \hat{B}_{t-1} - (R_t - 1 + \delta) \int_i \tilde{a}_{i,t-1} \ell(di) - W_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di). \end{aligned}$$

The quantity $\mathcal{B}_t$ is the budget constraint of the government, whereas $\mathcal{W}_t$ is the aggregate welfare taking into account the possible general-equilibrium effects generated by each agent's choice and captured by individual Lagrange multipliers $\lambda_{c,i,t}$ and $\lambda_{l,i,t}$. Note that if these multipliers were zero, then $\mathcal{W}_t$ would only be the weighted welfare.

Considering an instrument I_k in period k (I_k can be public debt, the interest rate, the labor tax, or labor-tax progressivity), the FOC of the Lagrangian with respect to I_k implies:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \frac{d\mathcal{B}_t}{dI_k} + \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int \frac{\partial \mathcal{W}_t}{\partial I_k} \ell(di) = -\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int \frac{\partial \mathcal{W}_t}{\partial c_{i,t}} \frac{\partial c_{i,t}}{\partial I_k} \ell(di). \tag{83}$$

Since W_t, R_t, τ_t only affect the current value of welfare and the budget constraint (for $I_t \in \{W_t, R_t, \tau_t\}$, $\frac{\partial \mathcal{W}_t}{\partial I_k} = \frac{\partial \mathcal{B}_t}{\partial I_k} = 0$ if $k \neq t$), we have for any $I_t \in \{W_t, R_t, \tau_t\}$:

$$\mu_t = \int \frac{\partial \mathcal{W}_t}{\partial c_{i,t}} \frac{-\frac{\partial c_{i,t}}{\partial I_t}}{\frac{d\mathcal{B}_t}{dI_t} + \frac{1}{\mu_t}\frac{\partial \mathcal{W}_t}{\partial I_t}} \ell(di). \tag{84}$$

It is then straightforward to compute the partial derivatives and verify that we obtain the same expressions as in Section C.2. For instance, we have $\frac{\partial \mathcal{W}_t}{\partial c_{i,t}} = \psi_{i,t}$ and when the fiscal instrument is the post-tax rate R_t: $\frac{d\mathcal{B}_t}{dR_t} = -\int_i \tilde{a}_{i,t-1}\ell(di)$, $\frac{\partial \mathcal{W}_t}{\partial R_t} = \int_i R_t \lambda_{c,i,t-1} u'(c_{i,t})$, and $\frac{\partial c_{i,t}}{\partial R_t} = \tilde{a}_{i,t-1}$. Then, (84) becomes:

$$\mu_t = \int \psi_{i,t} \frac{-\tilde{a}_{i,t-1}}{-\int_i \tilde{a}_{i,t-1}\ell(di) + \frac{1}{\mu_t}\int_i R_t \lambda_{c,i,t-1} u'(c_{i,t})} \ell(di),$$

which can be written as $\int (\psi_{i,t} - \mu_t)\tilde{a}_{i,t-1}\ell(di) + \int_i R_t \lambda_{c,i,t-1} u'(c_{i,t}) = 0$, which is the FOC (81) with $\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t$. The same derivations can be obtained for R_t, τ_t.

The instrument $\hat{B}_t$ is a fiscal instrument that affects current and future budget constraints but not welfare: $\frac{\partial \mathcal{W}_t}{\partial \hat{B}_k} = 0$, for all k and $\frac{\partial \mathcal{B}_t}{\partial \hat{B}_t} = -1$, $\frac{\partial \mathcal{B}_{t+1}}{\partial \hat{B}_t} = 1 + \tilde{r}_{t+1}$, and $\frac{\partial \mathcal{B}_\tau}{\partial \hat{B}_t} = 0$ if $\tau \notin \{t, t+1\}$. As a consequence, the FOC (83) simplifies into $\mathbb{E}_0[\beta^t \mu_t \frac{d\mathcal{B}_t}{d\hat{B}_t} + \beta^{t+1}\mu_{t+1}\frac{d\mathcal{B}_{t+1}}{d\hat{B}_t}] = 0$, or $-\mu_t + \beta \mathbb{E}_t[\mu_{t+1}(1 + \tilde{r}_{t+1})] = 0$, which is the FOC (78).

D Truncating the model and identification of Pareto Weights

D.1 The truncated model

The key step of the aggregation consists of assigning the same wealth and allocation to all agents sharing the same idiosyncratic history over the recent past. The recent past is characterized by a number of periods, called the *truncation length* and denoted N, it is a parameter of the model. This N-period history will be referred to as a *truncated history*. For a history $y^t = \{\ldots, y^t_{t-N}, y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t\}$, this corresponds to the N-length vector denoted $y^N := \{y^t_{t-N}, y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t\}$. To sum up, we can represent the truncated history of an agent i whose idiosyncratic history is y^t as:

$$y^t = \{\underbrace{\ldots, y^t_{t-N-2}, y^t_{t-N-1}, y^t_{t-N}}_{\xi_{y^N}}, \underbrace{y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t}_{=y^N}\},$$

where the parameter ξ_{y^N} captures the residual heterogeneity for the truncated history y^N, and y^t_{t-k} represents the idiosyncratic variable (at date t) k periods in the past. The method to compute the set of parameters $(\xi_{y^N})_{y^N}$ will be discussed further below. In what follows, we will discuss the various elements needed to apply the aggregation procedure.

First, we need to compute the measure of agents with the same history y^N. An agent with history $\tilde{y}^N$ at $t-1$ will have a different truncated history in period t depending on the realization of the idiosyncratic variable at date t. The probability to transit from truncated history $\tilde{y}^N$ to truncated history y^N will be denoted by $\Pi_{\tilde{y}^N y^N}$ (with $\sum_{y^N \mathcal{Y}^{\infty N}} \Pi_{\tilde{y}^N y^N} = 1$) and can be computed from the transition probabilities for the productivity process as:

$$\Pi_{\tilde{y}^N y^N} = 1_{y^N \succeq \tilde{y}^N} \Pi_{\tilde{y}^N_0 y^N_0} \geq 0,$$

where $1_{y^N \succeq \tilde{y}^N}$ is equal to 1 if y^N is a possible continuation of $\tilde{y}^N$, and 0 otherwise. With these elements, we can compute the share of agents with truncated history y^N as S_{t,y^N}. This element is:

$$S_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} S_{t-1,\tilde{y}^N} \Pi_{\tilde{y}^N y^N}, \tag{85}$$

where the initial shares $(S_{-1,y^N})_{y^N \in \mathcal{Y}^N}$ are given with $\sum_{y^N \in \mathcal{Y}^N} S_{-1,y^N} = 1$.

The model aggregation then assigns to each truncated history the average choices (whether for consumption, saving, or labor supply) of the group of agents sharing the same truncated history. Let us consider a generic variable denoted by $X_t(y^t, s^t)$, and denote by X_{t,y^N} the average quantity of X assigned to truncated history y^N. Formally:

$$X_{t,y^N} = \frac{1}{S_{t,y^N}} \sum_{y^t \in \mathcal{Y}^{t+1} \big| (y^t_{t-N+1}, \ldots, y^t_{t-1}, y^t_t) = y^N} X_t(y^t, s^t) \mu_t(y^t), \tag{86}$$

where we recall that $\mu_t(y^t)$ is the measure of agents with history y^t. Definition (86) can be applied to consumption, savings, labor supply, and the credit-constraint Lagrange multiplier. This leads to the quantities c_{t,y^N}, $\tilde{a}_{t,y^N}$, l_{t,y^N}, and ν_{t,y^N} respectively. Note that applying (86) to beginning-of-period wealth involves accounting for the fact that agents with truncated history y^N at date t may come from various truncated histories at $t-1$. Specifically, this variable consists of the wealth of all agents with history y^N in period t but with any other possible history at $t-1$. Formally, the beginning-of-period wealth

$\tilde{\tilde{a}}_{t,y^N}$ for truncated history y^N is:

$$\tilde{\tilde{a}}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \tag{87}$$

We now define the various "ξs". First, we define $\xi^{u,0}_{y^N}$ as:

$$\sum_{y^t \in \mathcal{Y}^{t+1} \big| (y^t_{t-N+1},\ldots,y^t_{t-1},y^t_t)=y^N} u(c_t(y^t)) = \xi^{u,0}_{y^N} u(\sum_{y^t \in \mathcal{Y}^{t+1} \big| (y^t_{t-N+1},\ldots,y^t_{t-1},y^t_t)=y^N} c_t(y^t)),$$

or compactly as:

$$\sum_{y^t_i \in \mathcal{Y}^{t+1} | y^{t,N}_i = y^N} u(c_{i,t}) = \xi^{u,0}_{y^N} u(c_{t,y^N}). \tag{88}$$

The quantity $\xi^{u,0}_{y^N}$ reflects that aggregating utility levels is not equal to the utility of aggregated consumption. This comes from a combination of two factors. First, there is heterogeneity of consumption among the population of agents having the truncated history y^N, due to their history prior to date $t-N$. Second, the utility function is not affine in general.

The same procedure applied to the other variables for the Ramsey problem (69)–(75) yields:

$$\sum_{y^t_i \in \mathcal{Y}^{t+1} | y^{t,N}_i = y^N} v(l_{i,t}) := \xi^{v,0}_{y^N} v(l_{t,y^N}), \tag{89}$$

$$\sum_{y^t_i \in \mathcal{Y}^{t+1} | y^{t,N}_i = y^N} u'(c_{i,t}) := \xi^{u,1}_{y^N} u'(c_{t,y^N}), \tag{90}$$

$$\sum_{y^t_i \in \mathcal{Y}^{t+1} | y^{t,N}_i = y^N} (y_{i,t} l_{i,t})^{1-\tau_t} := \xi^{y}_{y^N} (y^N_0 l_{t,y^N})^{1-\tau_t}. \tag{91}$$

We can now proceed with the aggregation of the full-fledged model. First, the aggregation of individual budget constraints (64) yields:

$$c_{t,y^N} + \tilde{a}_{t,y^N} = W_t \xi^y_{y^N} (l_{t,y^N} y^N_0)^{1-\tau_t} + R_t \tilde{\tilde{a}}_{t,y^N}, \text{ for } y^N \in \mathcal{Y}^{\infty N}. \tag{92}$$

The aggregation of Euler equations for consumption (73) and labor (74) yields:

$$\xi^{u,E}_{y^N} u'(c_{t,y^N}) = \beta \mathbb{E}_t \left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \Pi_{t+1,y^N \tilde{y}^N} \xi^{u,E}_{\tilde{y}^N} u'(c_{t+1,\tilde{y}^N}) \right] + \nu_{t,y^N}, \tag{93}$$

$$\xi^{v,1}_{y^N} v'(l_{t,y^N}) := (1-\tau_t) W_t \xi^y_{y^N} (l_{t,y^N} y^N_0)^{1-\tau_t} \xi^{u,1}_{y^N} (u'(c_{t,y^N})/l_{t,y^N}), \tag{94}$$

where the coefficients $(\xi^{u,E}_{y^N})_{y^N}$ for the consumption Euler equations ensure that the aggregate Euler equations yield Euler equations with aggregate consumption levels. In other words, the $(\xi^{u,E}_{y^N})_{y^N}$ are determined such that the aggregated consumption levels (for truncated histories) satisfy the consumption Euler equation (93). These coefficients are necessary because Euler equations involve nonlinear marginal utilities. The same idea applies to the coefficients $(\xi^{v,1}_{y^N})_{y^N}$ for the FOC on labor.

Finally, the market clearing conditions can be expressed as:

$$K_t + \hat{B}_t = \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} \tilde{a}_{t,y^N}, \quad L_t = \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} y_{y^N} l_{t,y^N}. \tag{95}$$

Equations (92)–(95) exactly characterize the dynamics of the aggregated variables c_{t,y^N}, $\tilde{a}_{t,y^N}$, l_{t,y^N}, and ν_{t,y^N}, as well as aggregate quantities K_t, $\hat{B}_t$, and L_t.

Steady state and computation of the ξs. Steady-state allocations allow us to compute the parameters ξs as follows. We compute the policy functions and wealth distribution of the Bewley model, and identify the set of credit-constrained histories, denoted $\mathcal{C}$. Aggregation equations (86) and (87) can then be used to aggregate (steady-state) allocations c_{y^N}, $\tilde{a}_{y^N}$, l_{y^N}, and ν_{y^N}. We then invert the consumption Euler equations (93) to deduce the preference parameters $(\xi^{u,E}_{y^N})_{y^N}$. The other ξs are computed explicitly by equations (88), (89), (90), (91), and (94).

The truncated model in the presence of aggregate shocks. We state two further assumptions that enable us to use the truncation method in the presence of aggregate shocks, which results in the so-called truncated model.

Assumption B *We make the following two assumptions:*

1. *The preference parameters $(\xi_{y^N})_{y^N}$ remain constant and equal to their steady-state values.*

2. *The set of credit-constrained histories, denoted by $\mathcal{C} \subset \mathcal{Y}^{\infty N}$, is time-invariant.*

Finally, two properties are worth mentioning. First, a straightforward consequence of the construction of the ξs is that the steady-state allocations of the initial and truncated models are identical. Second, as the truncation length N becomes increasingly longer, truncated allocations (in the presence of aggregate shocks) can be shown to converge to those of the full-fledged equilibrium. Section 5 shows that from a quantitative standpoint, the ξs efficiently capture the heterogeneity within truncated histories, even when the truncation length remains limited.

D.2 Ramsey program

Program formulation. The finite state-space representation of the truncated model allows us to solve for the Ramsey program in the presence of aggregate shocks.[31] Let

[31] Our method involves deriving the FOCs of the truncated model, rather than truncating the FOCs of the full-fledged Ramsey model. This ensures numerical stability, as the truncated model is "well-defined" for the fiscal policy under consideration by construction.

$(\omega_y)_{y\in\mathcal{Y}}$ denote the period weights associated with each productivity level. The Ramsey program in the truncated economy can be written as follows:

$$\max_{\left(W_t,R_t,\tilde{w}_t,\tilde{r}_t,\tau_t^c,\tau_t^K,\tau_t,\kappa_t,\hat{B}_t,G_t,K_t,L_t,(c_{t,y^N},l_{t,y^N},\tilde{a}_{t,y^N},\nu_{t,y^N})_{y^N}\right)_{t\geq 0}} W_0, \tag{96}$$

where $W_0 := \mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t\sum_{y^N\in\mathcal{Y}^N}S_{t,y^N}\omega_{y^N}(\xi_{y^N}^{u,0}u(c_{t,y^N})-\xi_{y^N}^{v,0}v(l_{t,y^N})+u_G(G_t))\right]$ and subject to aggregate Euler equations (93) and (94), aggregate budget constraint (92), aggregate market clearing conditions (95), credit constraints $\tilde{a}_{t,y^N}\geq-\tilde{\bar{a}}$, as well as the governmental budget constraint (70), which is already present in the full-fledged Ramsey program.

First-order conditions. We define the net social value of liquidity of history y^N as in (79):

$$\begin{aligned}\hat{\psi}_{t,y^N} &= \omega_{y^N}\xi_{y^N}^{u,0}u'(c_{t,y^N})-\mu_t\\ &\quad-\left(\lambda_{c,t,y^N}\xi_{y^N}^{u,E}-R_t\tilde{\lambda}_{c,t,y^N}\xi_{y^N}^{u,E}-\lambda_{l,t,y^N}\xi_{y^N}^{y}(1-\tau_t)W_t(y_0^N)^{1-\tau_t}l_{t,y^N}^{-\tau_t}\xi_{y^N}^{u,1}\right)u''(c_{t,y^N}).\end{aligned} \tag{97}$$

FOC with respect to $\tilde{a}_{t,y^N}$:

$$\hat{\psi}_{t,y^N}=\beta\mathbb{E}_t\left[R_{t+1}\sum_{\tilde{y}^N\in\mathcal{Y}^{\infty N}}\Pi_{t,y^N\tilde{y}^N}\hat{\psi}_{t+1,\tilde{y}^N}\right]\text{ if }\nu_{y^N}=0\text{ and }\lambda_{c,t,y^N}=0\text{ otherwise.} \tag{98}$$

FOC with respect to l_{t,y^N}:

$$\begin{aligned}\frac{\omega_{y^N}\xi_{y^N}^{v,0}v'(l_{t,y^N})+\lambda_{l,t,y^N}\xi_{y^N}^{v,1}v''(l_{t,y^N})}{(1-\tau_t)W_t\xi_{y^N}^{y}(y_0^N)^{1-\tau_t}l_{t,y^N}^{-\tau_t}} &= \hat{\psi}_{t,y^N}-\lambda_{l,t,y^N}\tau_t\xi_{y^N}^{u,1}(u'(c_{t,y^N})/l_{t,y^N})\\ &\quad+\mu_t(1-\alpha)\frac{Y_t}{(1-\tau_t)W_t\xi_{y^N}^{y}(y_0^N)^{-\tau_t}l_{t,y^N}^{-\tau_t}L_t}.\end{aligned} \tag{99}$$

FOC with respect to W_t:

$$\sum_{y^N\in\mathcal{Y}^{\infty N}}S_{t,y^N}\xi_{y^N}^{y}(l_{t,y^N}y_{y^N})^{1-\tau_t}\left(\hat{\psi}_{t,y^N}+\lambda_{l,t,y^N}(1-\tau_t)\xi_{y^N}^{u,1}(u'(c_{t,y^N})/l_{t,y^N})\right)=0. \tag{100}$$

FOC with respect to R_t:

$$\sum_{y^N\in\mathcal{Y}^{\infty N}}S_{t,y^N}\left(\hat{\psi}_{t,y^N}\tilde{\bar{a}}_{t,y^N}+\tilde{\lambda}_{c,t,y^N}\xi_{y^N}^{u,E}u'(c_{t,y^N})\right)=0. \tag{101}$$

FOC with respect to τ_t:

$$\begin{aligned}&\sum_{y^N\in\mathcal{Y}^{\infty N}}S_{t,y^N}\left(\hat{\psi}_{t,y^N}+\lambda_{l,t,y^N}(1-\tau_t)\xi_{y^N}^{u,1}(u'(c_{t,y^N})/l_{t,y^N})\right)\ln\left(l_{t,y^N}y_{y^N}\right)\xi_{y^N}^{y}(l_{t,y^N}y_{y^N})^{1-\tau_t}\\ &=-\sum_{y^N\in\mathcal{Y}^{\infty N}}S_{t,y^N}\lambda_{l,t,y^N}\xi_{y^N}^{y}(l_{t,y^N}y_{y^N})^{1-\tau_t}\xi_{y^N}^{u,1}(u'(c_{t,y^N})/l_{t,y^N}).\end{aligned} \tag{102}$$

FOC with respect to $\hat{B}_t$:

$$\mu_t = \beta \mathbb{E}\left[\mu_{t+1}\left(1+\alpha\frac{Y_{t+1}}{K_t}-\delta\right)\right]. \tag{103}$$

We must furthermore have:

$$\tilde{\lambda}_{c,t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \lambda_{c,t-1,\tilde{y}^N}, \tag{104}$$

$$\tilde{a}_{t,y^N} \geq 0 \text{ and } \tilde{\tilde{a}}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \tag{105}$$

D.3 Matrix expression

In this section, we provide closed-form formulas for preference multipliers ξs (Section D.1) and the weights ωs. We start with some notation:

$\circ$ is the Hadamard product, $\otimes$ is the Kronecker product, $\times$ is the usual matrix product.

For any vector V, we denote by $\text{diag}(V)$ the diagonal matrix with V on the diagonal.

The matrix representation consists in stacking together the equations characterizing the steady state, so as to provide a convenient matrix notation for solving the steady state. Truncated histories are simply indexed by y^N (the precise index does not matter as long as it remains the same). This also provides an efficient solution to compute the values for the coefficients (ξ_{y^N}) and (ω_{y^N}).

D.3.1 A closed-form formula for the ξs

Let $\boldsymbol{S} = (S_{y^N})_{y^N}$ be the N_{tot}-vector of steady-state history sizes (where N_{tot} is the number of truncated histories). Similarly, let $\boldsymbol{a}$, $\boldsymbol{c}$, $\boldsymbol{l}$, $\boldsymbol{\nu}$, $u'(\boldsymbol{c})$, $\boldsymbol{v'(l)}$ $\boldsymbol{u''(c)}$, $\boldsymbol{v''(l)}$ be the N_{tot}-vectors of end-of-period wealth, consumption, labor supply, Lagrange multipliers, marginal utilities, and derivatives of the marginal utility, respectively. These vectors are known from the steady-state equilibrium of the Bewley model. Each element is defined as the truncation of the relevant variable computed using equation (86). We also define by $\boldsymbol{y} = (y_0^N)_{y^N}$ the N_{tot}-vector of current productivity levels of truncated histories, and by $\boldsymbol{P}$ the diagonal matrix having 1 on the diagonal at y^N if and only if the history y^N is not credit-constrained (i.e., $\nu_{y^N} = 0$), and 0 otherwise. Finally, $\boldsymbol{I}$ is the $(N_{tot} \times N_{tot})$-identity matrix, and $\boldsymbol{\Pi}$ is the transition matrix across truncated histories.

Writing (85), (92) and credit constraints at the steady state yield, respectively:

$$\boldsymbol{S} = \Pi \boldsymbol{S}, \tag{106}$$

$$\boldsymbol{S} \circ \boldsymbol{c} + \boldsymbol{S} \circ \tilde{\boldsymbol{a}} = R\Pi\left(\boldsymbol{S} \circ \tilde{\boldsymbol{a}}\right) + W\boldsymbol{S} \circ \boldsymbol{\xi}^y \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau}, \tag{107}$$

$$(\boldsymbol{I} - \boldsymbol{P})\tilde{\boldsymbol{a}} = \boldsymbol{0}_{N_{tot}\times 1}. \tag{108}$$

The Euler equation for consumption in (93) becomes:

$$\boldsymbol{\xi}^{u,E} \circ u'(\boldsymbol{c}) = \beta R \boldsymbol{\Pi}^\top \left(\boldsymbol{\xi}^{u,E} \circ u'(\boldsymbol{c})\right) + \boldsymbol{\nu},$$

where the transpose matrix $\boldsymbol{\Pi}^\top$ implies expectations about next-period histories. Equivalently:

$$\boldsymbol{D}_{u'(\boldsymbol{c})}\boldsymbol{\xi}^{u,E} = \beta R \boldsymbol{\Pi}^\top \boldsymbol{D}_{u'(\boldsymbol{c})}\boldsymbol{\xi}^{u,E} + \boldsymbol{\nu},$$

where $\boldsymbol{D}_{\boldsymbol{x}}$ stands for the diagonal matrix with the vector $\boldsymbol{x}$ on the diagonal. Finally:

$$\boldsymbol{\xi}^{u,E} = \left[\left(\boldsymbol{I} - \beta R \boldsymbol{\Pi}^\top\right) \boldsymbol{D}_{u'(\boldsymbol{c})}\right]^{-1} \boldsymbol{\nu}. \tag{109}$$

From the FOC on labor in (94), we obtain:

$$\boldsymbol{\xi}^{v,1} = (1-\tau) W (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau} \circ \boldsymbol{\xi}^{y} \circ \boldsymbol{\xi}^{u,1} \circ u'(\boldsymbol{c})./(\boldsymbol{l} \circ v'(\boldsymbol{l})). \tag{110}$$

The equations (88)–(91) yield:

$$\boldsymbol{\xi}^{u,0} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} u(c_{i,t})}{u(c_{t,y^N})}, \quad \boldsymbol{\xi}^{u,1} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} u'(c_{i,t})}{u'(c_{t,y^N})}, \tag{111}$$

$$\boldsymbol{\xi}^{v,0} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} v(l_{i,t})}{v(l_{t,y^N})}, \quad \boldsymbol{\xi}^{y} = \frac{\sum_{y^N \in \mathcal{Y}^{\infty N}} (y_{i,t} l_{i,t})^{1-\tau}}{\left(y_0^N l_{t,y^N}\right)^{1-\tau}}. \tag{112}$$

D.3.2 Matrix expressions for the FOCs

We define the following variables: $\bar{\boldsymbol{\lambda}}_{\boldsymbol{l}} := \boldsymbol{S} \circ \boldsymbol{\lambda}_{\boldsymbol{l}}$, $\bar{\boldsymbol{\psi}} := \boldsymbol{S} \circ \hat{\boldsymbol{\psi}}$, $\bar{\boldsymbol{\Pi}} := \boldsymbol{S} \circ \boldsymbol{\Pi}^\top \circ (1/\boldsymbol{S})$, $\bar{\boldsymbol{\omega}} := \boldsymbol{S} \circ \boldsymbol{\omega}$, $\bar{\boldsymbol{\lambda}}_{\boldsymbol{c}} := \boldsymbol{S} \circ \boldsymbol{\lambda}_{\boldsymbol{c}}$, $\tilde{\boldsymbol{\xi}}^{u,1} := \boldsymbol{\xi}^{u,1}./\boldsymbol{l}$, $\tilde{\boldsymbol{\xi}}^{v,1} := \boldsymbol{\xi}^{v,1}./((1-\tau)W\boldsymbol{\xi}^{y} \circ \boldsymbol{y}^{1-\tau} \circ \boldsymbol{l}^{-\tau})$, and $\tilde{\boldsymbol{\xi}}^{v,0} := \boldsymbol{\xi}^{v,0}./((1-\tau)W\boldsymbol{\xi}^{y} \circ \boldsymbol{y}^{1-\tau} \circ \boldsymbol{l}^{-\tau})$. and notice that $\boldsymbol{S} \circ \tilde{\boldsymbol{\lambda}}_{\boldsymbol{c}} = \boldsymbol{\Pi}\bar{\boldsymbol{\lambda}}_{\boldsymbol{c}}$. The FOCs (97)–(103) become:

$$\bar{\boldsymbol{\psi}} = \bar{\boldsymbol{\omega}} \circ \boldsymbol{\xi}^{u,0} \circ u'(\boldsymbol{c}) - \mu \boldsymbol{S} \tag{113}$$

$$-\left(\bar{\boldsymbol{\lambda}}_c \circ \boldsymbol{\xi}^{u,E} - R\boldsymbol{\Pi}\bar{\boldsymbol{\lambda}}_c \circ \boldsymbol{\xi}^{u,E} - (1-\tau)W\bar{\boldsymbol{\lambda}}_l \circ \boldsymbol{\xi}^{y} \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1}\right) \circ u''(\boldsymbol{c}),$$

$$\boldsymbol{P}\bar{\boldsymbol{\psi}} = \beta R \boldsymbol{P}\bar{\boldsymbol{\Pi}}\bar{\boldsymbol{\psi}}, \tag{114}$$

$$(\boldsymbol{I} - \boldsymbol{P})\bar{\boldsymbol{\lambda}}_c = 0, \tag{115}$$

$$\left(\boldsymbol{\xi}^{y} \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau}\right)^\top \bar{\boldsymbol{\psi}} = -(1-\tau)\left(\boldsymbol{\xi}^{y} \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\boldsymbol{c})\right)^\top \bar{\boldsymbol{\lambda}}_l, \tag{116}$$

$$\tilde{\boldsymbol{a}}^\top \bar{\boldsymbol{\psi}} = -\left(\boldsymbol{\xi}^{u,E} \circ u'(\boldsymbol{c})\right)^\top \Pi \bar{\boldsymbol{\lambda}}_c, \tag{117}$$

$$\bar{\boldsymbol{\omega}} \circ \tilde{\boldsymbol{\xi}}^{v,0} \circ v'(\boldsymbol{l}) + \bar{\boldsymbol{\lambda}}_l \circ \tilde{\boldsymbol{\xi}}^{v,1} \circ v''(\boldsymbol{l}) = \bar{\boldsymbol{\psi}} - \tau \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\boldsymbol{c}) \circ \bar{\boldsymbol{\lambda}}_l + \mu F_L \boldsymbol{S}./((1-\tau)W\boldsymbol{\xi}^{y} \circ \boldsymbol{y}^{-\tau} \circ \boldsymbol{l}^{-\tau}), \tag{118}$$

$$\left(\ln(\boldsymbol{y} \circ \boldsymbol{l}) \circ \boldsymbol{\xi}^{y} \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau}\right)^\top \bar{\boldsymbol{\psi}} = -\left((\boldsymbol{1} + (1-\tau)\ln(\boldsymbol{y} \circ \boldsymbol{l})) \circ \boldsymbol{\xi}^{y} \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\boldsymbol{c})\right)^\top \bar{\boldsymbol{\lambda}}_l. \tag{119}$$

D.3.3 Solving the system

Equation (118) yields:

$$\begin{aligned} \boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,1}\circ v''(\boldsymbol{l})+\tau\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})}\bar{\boldsymbol{\lambda}}_l &= \mu F_L \boldsymbol{S}./((1-\tau)W\boldsymbol{\xi}^y \circ \boldsymbol{y}^{-\tau}\circ \boldsymbol{l}^{-\tau}) + \bar{\boldsymbol{\psi}} - \boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,0}\circ v'(\boldsymbol{l})}\bar{\boldsymbol{\omega}}, \\ \bar{\boldsymbol{\lambda}}_l &= \boldsymbol{M}_0\bar{\boldsymbol{\omega}} + \boldsymbol{M}_1\bar{\boldsymbol{\psi}} + \mu\boldsymbol{V}_0, \end{aligned} \tag{120}$$

with: $\boldsymbol{M}_0 := -\boldsymbol{M}_1\boldsymbol{D}_{\tilde{\boldsymbol{\xi}}^{v,0}\circ v'(\boldsymbol{l})}$, $\boldsymbol{M}_1 := \boldsymbol{D}^{-1}_{\tilde{\boldsymbol{\xi}}^{v,1}\circ v''(\boldsymbol{l})+\tau\tilde{\boldsymbol{\xi}}^{u,1}\circ u'(\boldsymbol{c})}$, and $\boldsymbol{V}_0 := F_L\boldsymbol{M}_1\boldsymbol{S}./((1-\tau)W\boldsymbol{\xi}^y\circ\boldsymbol{y}^{-\tau}\circ\boldsymbol{l}^{-\tau})$.

Equation (113) then implies:

$$\bar{\boldsymbol{\psi}} = \hat{\boldsymbol{M}}_0\bar{\boldsymbol{\omega}} + \hat{\boldsymbol{M}}_1\bar{\boldsymbol{\lambda}}_c + \hat{\boldsymbol{M}}_2\bar{\boldsymbol{\lambda}}_l - \mu\boldsymbol{S}, \tag{121}$$

with: $\hat{\boldsymbol{M}}_0 := \boldsymbol{D}_{\boldsymbol{\xi}^{u,0}\circ u'(\boldsymbol{c})}$, $\hat{\boldsymbol{M}}_1 := -\boldsymbol{D}_{\boldsymbol{\xi}^{u,E}\circ u''(\boldsymbol{c})}(\boldsymbol{I}-R\boldsymbol{\Pi})$, $\hat{\boldsymbol{M}}_2 := (1-\tau)W\boldsymbol{D}_{\boldsymbol{\xi}^y\circ(\boldsymbol{y}\circ\boldsymbol{l})^{1-\tau}\circ\tilde{\boldsymbol{\xi}}^{u,1}\circ u''(\boldsymbol{c})}$.

We obtain using (121) and (120):

$$\bar{\boldsymbol{\psi}} = \boldsymbol{M}_3\bar{\boldsymbol{\omega}} + \boldsymbol{M}_4\bar{\boldsymbol{\lambda}}_c + \mu\boldsymbol{V}_1, \tag{122}$$

where $\boldsymbol{M}_2 := \boldsymbol{I} - \hat{\boldsymbol{M}}_2\boldsymbol{M}_1$, $\boldsymbol{M}_3 := \boldsymbol{M}_2^{-1}(\hat{\boldsymbol{M}}_0 + \hat{\boldsymbol{M}}_2\boldsymbol{M}_0)$, $\boldsymbol{M}_4 := \boldsymbol{M}_2^{-1}\hat{\boldsymbol{M}}_1$, $\boldsymbol{V}_1 := \boldsymbol{M}_2^{-1}(\hat{\boldsymbol{M}}_2\boldsymbol{V}_0 - \boldsymbol{S})$.

Furthermore, equations (114), (115), and (122) imply:

$$\bar{\boldsymbol{\lambda}}_c = \boldsymbol{M}_5\bar{\boldsymbol{\omega}} + \mu\boldsymbol{V}_2, \tag{123}$$

where $\tilde{\boldsymbol{R}}_5 := -((\boldsymbol{I}-\boldsymbol{P}) + \boldsymbol{P}(I-\beta R\bar{\boldsymbol{\Pi}})\boldsymbol{M}_4)^{-1}\boldsymbol{P}(I-\beta R\bar{\boldsymbol{\Pi}})$, $\boldsymbol{M}_5 := \tilde{\boldsymbol{R}}_5\boldsymbol{M}_3$, and $\boldsymbol{V}_2 := \tilde{\boldsymbol{R}}_5\boldsymbol{V}_1$.

Substituting (122) and (123) into (117), we deduce:

$$\mu = -\boldsymbol{L}_0\bar{\omega}, \tag{124}$$

where $\boldsymbol{C}_1 := \tilde{\boldsymbol{a}}^\top(\boldsymbol{V}_1 + \boldsymbol{M}_4\boldsymbol{V}_2) + (\boldsymbol{\xi}^{u,E}\circ u'(\boldsymbol{c}))^\top\boldsymbol{\Pi}\boldsymbol{V}_2$ and $\boldsymbol{L}_0 := (\tilde{\boldsymbol{a}}^\top(\boldsymbol{M}_3 + \boldsymbol{M}_4\boldsymbol{M}_5) + (\boldsymbol{\xi}^{u,E}\circ u'(\boldsymbol{c}))^\top\boldsymbol{\Pi}\boldsymbol{M}_5)/\boldsymbol{C}_1$.

We deduce from (122) and (123):

$$\bar{\boldsymbol{\lambda}}_c = (\boldsymbol{M}_5 - \boldsymbol{V}_2\boldsymbol{L}_0)\bar{\boldsymbol{\omega}}, \tag{125}$$

$$\bar{\boldsymbol{\psi}} = \boldsymbol{M}_6\bar{\boldsymbol{\omega}}, \tag{126}$$

and from (120):

$$\bar{\boldsymbol{\lambda}}_l = \hat{\boldsymbol{M}}_6\bar{\boldsymbol{\omega}}. \tag{127}$$

We have defined $\hat{\boldsymbol{M}}_6 := \boldsymbol{M}_0 + \boldsymbol{M}_1\boldsymbol{M}_6 - \boldsymbol{V}_0\boldsymbol{L}_0$ and $\boldsymbol{M}_6 := \boldsymbol{M}_3 + \boldsymbol{M}_4(\boldsymbol{M}_5 - \boldsymbol{V}_2\boldsymbol{L}_0) - \boldsymbol{V}_1\boldsymbol{L}_0$.

Constructing the constraints. The constraint of equation (119) becomes after substituting the expressions (126) of $\bar{\boldsymbol{\psi}}$ and (127) of $\bar{\boldsymbol{\lambda}}_l$:

$$\tilde{\boldsymbol{L}}_1 \bar{\boldsymbol{\omega}} = 0, \tag{128}$$

where:

$$\begin{aligned}\tilde{\boldsymbol{L}}_1 := &\left(\ln(\boldsymbol{y} \circ \boldsymbol{l}) \circ \boldsymbol{\xi}^y \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau}\right)^\top \boldsymbol{M}_6 \\ &+ \left((\boldsymbol{1} + (1-\tau)\ln(\boldsymbol{y} \circ \boldsymbol{l})) \circ \boldsymbol{\xi}^y \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\boldsymbol{c})\right)^\top \hat{\boldsymbol{M}}_6.\end{aligned}$$

The constraint (116) becomes after substituting the expressions (126) of $\bar{\boldsymbol{\psi}}$ and (127) of $\bar{\boldsymbol{\lambda}}_l$:

$$\tilde{\boldsymbol{L}}_2 \bar{\boldsymbol{\omega}} = 0, \tag{129}$$

where:

$$\tilde{\boldsymbol{L}}_2 := \left(\boldsymbol{\xi}^y \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau}\right)^\top \boldsymbol{M}_6 + (1-\tau)\left(\boldsymbol{\xi}^y \circ (\boldsymbol{y} \circ \boldsymbol{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(\boldsymbol{c})\right)^\top \hat{\boldsymbol{M}}_6.$$

The two constraints imposed on the history weights are $\tilde{\boldsymbol{L}}_1 \bar{\boldsymbol{\omega}} = 0$ and $\tilde{\boldsymbol{L}}_2 \bar{\boldsymbol{\omega}} = 0$. However, $\bar{\boldsymbol{\omega}}$ is a vector of length N_{tot}, while we care in Definition 2 about a vector $\boldsymbol{\omega}^Y = (\omega_y)_y$ of length Y. We define the $N_{tot} \times Y$-matrix $\boldsymbol{R}_0$ that maps a Y-vector into an N_{tot}-vector (where $\boldsymbol{1}_{S_y} \in \mathbb{R}^{S_y}$ is an S_y-vector of 1):

$$\boldsymbol{R}_0 := \begin{bmatrix} \boldsymbol{1}_{S_{y_1}} & 0 & 0 & 0 \\ 0 & \boldsymbol{1}_{S_{y_2}} & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & S_{y_Y} \end{bmatrix},$$

and the $N_{tot} \times Y$-matrix $\boldsymbol{R}_1 := \boldsymbol{D_S R}_0$ that maps a Y-vector into an N_{tot}-vector, but where history sizes have been accounted for. To obtain dimensions compatible with other vectors and matrices, we define $\boldsymbol{\omega} = \boldsymbol{R}_0 \boldsymbol{\omega}^Y$ and $\bar{\boldsymbol{\omega}} = \boldsymbol{R}_1 \boldsymbol{\omega}^Y$.

In conclusion, the two constraints of Definition 2 on the weights $\boldsymbol{\omega}^Y = (\omega_y)_y$ are:

$$\boldsymbol{L}_1 \boldsymbol{\omega}^Y = \boldsymbol{L}_2 \boldsymbol{\omega}^Y, \tag{130}$$

with $\boldsymbol{L}_1 = \tilde{\boldsymbol{L}}_1 \boldsymbol{R}_1$ and $\boldsymbol{L}_2 = \tilde{\boldsymbol{L}}_2 \boldsymbol{R}_1$.

E Changes in the fiscal system

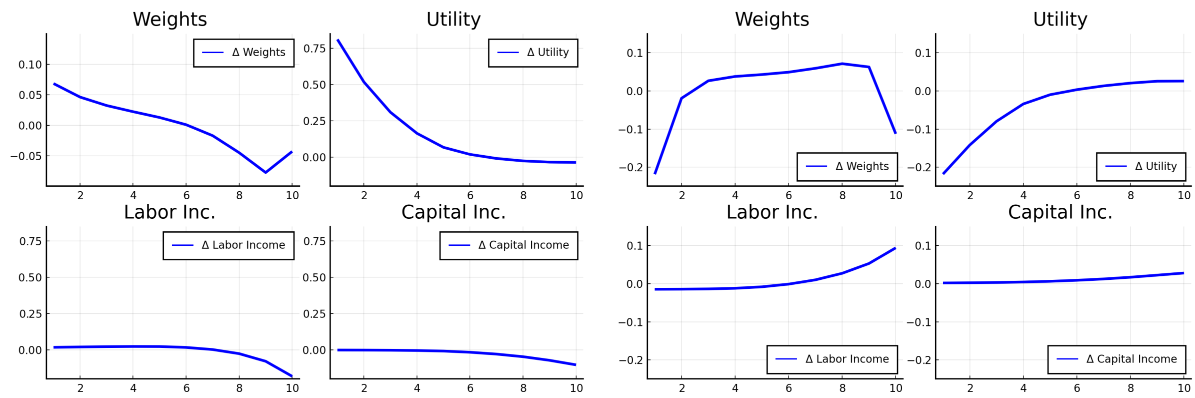
To identify the effect of the fiscal system in the identification of weights, we now assume that fiscal systems are swapped between the two countries: France adopts the US fiscal system and vice versa. Table 6 summarizes the new fiscal system for each country.

	United States	**France**
τ_k	0.35	0.36
τ_c	0.18	0.05
κ	0.98	0.65
τ	0.23	0.16
B/Y	0.21	0.91

Table 6: New fiscal system for the United States and France in the current experiment.

We use our estimation strategy to compute the new SWF weights with the updated fiscal system of Table 6. We report in Figure 7 the differences implied by the new fiscal system compared to the benchmark for some key variables. These variables are SWF weights, utility level, labor supply, and capital income change. The results are averaged for each productivity level.

Considering the United States in panel (a), we observe that the change in the fiscal system increases the weights of low-productivity agents and decreases those of high-productivity agents. This results from low-productivity agents benefiting from the new fiscal system, as can be seen from the period utility, which decreases with productivity. We also observe that the new fiscal system makes both labor and capital income more progressive. Considering France in panel (b), we first observe the opposite variations. As can be seen from the decreasing utility, low-productivity agents suffer from the new fiscal system, which contributes to lowering the SWF weights of low-productivity agents. The hump-shaped weights arise because high-productivity agents benefit from the new fiscal system, which makes the labor and capital income less progressive.



(a) United States (b) France

Figure 7: Difference in weights between the US and French fiscal systems.

F Robustness checks for SWF weights

We conduct three robustness checks regarding our estimation of SWF and IWF weights. First, we relax two assumptions of our identification strategy of SWF weights: (i) that the weights are determined exactly by the Ramsey constraints by imposing a parametric functional form (Section F.1); (ii) that the weights depend only on the current productivity level (Section F.2). Finally, we assess the sensitivity of IWF weights to political weights (G).

F.1 Non-parametric weights

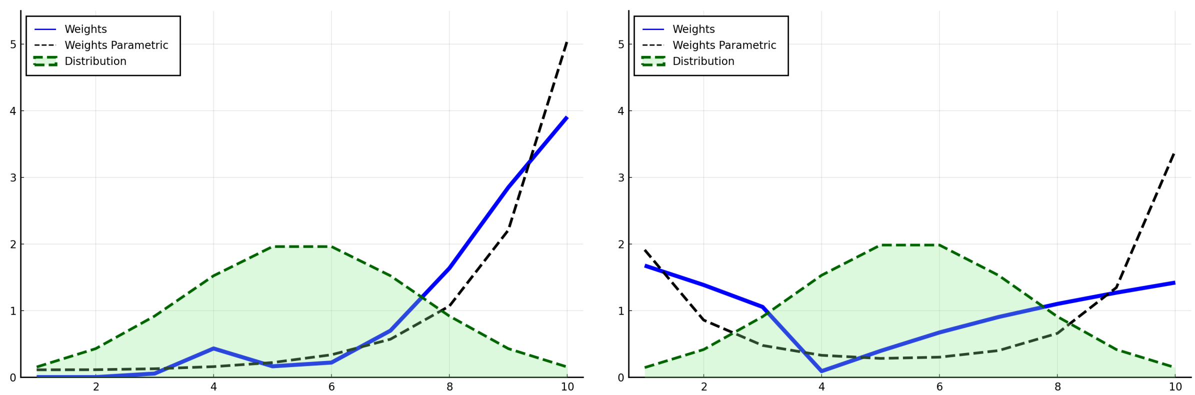
In the baseline quantitative exercise, we estimated parametric weights by imposing a functional relationship between the weights and productivity to obtain an exact identification (see Definition 2). We relax here this assumption and estimate non-parametric weights, by choosing, among all the weights verifying the Ramsey constraints, those with the lowest variance.

As explained in Section 3.4 (see equation (130)), the Ramsey FOCs impose two constraints: $\sum_{y\in\mathcal{Y}} L_{k,y}\omega_y = 0$, where $L_{k,y} \in \mathbb{R}$ ($k = 1, 2$ and $y \in \mathcal{Y}$). The variance-minimizing weights are characterized by the vector $(\hat{\omega}_y)_y$, solving the following program:

$$(\hat{\omega}_y)_y = \text{argmin}_{(\omega_y)_y} \sum_{y\in\mathcal{Y}} \pi_y(\omega_y - 1)^2, \tag{131}$$

$$\text{s.t. } 0 = \sum_{y\in\mathcal{Y}} L_{k,y}\omega_y \text{ for all } k = 1, 2, \tag{132}$$

$$1 = \sum_{y\in\mathcal{Y}} \pi_y\omega_y. \tag{133}$$



(a) United States (b) France

Figure 8: Non-parametric weights (solid line) as a function of productivity levels for the US and France.

Figure 8 plots the non-parametric weights (blue solid lines) along the productivity dimension for the agents. We also report the parametric weights discussed in Section 5.3

(black dashed lines). Parametric and non-parametric weights are quite close to each other and exhibit a similar shape. The weights are increasing in the US and have a U-shape in France, with a high value of weights for low-productivity agents. From this experiment, we conclude that the shape of the weights is robust to the identification strategy, even if the value of the weights for high-productivity agents is not exactly identified in France.

F.2 Weights per truncated history

We relax the assumption here that the SWF weights depend solely on the current productivity level. We assume that weights possibly depend on the whole truncated history. We thus need to compute Y^N weights instead of Y. These weights are thus strongly under-identified. We use the same identification strategy as for non-parametric weights in Section F.1. We select the minimal-variance weights verifying the constraints imposed by the Ramsey program. Formally, the weights $(\hat{\omega}_{y^N})_{y^N}$ are determined as follows:

$$(\hat{\omega}_{y^N})_{y^N} = \operatorname{argmin}_{(\omega_{y^N})_{y^N}} \sum_{y^N \in \mathcal{Y}^N} S_{y^N}(\omega_{y^N} - 1)^2, \tag{134}$$

$$\text{s.t. } 0 = \sum_{y^N \in \mathcal{Y}^N} \tilde{L}_{k,y^N}\omega_{y^N} \text{ for all } k = 1, 2, \tag{135}$$

$$1 = \sum_{y^N \in \mathcal{Y}^N} S_{y^N}\omega_{y^N}, \tag{136}$$

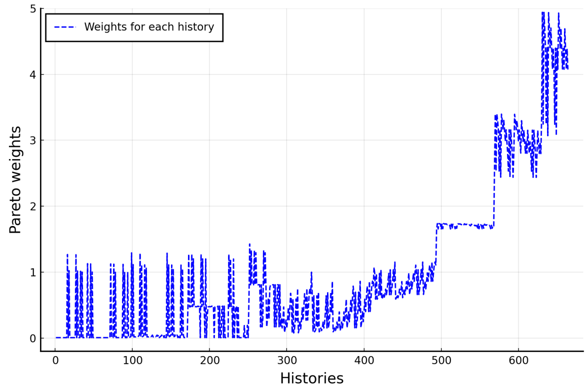
where $\tilde{L}_1$ and $\tilde{L}_2$ are defined in (128) and (129).

In Figure 9, we plot these history weights for the US in panel (a) and France in panel (b). We restrict ourselves to histories with a positive mass. To make them comparable with previous parametric weights, we compute an average weight by summing the weights of truncated histories that have the same productivity level in the first period, and taking into account the size of each truncated history. This results in 10 weights, as for the initial weights. The results are plotted in Figure 10, where we report the non-parametric Pareto weights as a solid blue line and the average history weights as a dashed red line (averaged over histories having the same current productivity level). We can observe that the average history weights (red dashed line) closely approximate the non-parametric ones (blue line). Despite small differences, the two methods imply very similar weights.

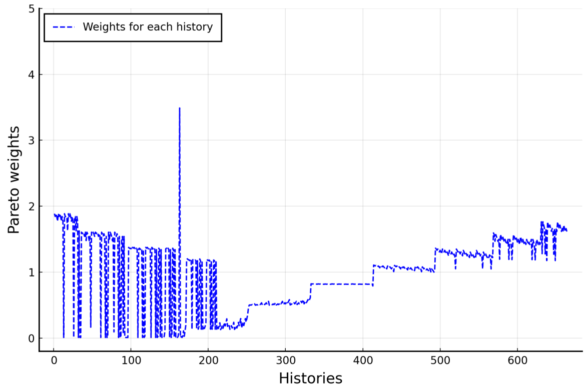
G Outcomes with other political weights

As a robustness check, we provide estimates of the IWFs using different shapes of political weights and using the same estimation strategy. We consider three cases. First, we estimate the SWF assuming that the political weights are the same for all agents within France and within the US.

The results are provided in Figure 11. One can observe that the weights are very

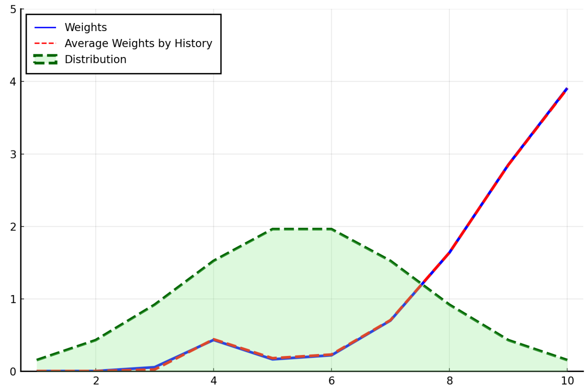


(a) United States

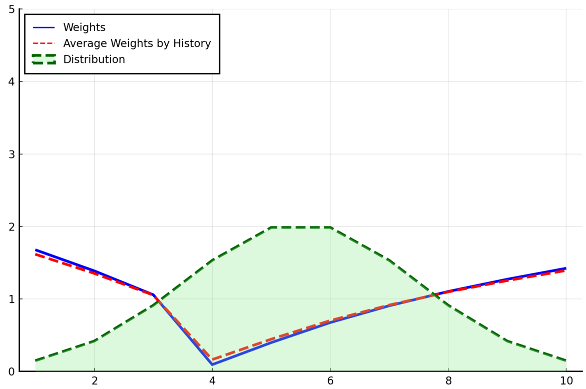


(b) France

Figure 9: History weights for the US and France.
Histories are arranged in the ascending order of the first-period productivity level.

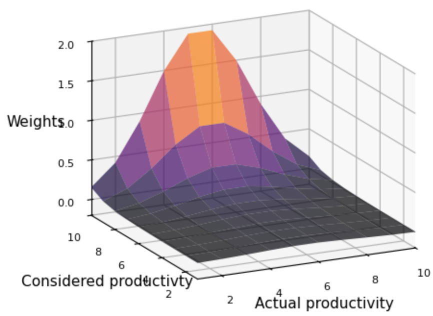


(a) United States

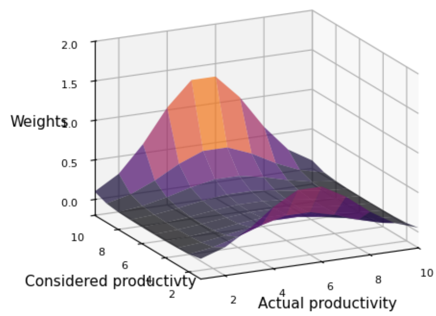


(b) France

Figure 10: Comparison between non-parametric weights and average history weights for the US and France.



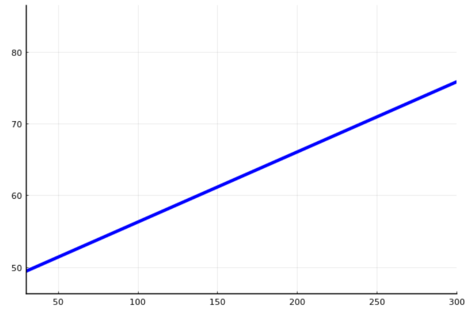
IWFs: US



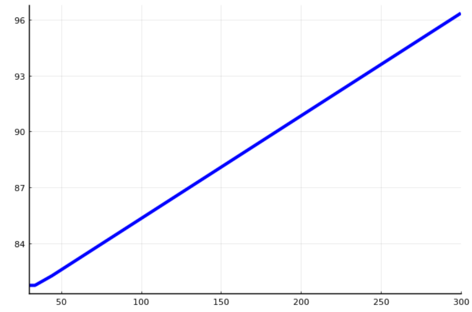
IWFs : France

Figure 11: Implied IWFs when political weights are the same for productivity levels, for the US and France.

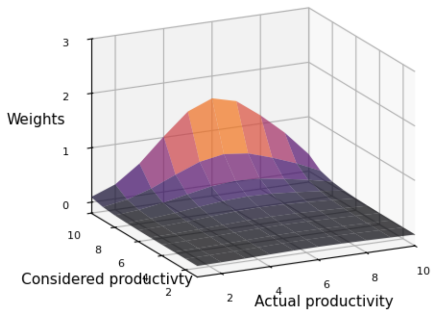
similar to those provided in the benchmark case. The only small difference is that the social weights are a little smaller for high productivity agents, as they have less political power compared to the benchmark case.



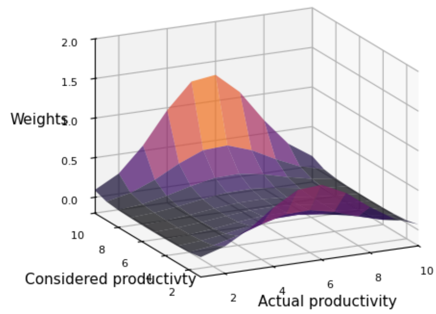
(a) Political weights : US



(b) Political weights : France



(c) IWFs: US



(d) IWFs : France

Figure 12: Linear political weights for the US and France and implied IWFs.

In addition, in Figure 12, we consider linear political weights (instead of concave) for the US and France given in panel (a) and (b). One can observe in panel (c) and (d) that the estimated IWFs are close to the ones found in the benchmark case. The only difference is that the weights given by intermediate-productivity levels to the high-productivity levels are a little lower.

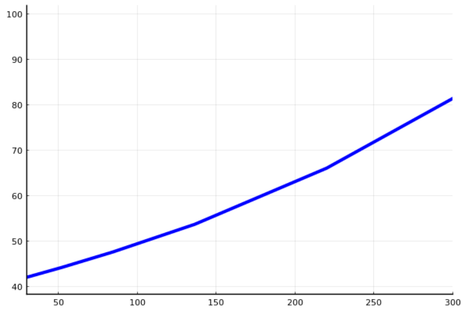
In Figure 13, we consider convex political weights for the US and France given in panel (a) and (b). Once again, the estimated weights are close to the benchmark outcome with even lower weights given to high-productivity levels by intermediate-productivity levels.

H Results with self-interested priors

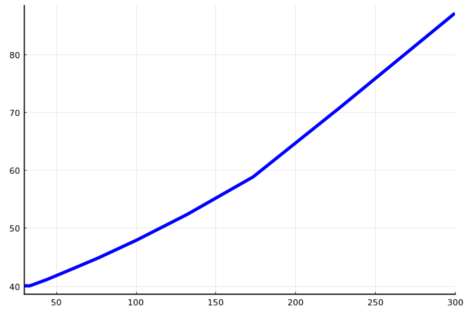
With the political weights of Section 6, we can calculate the IWFs of equation (46) in Definition 3, using the self-interested prior

$$\omega_{\tilde{y},y}^{prior} = 1_{y=\tilde{y}}$$

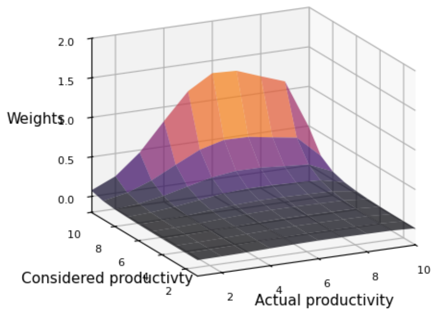
The results are plotted in Figure 14, where the left panel (a) is for the US and the right



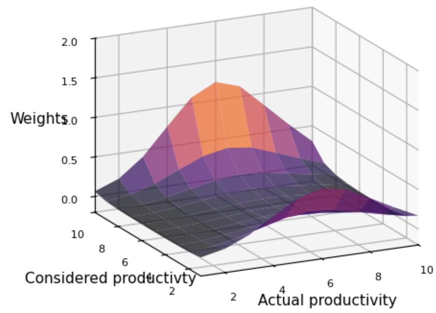
(a) Political weights : US



(b) Political weights : France



(c) IWFs: US



(d) IWFs : France

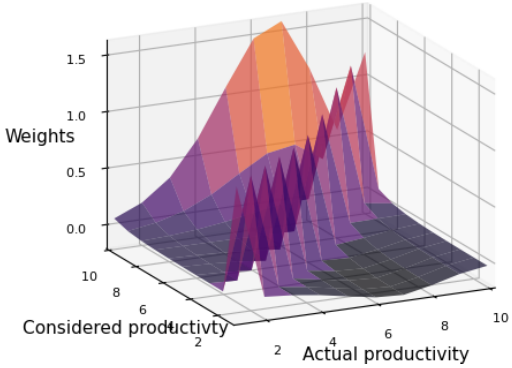
Figure 13: Convex political weights for the US and France and implied IWFs.

panel (b) is for France. More precisely, we report the politically-weighted factors given by agents of productivity y (*Actual productivity*) to agents of productivity $\tilde{y}$ (*Considered productivity*). The politically-weighted factors on the z-axis are equal to the value of $\omega_{P,y}\pi_y\omega_{y,\tilde{y}}$ for all $y, \tilde{y} = 1, \ldots, 10$. As explained in Section 3.4, these factors have a simple interpretation: the larger the factors $\omega_{P,y}\pi_y\omega_{y,\tilde{y}}$, the more agents y affect the valuation of the planner for agents $\tilde{y}$.

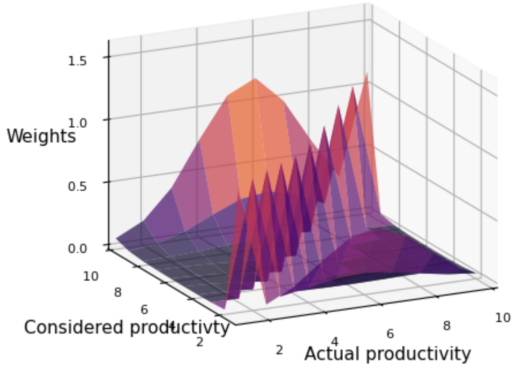
First, in both countries, the diagonal shows high weights, reflecting that the self-interest motive dominates the altruistic one: agents mostly care about their own productivity.[32] The diagonal weights are also increasing with productivity in both countries, which mirrors the higher political weight of high-productivity agents. In the US, the increase is steeper than in France, and diagonal weights reach higher values than in France because SWF weights are also higher. In consequence, the most productive agents have the highest impact on social preferences, and this is especially true for the US. This is consistent with the results of Section 5.3.

Second, out of the diagonal, the US weights exhibit an increasing pattern in $\tilde{y}$ for each productivity level y. This is especially true for middle-class agents, corresponding to intermediate values of y, who have the largest share in the population. Such a shape generates a low level of redistribution, as discussed in Section 2.3: higher weights are

[32]We recall that equation (46) implies: $\pi_y\omega_{P,y}\hat{\omega}_{yy} = 1 + \frac{\pi_y(\omega_{P,y})^2}{\sum_{\tilde{y}\in\mathcal{Y}^\infty}\pi_{\tilde{y}}(\omega_{P,\tilde{y}})^2}(\omega_y - 1)$.



(a) US



(b) France

Figure 14: IWF weights for the US and France.
The IWF weights are measured as politically-weighted factors $(\omega_{P,y}\pi_y\omega_{y,\tilde{y}})_{y,\tilde{y}}$. *Actual productivity* is the productivity y of the agents, while *Considered productivity* is the productivity $\tilde{y}$ of the agents under consideration.

attributed to the most productive agents. Third, out of the diagonal, the French weights exhibit a U-shape pattern, consistent with the findings of Section 5.3 for the French SWF.